EDO Board Prep Study Guide

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Guide Updated: 2025-12-27

EDO Coursebook Version: 2025-09-19

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1 Organization

1.1 Command Chains and Governance

1.1.1 Operations Chain (Title 10)

Chain flow. President → Secretary of War (SECWAR) → Combatant Commander (CCDR). The Chairman of the Joint Chiefs of Staff (CJCS) is the principal military adviser; service chiefs are *not* in the operational chain of Title 10.

Board cue. Boards love the contrast with acquisition authority; avoid conflating operational and acquisition chains.

Echelons. Counted from the Chief of Naval Operations (CNO) (e.g., CNO is Echelon I, Systems Commands (SYSCOMs) are Echelon II, etc.).

Figure 1.1 shows the Title 10 chain of command. Figure 1.2 shows the Chain of Command from the President to Echelon III commands.

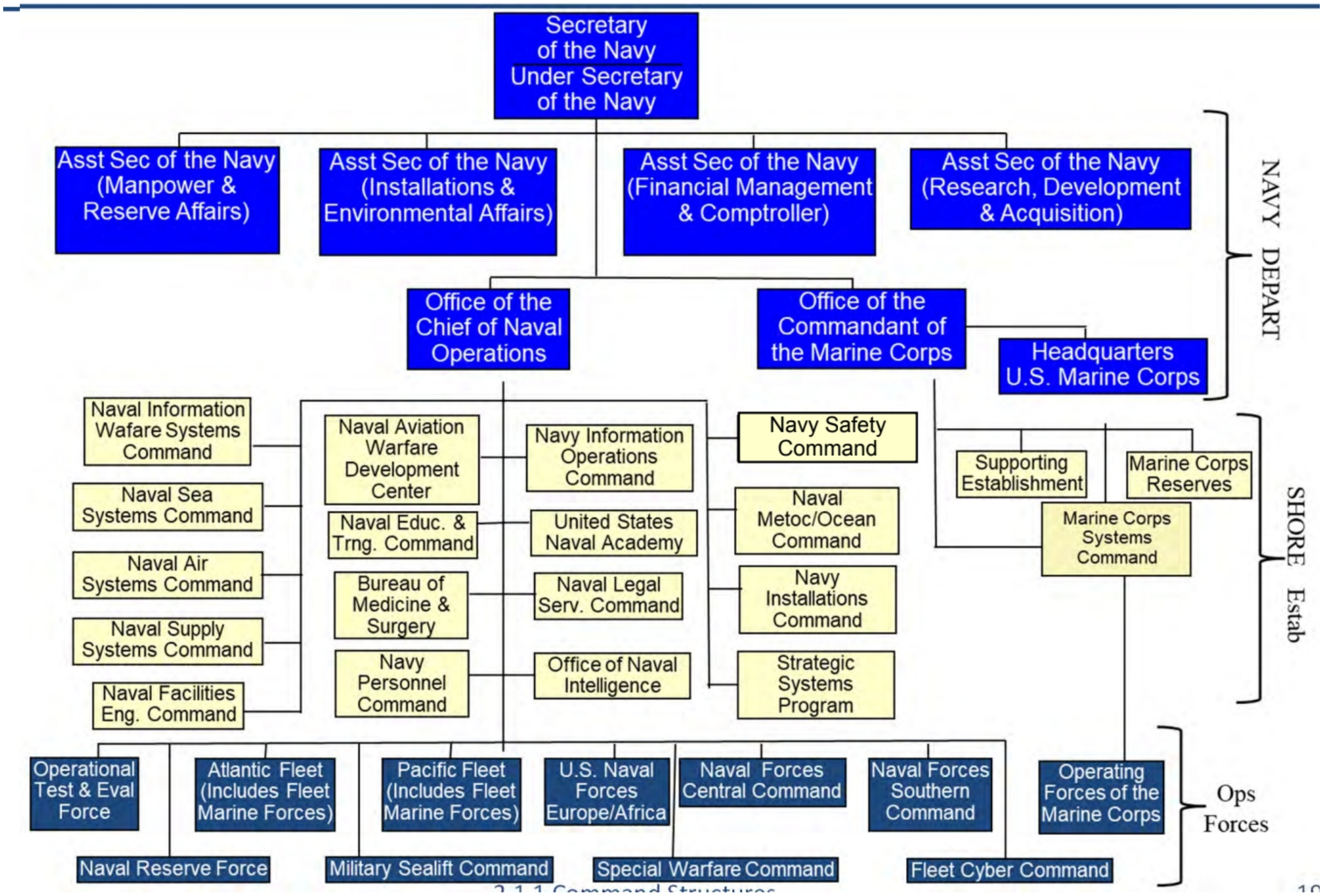


Figure 1.1. *Title 10 Chain of Command. Source:* 2.1.1. Command Structures, *2025 [1].*

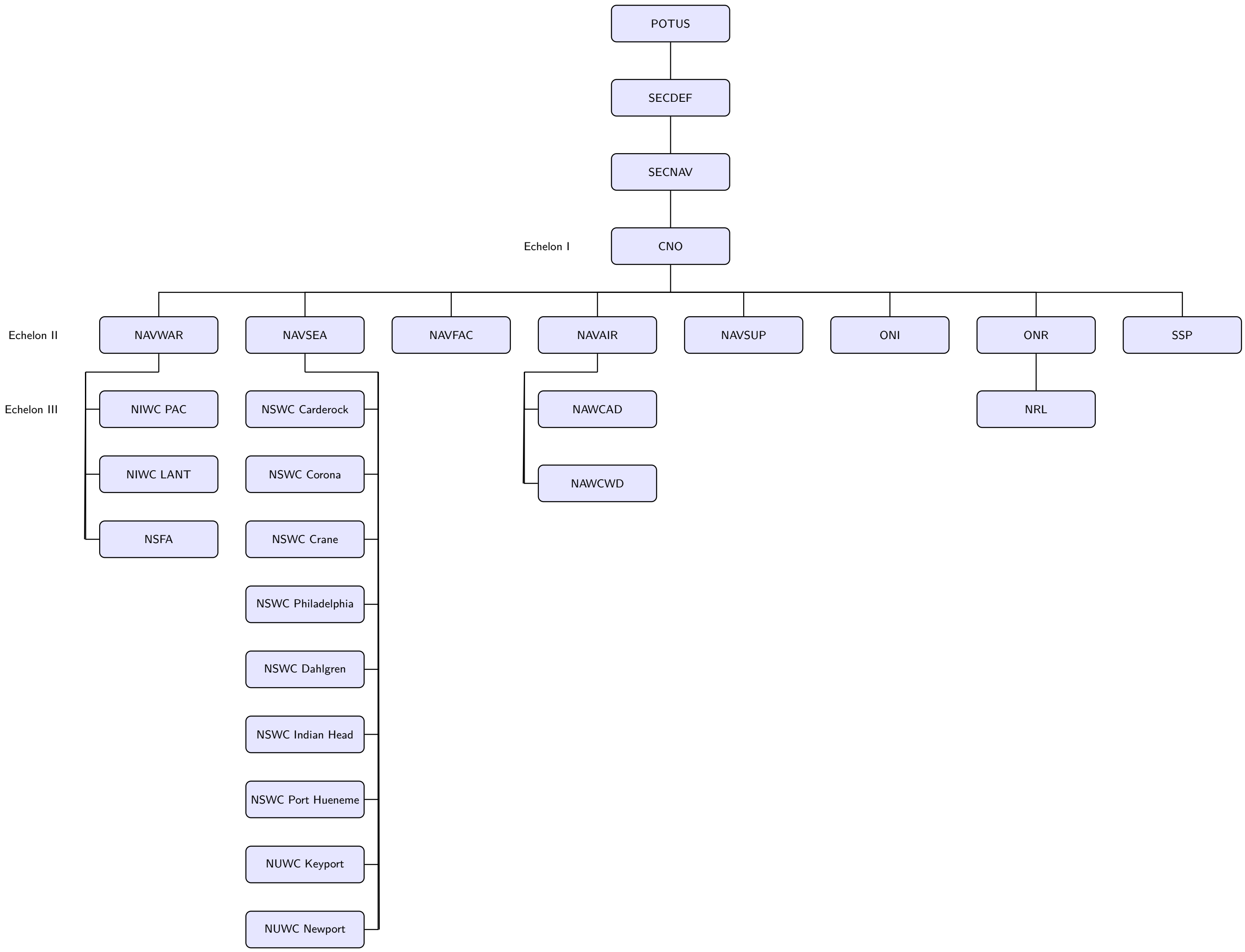


Figure 1.2. *Organization Command Structure*

1.1.2 Acquisition Governance (Department of the Navy)

Policy authority. Assistant Secretary of the Navy for Research, Development and Acquisition (ASN(RD&A)) is the Service Acquisition Executive (SAE) for the Department of the Navy (DON) and sets policy for Program Executive Offices (PEOs) and Direct Reporting Program Managers (DRPMs) [2, 3].

Matrix execution. PEOs hold program authority; SYSCOMs (Naval Sea Systems Command (NAVSEA), Naval Information Warfare Systems Command (NAVWAR), Naval Air Systems Command (NAVAIR)) provide the Technical Authority (TA) warrants and workforce; Naval Warfare Centers (NWCs) deliver research, development, test, and engineering support [2].

Flag/PEO billets (current). **EDO-led today:** ASN(RD&A)'s Principal Military Deputy (PMILDEP) (Vice Admiral (VADM) Seiko Okano), Commander, Naval Information Warfare Systems Command (COMNAVWAR) (Rear Admiral (Upper Half) (RADM) J. Kurt Rothenhaus), Program Executive Office, Ships (PEO Ships) (RADM Brian Metcalf), Program

Executive Office, Aircraft Carriers (PEO CVN) (RADM Casey Moton), and Program Executive Office, Attack Submarines (PEO SSN) (RADM Jonathan Rucker) anchor TA and acquisition governance for the community [4, 5, 6, 7, 8].

Non-Engineering Duty Officer (EDO) leaders (awareness): NAVSEA 08 is currently ADM William J. Houston (Submarine officer), Program Executive Office, Integrated Warfare Systems (PEO IWS) is RDML Thomas J. Dickinson (Surface Warfare Acquisition Professional), Program Executive Office, Manpower, Logistics and Business IT (PEO MLB) is RDML Kevin P. Biehn (Surface Warfare Acquisition Professional), NAVSEA 05 is RDML Jay A. Seif (Submarine Acquisition Professional), and Program Executive Office, Unmanned and Small Combatants (PEO USC) is filled by an SES (formerly RDML Kevin R. Smith) [9].

(Appendix C.2: See Appendix C.2 for the current flag billet roster and non-EDO billets.)

Figure 1.3 shows the Navy acquisition governance chain of command.

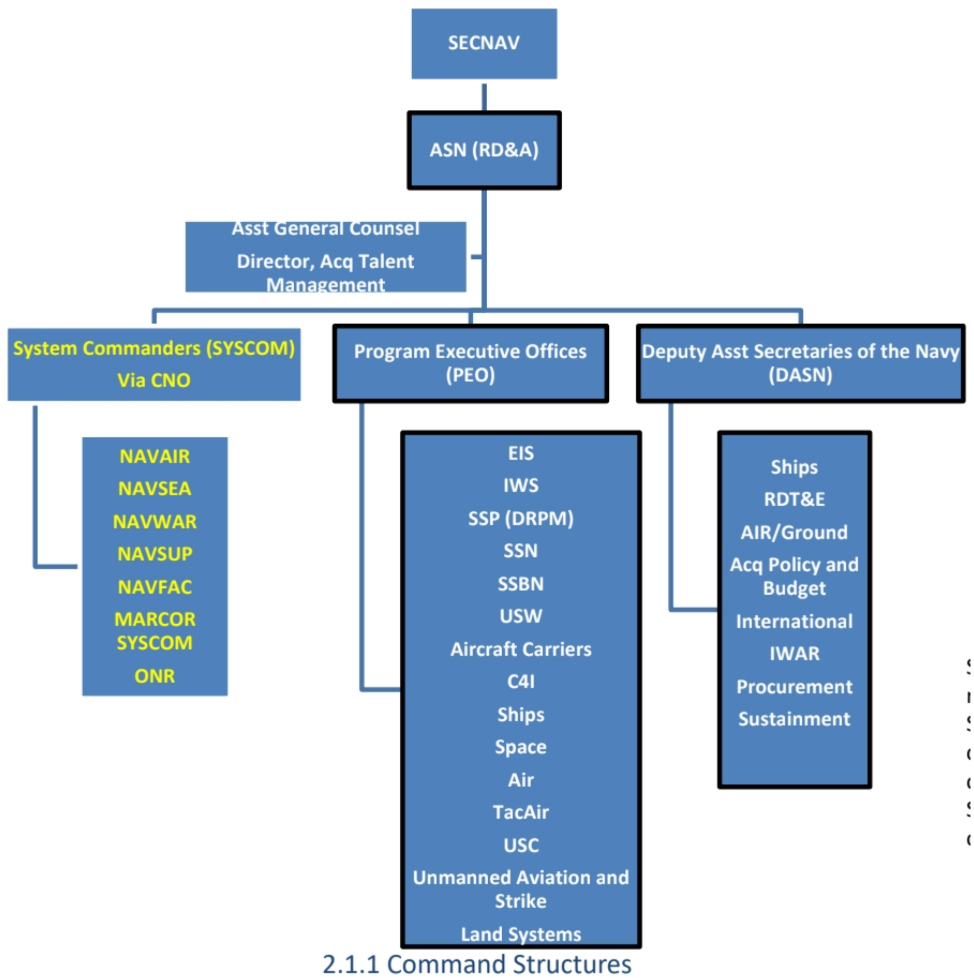


Figure 1.3. *Acquisition Governance Chain of Command. Source:* 2.1.1. Command Structures, *2025 [1].*

1.1.3 ASN (RD&A) Organization

Current as of 2025-12-18, the ASN(RD&A) team is led by Mr. Jason L. Potter (Performing the Duties) with PMILDEP VADM Seiko Okano supporting Department of the Navy acquisition governance. Figure 1.4 shows the current office alignment, and Tables I and II capture roles and responsibilities for the ASN(RD&A), PMILDEP, and Deputy Assistant Secretary of the Navys (DASNs).

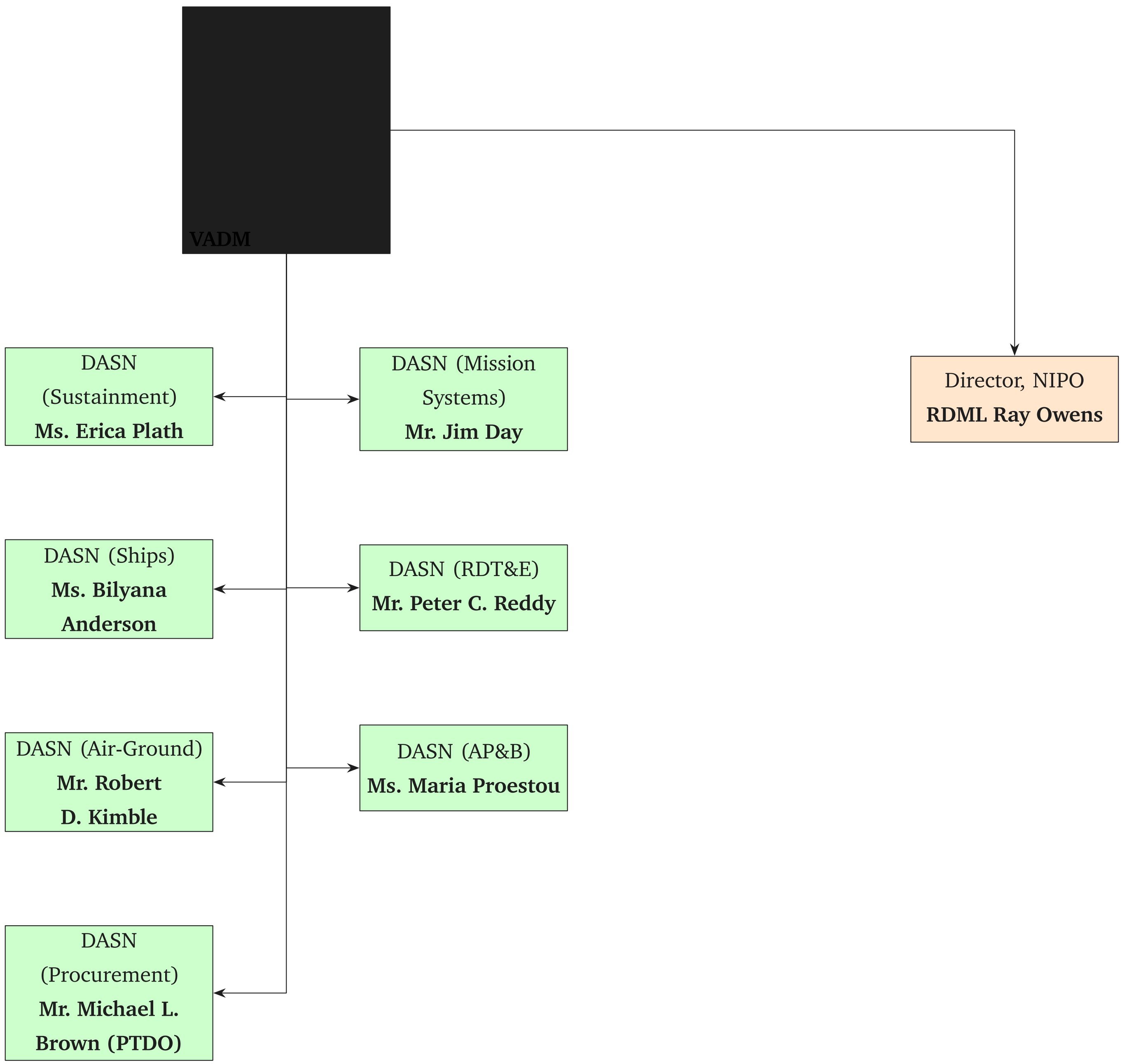


Figure 1.4. ASN (RD&A) organization chart (current as of 2025-12-18). Source: ASN (RDA) Organization Chart, 2025 [10]

Table I lists the ASN(RD&A) and PMILDEP roles.

Table I: ASN (RD&A) leadership roles

Role	Person	Roles and responsibilities
Performing the Duties of ASN(RD&A)	Jason L. Potter	Serves as the DON SAE with authority and accountability for acquisition functions, policy, and program oversight; represents the DON to OSD and Congress; and acts as Milestone Decision Authority for ACAT IC programs.
PMILDEP	VADM Seiko Okano	Provides senior military advice to the ASN(RD&A), helping manage DON research, development, and acquisition activities and delivering operational perspective across the enterprise.

Source: *ASN (RDA) Organization Chart*, 2025 [10]

Table II summarizes the DASNs portfolio leads and their responsibilities.

Table II: Deputy Assistant Secretaries of the Navy (ASN (RD&A))

Office	Person	Roles and responsibilities
DASN (Sustainment)	Ms. Erica Plath	Principal advisor on sustainment; leads the enterprise Naval Sustainment System framework, sets sustainment policy, reviews program performance, and modernizes infrastructure to reduce costs and strengthen the supply chain.
DASN (Ships)	Ms. Bilyana Anderson	Principal advisor for aircraft carriers, surface ships, and submarines; monitors NAVSEA and PEOs for ships and submarines; and advises on the shipbuilding industrial base, ship disposal, and donation.
DASN (Air-Ground Programs)	Mr. Robert D. Kimble	Principal advisor for aircraft, ground systems, cruise missiles, and air-launched weapons; oversees programs managed by PEO (Air), Tactical Aircraft, Unmanned Aviation, Joint Strike Fighter, and Land Systems.
DASN (Procurement)	Mr. Michael L. Brown (PTDO)	Leads the DON contracting community; serves as the DON Competition Advocate General; establishes contracting, services acquisition, and government property policy; and drives improvements to the procurement system.
DASN (Mission Systems)	Mr. Jim Day	Principal advisor for C4I, enterprise IT, and business systems; dual-hatted as the DON CIO Chief Technology Officer; and provides acquisition oversight for PEO C4I, PEO Digital, and PEO MLB.
DASN (RDT&E)	Mr. Peter C. Reddy	Principal advisor for Navy science, technology, advanced RDT&E, and system prototype programs; leads the Department Chief Systems Engineer and the Deputy for Test and Evaluation.

Continued on next page

Table II: Deputy Assistant Secretaries of the Navy (ASN (RD&A)) **(Continued)**

Office	Person	Roles and responsibilities
DASN (AP&B)	Ms. Maria Proestou	Principal advisor for acquisition policy (including Earned Value Management), the PPBE process, and Congressional affairs; directs Navy ACAT program analyses; manages the ASN(RD&A) information system; and oversees corporate operations.
Director, Navy International Programs Office	RDML Ray Owens	Leads NIPO in managing and implementing international security assistance programs, cooperative development, and technology security policy to build integrated capability with allies and partners.

Source: *ASN (RDA) Organization Chart*, *Deputy Assistant Secretary of the Navy (Ships)*, *Deputy Assistant Secretary of the Navy (Research, Development, Test and Evaluation)*, *Deputy Assistant Secretary of the Navy (Air-Ground Programs)*, *Deputy Assistant Secretary of the Navy (Acquisition Policy and Budget)*, *Navy International Programs Office (NIPO)*, *Deputy Assistant Secretary of the Navy (Information Warfare)*, *Deputy Assistant Secretary of the Navy (Procurement)*, *Deputy Assistant Secretary of the Navy (Sustainment)*, 2025, 2025, 2025, 2025, 2025, 2025, 2025, 2025, 2025 [10, 11, 12, 13, 14, 15, 16, 17, 18]

1.1.4 Approving Authorities

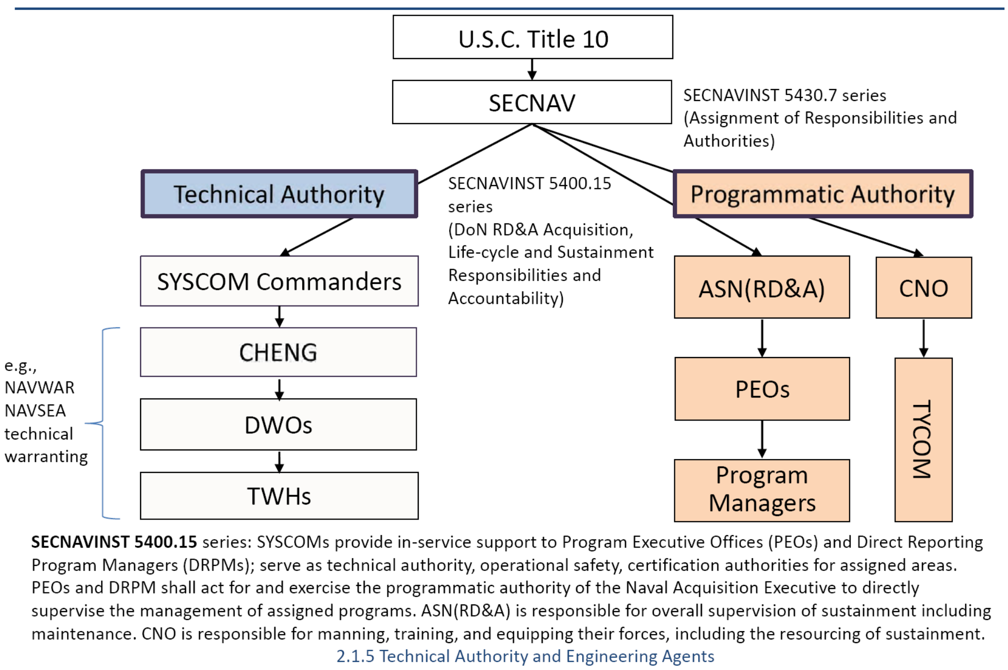


Figure 1.5. *Technical Authority v.s. Program Authority alignment. Source:* 2.1.5. Technical Authority and Engineering Agents, *2025 [19]*

Figure 1.5 contrasts the roles of TA and Program Authority (PA) in Navy acquisition.

1.2 Technical Authority and Engineering Agents

1.2.1 What Technical Authority Is

TA. Independent engineering authority to set, maintain, and certify conformance to technical standards and baselines across the lifecycle [2].

Ultimate Technical Authority (UTA). Executed at the SYSCOM level (e.g., NAVSEA); resides with the Commander and flows through the Chief Engineer via a formal warranting system [19].

Delegated TA. Technical Warrant Holders (TWHs) (*Lead/Local/Warranted* TA) receive domain-specific authority across hull, mechanical, and electrical systems, combat systems, cybersecurity, and airworthiness or seaworthiness [19].

Core TA duties. Must-know responsibilities:

1. Establish and approve technical requirements, standards, and certification criteria;
2. Control the authoritative technical baseline (drawings, specs, interface control);
3. Adjudicate departures and waivers from specification and safety-critical changes; and
4. Certify readiness, readiness for service, and technical acceptability.

Independence. TA remains separate from cost, schedule, and programmatic authority to protect warfighter safety and mission assurance [2].

1.2.2 What EAs Do (and Do Not Do)

Engineering Agents (EAs) perform lifecycle engineering under TA governance. They are *not* TA unless they hold a TA warrant [19]. Common EA roles include:

In-Service Engineering Agent (ISEA). Lifecycle systems engineering for *fielded* systems (e.g., fleet introduction, distance support, troubleshooting/Casualty Report (CASREP) support, maintenance planning, obsolescence, technical manuals/data, configuration of the in-service baseline).

Design Agent (DA). Develops detailed design and configuration documentation for a system or platform (e.g., ship detail design, Initial Capabilities Documents (ICDs), drawings, Technical Data Packages (TDPs)) to TA-approved standards.

Alteration Engineering Agent (AEA). Produces alteration packages (e.g., installation drawings, test procedures, logistics updates) and supports fleet introduction for ship or system changes.

Systems Integration Agent (SIA). Orchestrates system-of-systems integration (e.g., interfaces, interoperability, cybersecurity in the integration space) across combat system elements and platforms.

Technical Direction Agent (TDA). Issues and maintains technical work direction for installations and maintenance (e.g., Technical Work Document (TWD), local instructions) consistent with the TA-approved baseline.

1.2.3 Where Engineering Agent Authority Comes From

Statutory basis. [2] assigns DON acquisition responsibilities, directing SYSCOMs to execute TA and to designate Warfare Centers as EAs that support PEOs and Program Managers (PMs).

Delegations from TA. NAVSEA, NAVAIR, and NAVWAR Chief Engineers issue written warrants or designation letters (Lead/Lab/Local TA) that flow requirements to specific Warfare Center codes; EA charters reference those warrants [19].

Execution orders. PEOs and PMs provide technical direction letters, project orders, and statements of work that scope the EA's tasks while preserving independence for specification compliance and certification [20].

Accountability. EAs report to the SYSCOM Chief Engineer for technical rigor and to the sponsoring PM for cost and schedule; loss of a warrant or charter terminates their authority to issue technical documentation [19].

1.2.4 Waterfront Triage: What Engages Whom

Spec/drawing nonconformance (build or repair). Generate a departure or waiver request for the proper TA warrant holder; minor deviations may be approved by Local or Lead TA, while major or safety-critical issues go to the Chief Engineer/UTA.

In-service failure (CASREP/technical assist). ISEA opens a support case, provides immediate workarounds and troubleshooting, coordinates root-cause analysis, updates technical data, and recommends permanent fixes (engineering change or alteration).

Change to configuration. If a permanent change is needed, the AEA develops the alteration package; SIA validates interfaces; DA updates drawings and TDPs; TA approves and certifies.

Principle to remember. EAs execute engineering; TA sets the rules and certifies compliance.

1.3 NAVSEA Organization and Warfare Centers

1.3.1 HQ Codes (must-know one-liners)

NAVSEA headquarters staff code (SEA) 01 – Comptroller provides financial policy, advice and quality services to ensure NAVSEA's customers' budgets are efficiently and effectively executed. SEA 01 manages appropriation areas as well as providing cost engineering, and industrial analysis.

SEA 02 – Contracts and its field contracting offices under the Contracts Competency award approximately $36 billion in contracts annually for new construction ships and submarines, ship repair, major weapon systems and services.

SEA 03 – Cyber Engineering & Digital Transformation has the responsibility for providing the infrastructure and support services for the programs and developers in the NAVSEA community. SEA 03 delivers combat power to the fleet through enterprise digital capabilities, infrastructure for cyber-secure digital work and innovation, and enhanced enterprise user's experience. *(Appendix C.2: Rear Admiral (Lower Half) (RDML) Huan Nguyen serves as NAVSEA Deputy Commander for Cyber Engineering)*

- Cyber Security: SEA 03 provides the NAVSEA Enterprise with guidance on cyber issues and ensures that all ships are able and ready to detect, defend and recover from cybersecurity attacks.
- IT Services: SEA 03 maintains and facilitates Network Operations and IT Service Delivery for Headquarters NAVSEA end users to enable maximum productivity for NAVSEA civilians and military personnel in the support of the warfighter.
- Digital Transformation: SEA 03 transforms NAVSEA into a digital capability adopting new digital technologies, capitalizing on data, increasing digital skills, and modernizing business processes.

SEA 04 – Industrial Operations has the important mission of getting ships to sea and keeping them ready. SEA 04 is the preferred integrator of maintenance and industrial operations for its Enterprise customers. SEA 04 manages the four Naval Shipyards and the four Supervisor of Shipbuildings (SUPSHIPs) *(Appendix C.2: RDML Scott M. Brown is the NAVSEA Deputy Commander for Industrial Operations)*

SEA 05 – Naval Systems Engineering & Logistics Directorate (TA) is responsible for providing the engineering and scientific expertise, knowledge, and technical authority necessary to design, build, maintain, repair, modernize, certify, and dispose of the Navy's ships, submarines, and associated warfare systems. SEA 05 is organized into 16 groups. *(Appendix C.2: RDML Peter D. Small is the NAVSEA Chief Engineer and Deputy Commander for Naval Systems Engineering (also Commander, Naval Surface Warfare Center (NSWC)/Naval Undersea Warfare Center (NUWC)))*

- Office of the Chief Engineer (SEA 05B)
- Cost Engineering and Industrial Analysis (SEA 05C)
- Explosive Ordnance Engineering (SEA 05E)
- Integrated Warfare Systems Engineering (SEA 05H)
- Logistics and Maintenance Warfare Systems Engineering (SEA 05M)
- Undersea Warfare Systems Engineering (SEA 05N)
- Ship Integrity and Performance Engineering (SEA 05P)
- Industrial Engineering, Technical Policy and Standards (SEA 05S)
- Technology Office (SEA 05T)
- Submarine/Submersible Design and Systems Engineering (SEA 05U)
- Aircraft Carrier Design and Systems Engineering (SEA 05V)
- Surface Warfare Systems Engineering (SEA 05W)
- Marine Engineering (SEA 05Z)

SEA 06 – Sustainment. Life-cycle product support.

SEA 07 – Undersea Warfare. Dual-hatted as Program Executive Office, SSBN (PEO SSBN); provides a full spectrum of research, development, test and evaluation, Hull, Mechanical, and Electrical (HM&E) systems engineering and fleet support services to the in-service submarine and undersea forces. Submarine/Undersea Warfare Technology (SUBTECH) coordinates the development of technologies to fulfill undersea warfare capability requirements.

SEA 08 – Nuclear Propulsion. Manages all technical matters pertaining to naval reactors. Triple-hat duties:

SEA 08 Deputy Commander. Leads nuclear propulsion activities within NAVSEA.

Director, Naval Nuclear Propulsion (Office of the Chief of Naval Operations (OPNAV) 00N). Serves as the Navy's principal nuclear propulsion authority.

Deputy Administrator for Naval Reactors. Interfaces with the National Nuclear Security Administration and the Department of Energy.

SEA 09 – Safety and Regulatory Compliance. Strengthens and aligns safety oversight and reporting.

SEA 10 – Total Force and Corporate Operations performs all operations support for NAVSEA directorates and field activities as well as PEOs. Support includes administrative products and services, career planning, employee development, facilities, foreign military sales coordination, human resources, security, and university research assistance.

SEA 21 – Surface Ship Maintenance, Modernization and Sustainment manages the complete lifecycle support for all non-nuclear surface ships and is the principal interface with the Surface Warfare Enterprise. The directorate is responsible for the maintenance and modernization of non-nuclear surface ships currently operating in the Fleet. Through planned modernization and upgrade programs, SEA 21 will equip today's surface ships with the latest technologies and systems to keep them in the Fleet through their service lives. Additionally, SEA 21 oversees the ship inactivation process, including ship transfers or sales to friendly foreign navies, inactivation and/or disposal.

Program Manager, Ships (PMS) 321. Unmanned small combatants and amphibious ships.

PMS 326. International fleet support.

PMS 339. Surface training systems.

PMS 421. Large surface combatant modernization and sustainment.

PMS 443. Bridge integration and HM&E sustainment.

PMS 451. Destroyer Modernization 2.0 portfolio.

SEA 21I. Inactive Ships Directorate.

Surface Maintenance Engineering Planning Program (SURFMEPP) Activity. Surface Maintenance Engineering Planning Program.

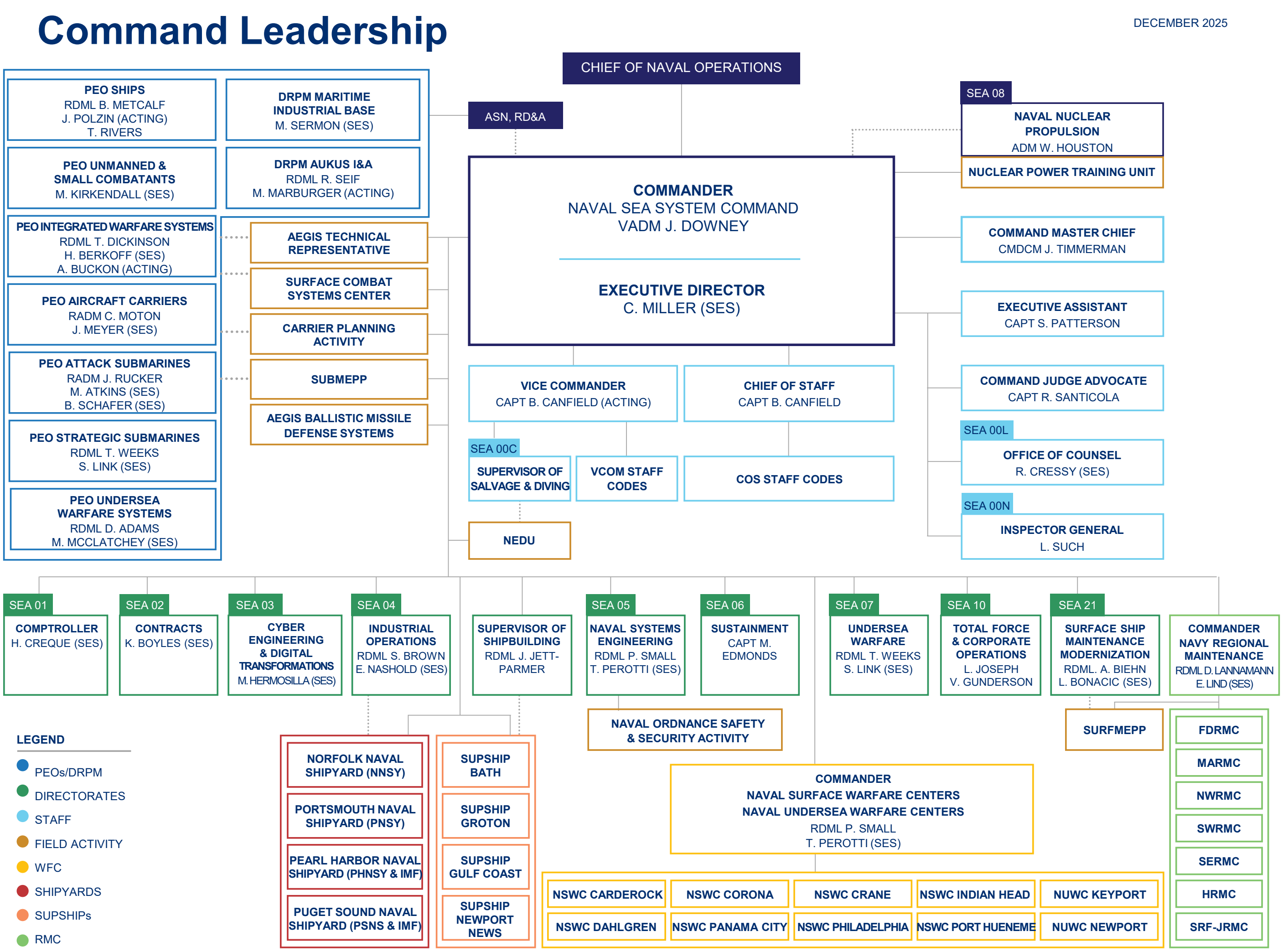


Figure 1.6. *NAVSEA Organization Chart, December 2025.Source:* NAVSEA Organization Chart (December 2025), *2025 [9]*

1.3.2 Warfare Centers (alphabetical)

Table III lists NAVSEA Warfare Centers with locations and descriptions. Figure 1.7 shows the geographic locations of NWCs. For the EDO board, most recommend drawing the US to show where the NWCs are located. The other option is to list them out, but a map is more visual.

Table III: NAVSEA Warfare Centers with locations and descriptions.

Center	Location	Description
NSWC Carderock	West Bethesda, MD	Carderock Division is the U.S. Navy's state-of-the-art research, engineering, modeling, and test center for ships and ship systems; the Navy and maritime communities rely on it for advanced platforms, analytics, and cost-conscious performance improvements.
NSWC Corona	Corona, CA	NSWC Corona Division is the Navy's independent assessment agent, leading NAVSEA data analytics and providing transparency for warfighting readiness, fleet LVC training integration, and metrology/calibration accuracy.
NSWC Crane	Crane, IN	NSWC Crane provides technical engineering solutions and total lifecycle leadership for many of the systems that protect and enable the Warfighter. Every decision we make, technology we create or business relationship we foster is performed with the fighting man and woman in mind. NSWC Crane has concentrated its resources and core competencies in the 3 mission areas which best support the Warfighter. The misson areas are Strategic Mission, Electromagnetic Warfare, and Exepeditionary Warfare.
NSWC Dahlgren	Dahlgren, VA	The Naval Surface Warfare Center Dahlgren Division originally established itself as the major testing area for naval guns and ammunition. Today, it continues to provide the military with the development and integration of warfare systems for the warfighter, warfighting and the future fleet.
NSWC Indian Head	Indian Head, MD	As a field activity of the Naval Sea Systems Command, and part of the Navy's Science and Engineering Establishment, NSWC IHD is the Navy's premier facility for ordnance, energetics and explosive ordnance disposal solutions. We support the warfighter of today and tomorrow through discoveries that anticipate the next generation's future needs.
NSWC Panama City	Panama City, FL	The mission of NSWC, Panama City Division is to conduct research, development, test and evaluation, and in-service support of mine warfare systems, mines, naval special warfare systems, diving and life support systems, amphibious and expeditionary maneuver warfare systems, and other missions that occur primarily in coastal (littoral) regions.
NSWC Philadelphia	Philadelphia, PA	Provide research, development, test and evaluation, acquisition support, engineering, systems integration, in-service engineering and fleet support with cyber-security, comprehensive logistics, and life-cycle savings through commonality for surface and undersea vehicle machinery, ship systems, equipment and material.
NSWC Port Hueneme	Port Hueneme, CA	Integrate, Test, Evaluate, and Provide Life-Cycle Engineering and Product Support for Warfare Systems.

Continued on next page

Table III: NAVSEA Warfare Centers with locations and descriptions. **(Continued)**

Center	Location	Description
NUWC Keyport	Keyport, WA	Keyport's mission is focused on developing and applying advanced technical capabilities to test, evaluate, field, and maintain undersea warfare systems and related defense assets. These advanced technical capabilities directly support the full spectrum of Navy undersea programs.
NUWC Newport	Newport, RI	The Naval Undersea Warfare Center Division Newport provides research, development, test and evaluation, engineering, analysis, and assessment, and fleet support capabilities for submarines, autonomous underwater systems, and offensive and defensive undersea weapon systems, and stewards existing and emerging technologies in support of undersea warfare.

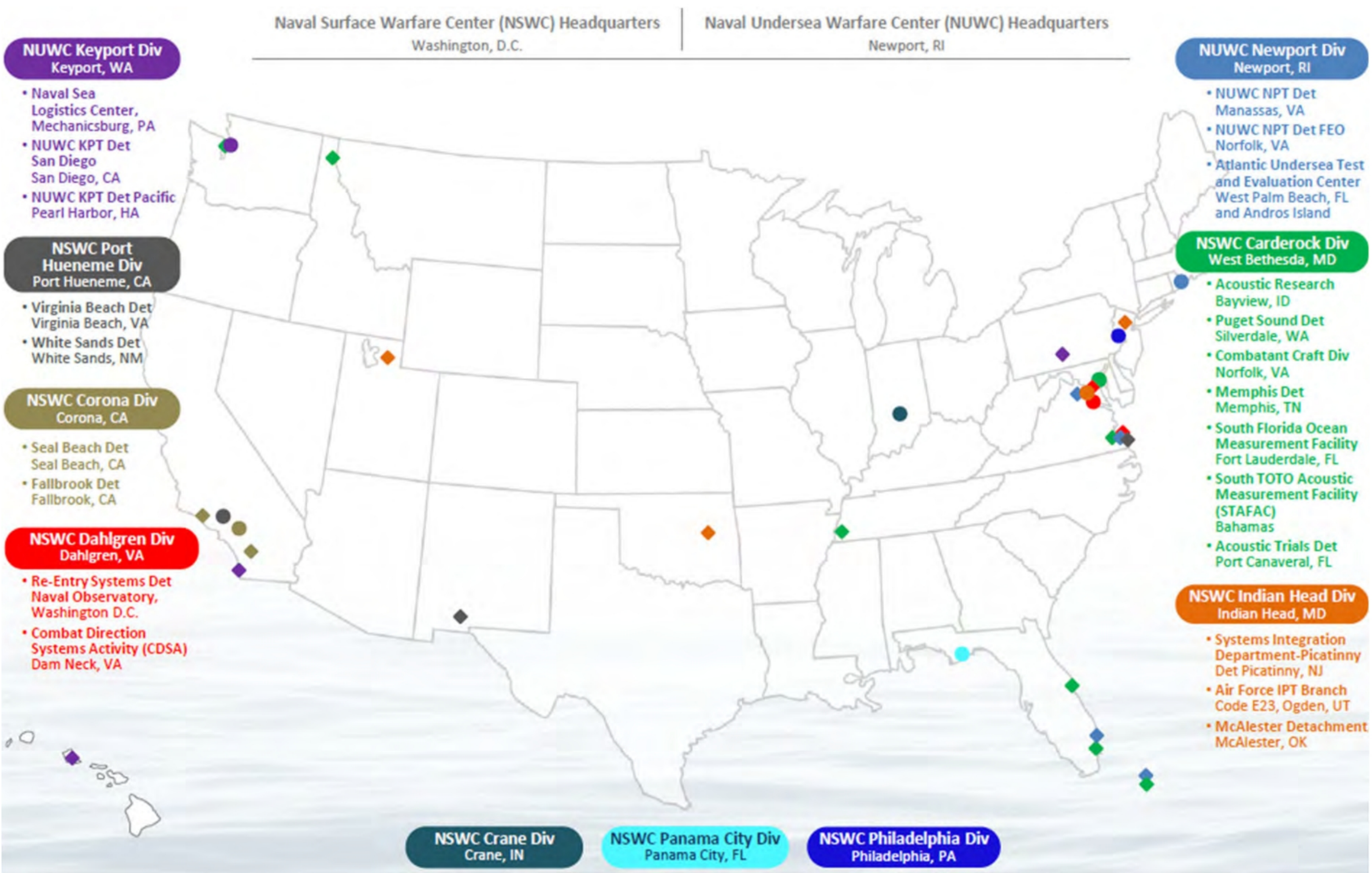


Figure 1.7. *Geographic locations of Navy Warfare Centers.Source:* NAVSEA Organization Chart (December 2025), 2.1.4. Naval Warfare Centers, *2025, 2024 [9, 20].*

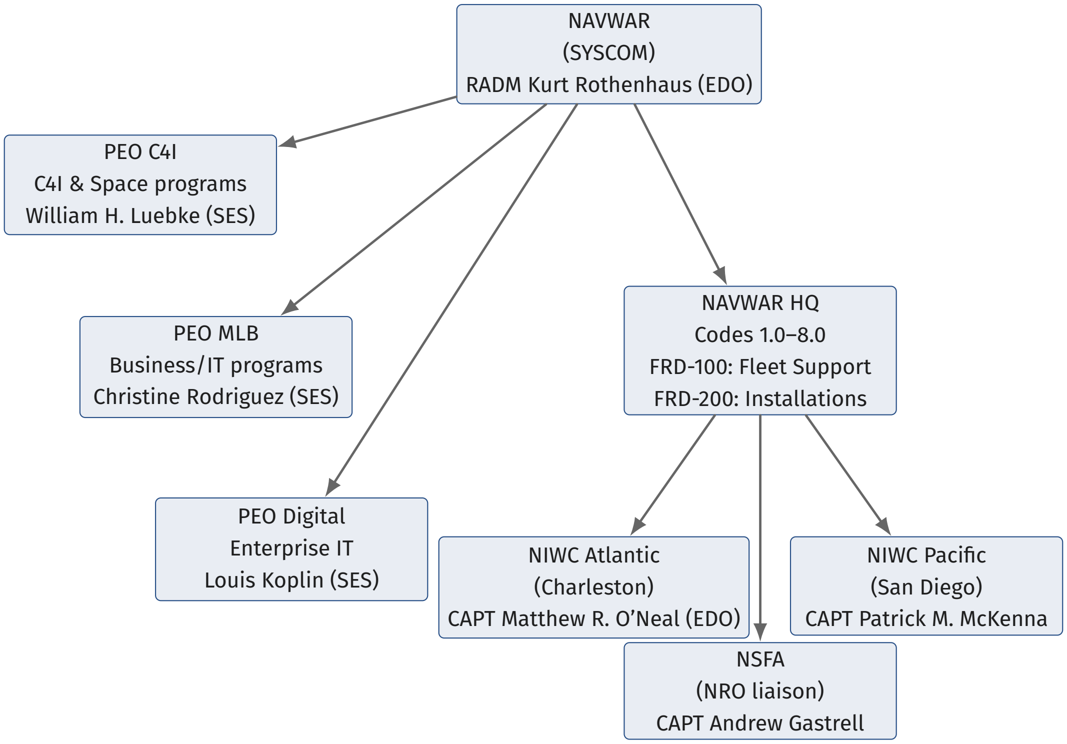


Figure 1.8. *NAVWAR enterprise structure showing the SYSCOM commander with PEO portfolios, headquarters directorates, Field Readiness Directorates, and warfare centers.*

1.4 NAVWAR Enterprise and Warfare Centers

1.4.1 HQ Directorates and FRD

Table IV lists NAVWAR HQ directorates and Fleet Readiness Directorates (FRDs) with one-line roles.

Table IV: NAVWARHQ directorates and Field Readiness Directorates – one-line roles.

Code	Office	One-liner
1.0	Comptroller	Budget formulation/execution; BSO duties; funds control and financial reporting.
2.0	Contracts	Contracting policy/oversight; acquisition strategy; award/administration of instruments.
3.0	Counsel	Legal counsel for acquisition/ethics; protests, claims, agreements, and compliance.
4.0	Logistics & Fleet Support	Life-cycle logistics; sustainment planning; supply/tech data; fleet support integration.
5.0	Chief Engineer / (TA for IT & Cyber)	Technical standards/architecture; interoperability; cyber TA; certification/air-gapping where required.
6.0	Program Management	Program governance; milestone readiness; portfolio integration; interface with PEOs.
7.0	Science & Technology	S&T portfolio; experimentation/prototyping; tech transition to programs of record.

Continued on next page

Table IV: NAVWARHQ directorates and Field Readiness Directorates – one-line roles. **(Continued)**

Code	Office	One-liner
8.0	Corporate Operations	Workforce, security, facilities, enterprise information-technology/chief-information-officer services, PAO, and other corporate enablers.
FRD-100	Fleet Support	Sustainment services and distance support; field engineering and readiness assistance.
FRD-200	Installations	C4I installation planning/execution; shore/afloat integration and cutover support.

Source: *2.1.3. NAVWAR Enterprise*, 2024 [21]

1.4.2 Warfare Centers (alphabetical)

Table V lists NAVWAR Warfare Centers with locations and descriptions.

Table V: NAVWAR Warfare Centers with locations and descriptions.

Center	Location	Description
NIWC LANT	Charleston, SC	Conduct research, development, prototyping, engineering, test and evaluation, installation, and sustainment of integrated information warfare capabilities and services across all warfighting domains with an emphasis on **Expeditionary Tactical Capabilities & Enterprise IT and Business Systems** in order to drive innovation and warfighter information advantage
NIWC PAC	San Diego, CA	Conduct research, development, prototyping, engineering, test and evaluation, installation, and sustainment of integrated information warfare capabilities and services across all warfighting domains with emphasis on **Basic and Applied Research and Tactical Systems Afloat and Ashore** in order to drive innovation and warfighter information advantage.
NSFA	Chantilly, VA	Navy link to the NRO; Provide naval warfare and acquisition expertise to national reconnaissance programs; Coordinate naval space research, development, and acquisition activities with national reconnaissance programs; Provide and coordinate training and tools to enable the fleet to use national space capabilities; Conducts intelligence-related activities essential for US national security.

Source: *2.1.3. NAVWAR Enterprise*, 2024 [21]

NIWC LANT detachments

- **HQ:** Charleston, SC.
- New Orleans, LA.
- Norfolk, VA.
- Tampa, FL.
- Washington, DC.
- Stuttgart, Germany.

- Naples, Italy.
- Rota, Spain.
- Manama, Bahrain.

Figure 1.9. NIWC LANT Locations

NIWC PAC DETACHMENTS

- **HQ:** San Diego, CA.
- Everett, WA.
- Philadelphia, PA.
- Pearl City, HI.
- Santa Rita, Guam.
- Yokosuka, Japan.

1.4.3 Project Overmatch

Mission.

Deliver a resilient Naval Operational Architecture that links platforms, weapons, sensors, and data fabrics for Distributed Maritime Operations and Combined Joint All-Domain Command and Control (CJADC2); NAVWAR provides the digital architecture, data transport, and cyber TA that underpin the hybrid fleet.

Figure 1.10. *NIWC PAC Locations*

Five Eyes coalition network.

Under an Oct 2024 Project Arrangement with Australia, Canada, New Zealand, and the United Kingdom, Cooperative Project Personnel embed in San Diego to co-develop kill-chain networks; Capt. Remil Capili (PM), VADM Seiko Okano (steering chair/PMILDEP), Shayna Bond (international deputy), and David Flowers (international lead) oversee the coalition lab and Rim of the Pacific Exercise (RIMPAC) 26 demonstrations *(Appendix C.2: VADM Seiko Okano is the PMILDEP flag billet holder in Appendix C.2)* [22].

Allied exercises.

Overmatch Five Eyes (Australia, Canada, New Zealand, United Kingdom, United States) (FVEY) solutions are being exercised at Trident Warrior 24, Talisman Sabre 25, and a capstone during RIMPAC 26 to prove resilient comms in contested electromagnetic environments[22].

Rapid onboarding.

Open Data and Applications Government-owned Interoperable Repositories (DAGIR) and Maven Smart Systems pipelines let NAVWAR ingest commercial Artificial Intelligence and Machine Learning (AI/ML) tools in weeks, accelerating fleet readiness visualizations and data-driven decision aids across the hybrid force [23].

Governance.

As PMILDEP and DRPM Overmatch, VADM Okano keeps digital architecture tied to Program Decision Meetings and PEO portfolio trades, ensuring Overmatch data standards propagate into new ship, sub, and aircraft programs [3, 22].

1.5 NAVAIR Warfare Centers

Table VI lists NAVAIR Warfare Centers with locations and descriptions. This is an optional item to know for the EDO board, not as critical as NAVSEA/NAVWAR.

Table VI: NAVAIR Warfare Centers with locations and descriptions.

Center	Location	Description
NAWCAD	Patuxent River, MD	Aircraft/engines/avionics RDT&E; Test Pilot School.
NAWCWD	China Lake / Point Mugu, CA	Weapons systems, ranges, guided-missle integration.

Source: *2.1.4. Naval Warfare Centers*, 2024 [20]

1.6 Navy Program Executive Offices (PEOs)

PEO CVN. Designs, builds, delivers, and sustains nuclear-powered aircraft carriers. *(Appendix C.2: RADM Casey J. Moton holds the PEO CVN billet)*

PMS 312. In-service aircraft carrier program management.

PMS 378. Nuclear-Powered Aircraft Carrier (CVN)-78 class program management.

PMS 379. CVN-79/80 program management.

PEO IWS. Develops and sustains ship and submarine combat systems.

> **Note**
>
> RDML T. Dickinson is currently PEO IWS. He is a Surface Warfare Acquisition Professional, not a EDO.

Integrated Warfare Systems (IWS) 1.0. Aegis combat system lead.

IWS 1.0F. Aegis fleet readiness.

IWS 2.0. Above-water sensors portfolio.

IWS 3.0. Surface ship weapons integration.

IWS 4.0. International and foreign military sales.

IWS 5.0. Undersea systems.

IWS 6.0. Command-and-control systems.

IWS 9.0. Guided Missile Destroyer (DDG)-1000, littoral combat ship, and patrol craft combat systems.

IWS 11.0. Terminal defense systems.

IWS 12.0. North Atlantic Treaty Organization (NATO) Seasparrow programs.

IWS 80.0. Atalanta combat systems.

IWS X. Integrated combat-system document center.

PEO Ships. Oversees surface combatant and amphibious ship construction and modernization. *(Appendix C.2: RADM Brian A. Metcalf serves as PEO Ships)*

PEO USC. Manages littoral combat ships, frigates, expeditionary platforms, and unmanned surface/undersea portfolios. *(Appendix C.2: RDML Kevin R. Smith serves as PEO USC)*

Team Submarines. Integrates strategic and attack submarine acquisition and sustainment.

PEO SSN. Attack submarine programs. *(Appendix C.2: RADM Jonathan E. Rucker serves as PEO SSN)*

PMS 351. New attack submarine acquisition.

PMS 390. Undersea special mission systems.

PMS 391. In-service submarine sustainment.

PMS 394. Advanced undersea system's development.

PMS 450. Virginia class program management.

PEO SSBN. Strategic deterrent submarine portfolio.

PMS 396. In-service Ballistic Missile Submarine, Nuclear (SSBN) sustainment.

PMS 397. Columbia class program management.

Submarine Maintenance Engineering Planning Program (SUBMEPP). Surface maintenance engineering planning program support to ballistic submarine availabilities.

Program Executive Office, Undersea Warfare Systems (PEO UWS). Submarine combat, cyber, and sensor systems.

PMS 401. Submarine acoustic systems.

PMS 404. Undersea weapons.

PMS 415. Undersea defensive warfare systems.

PMS 425. Combat and weapon control systems.

PMS 435. Electromagnetic systems.

PMS 485. Maritime surveillance systems.

PEO C4I. Delivers fleet C4I capabilities.

Program Management Warfare (NAVWAR PEO C4I/NAVAIR PMS series) (PMW) 120. Battlespace awareness and information operations.

PMW 130. Information Assurance (IA) and cybersecurity programs.

PMW 150. Navy command-and-control systems.

PMW 160. Tactical networks.

PMW 170. Communications and Global Positioning System (GPS) navigation.

PMW 740. International C4I integration.

PMW 750. Carrier strike and air integration.

PMW 760. Ship integration.

PMW 770. Undersea integration.

PMW 790. Shore and expeditionary integration.

PMW series focus. Two PMW series within PEO C4I have distinct roles:

PMW 1XX. Major capability development portfolios (new platforms and end-to-end networks) where PEO C4I owns the baseline capabilities document, acquisition strategy, and milestone execution; Warfare Centers supply lead systems integration, test ranges, and engineering agents via tailored project orders [2, 20].

PMW 7XX. Fleet integration portfolios synchronizing new capabilities into existing hulls, aircraft, and shore nodes; emphasizes installation planning, logistics, and interoperability packages coordinated through regional Warfare Centers and type commanders [2, 24].

PEO Digital. Provides enterprise digital services (e.g., Flank Speed collaboration environment).

PEO MLB. Modernizes manpower, logistics, and business information-technology systems.

1.6.1 PEO/Warfare Center Interaction Model

Role Designation. PEOs charter Warfare Centers as EAs (ISEA, DA, SIA) via technical direction letters aligned with [2]; the Warfare Center executes under Navy Working Capital Fund (NWCF) funding while reporting performance to the PM [20].

Requirements Translation. PM Integrated Product Teams (IPTs) decompose Capabilities Development Documents (CDDs) into technical requirements that Warfare Center TA holders validate and flow down through alteration packages, interface control documents, and test plans [20].

Funding Mechanisms. PEO Program Management Offices (PMOs) issue project orders or Economy Act orders using their appropriations; Warfare Centers accept into NWCF, schedule Work Breakdown Structures (WBSes), and recover rates through Stabilized Labor Rates (SLRs) while keeping the PM apprised of burn rates and Net Operating Result (NOR) impacts [24].

Governance Rhythm. Monthly technical reviews focus on requirements churn, test results, and configuration control; quarterly business reviews reconcile execution to SLR assumptions and assess workforce mix, drawing on Warfare Center cost visibility [2, 24].

Acquisition Decision Events. Warfare Centers provide independent readiness assessments (test status, technical risk) for PEO milestone decisions, often serving as certifying authorities for safety releases or flight clearances prior to fielding [2, 20].

2 PPBE

2.1 PPBE (Planning, Programming, Budgeting, and Execution)

2.1.1 What PPBE does

Aligns strategy to resources across the Future Years Defense Program (FYDP). *Planning* (strategy → guidance), *Programming* (balanced force/program → Program Objective Memorandum (POM)), *Budgeting* (validated request → President's Budget (PB)), *Execution* (obligations/outlays → performance feedback) [25].

2.1.2 Timeline (annual rhythm)

Figure 2.1 shows the PPBE timeline across multiple fiscal years with key milestones during each calendar year.

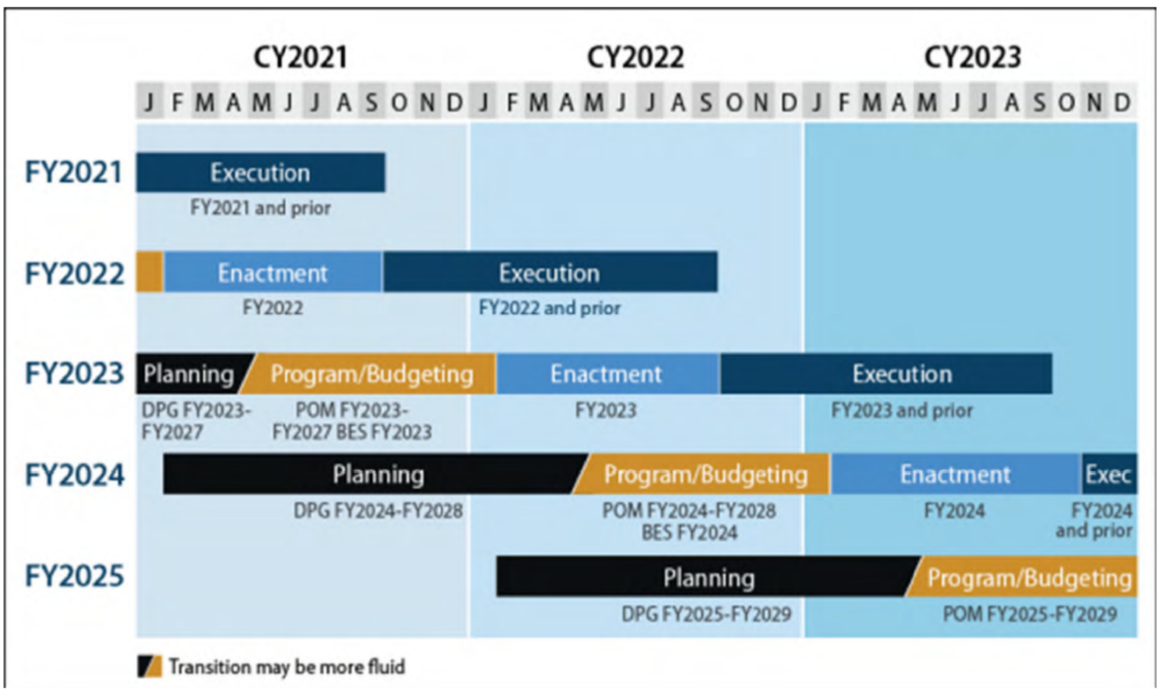


Figure 2.1. *PPBE Timeline for multiple fiscal years. Source:* 3.1.1. PPBE (Planning, Programming, Budgeting, and Execution), *2025 [25].*

The Platinum Card view in Figure 2.2 is a quick board sketch for the Planning phase (War aims → strategy → guidance) before the POM/PB forks.

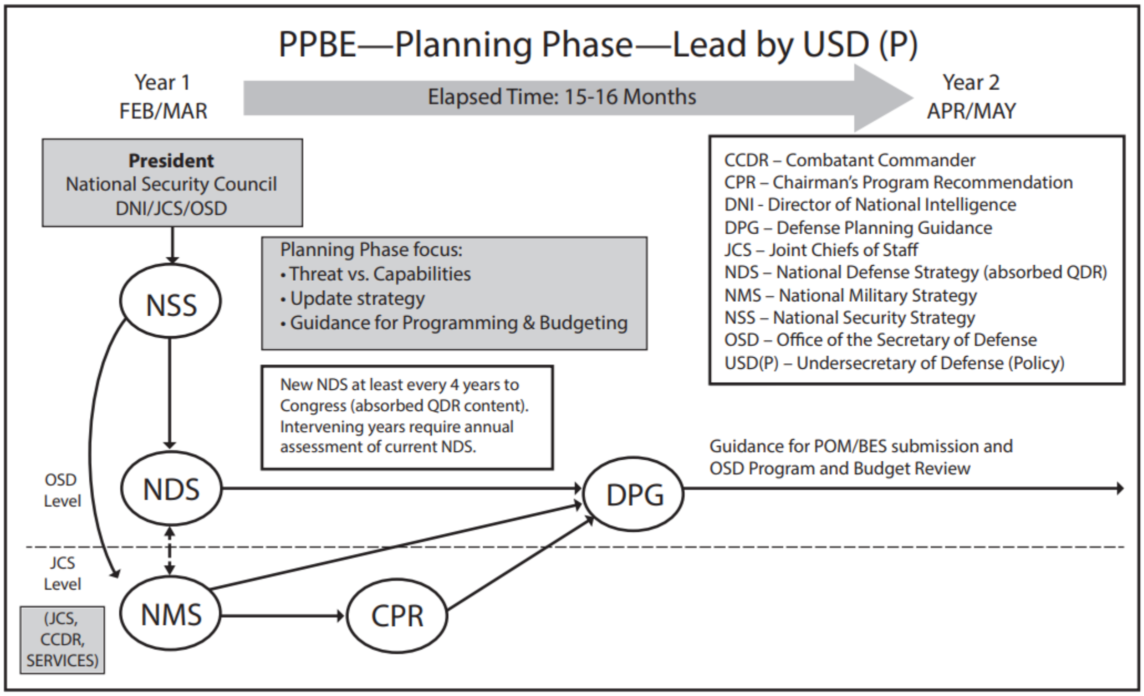


Figure 2.2. *Platinum Card view of the PPBE Planning phase (USDP-led strategic guidance path). Source:* 3.1.1. PPBE (Planning, Programming, Budgeting, and Execution), *2025 [25].*

2.1.3 Congressional Enactment (Regular Order)

Figure 2.3 highlights how Congress moves from the PB submission through authorization and appropriations when operating on-time without a Continuing Resolution (CR). Being able to walk this chart helps bridge PPBE milestones with Hill activity.

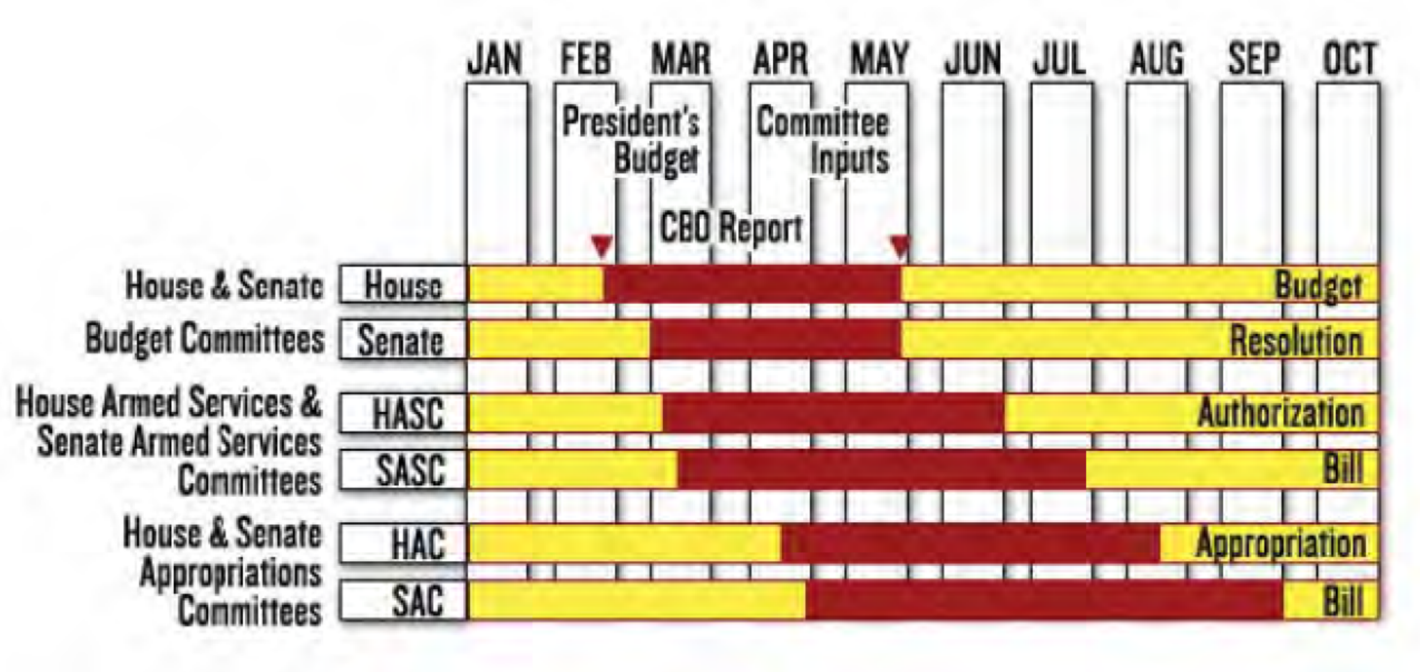


Figure 2.3. Congressional Enactment Timeline. Source: 3.1.2. Congressional Enactment, *2025 [26].*

2.1.4 Key Terms and Definitions

Table VII lists key terms and definitions.

Table VII: Key terms in PPBE

Term	One-liner
PY	Last completed FY: execution look-back and CR baselines.
CY	Ongoing FY: execution and mid-year reviews.
BY	Next FY in submission.
POM	Services balanced force/program across the FYDP.
FYDP	5-year program structure and resources.

Source: *DoW Financial Management Regulation 7000.14—R, Volume 3*, 2025 [27]

Other terms and definitions from [25].

Cost Assessment and Program Evaluation (CAPE). Independent analytic staff to the SECWAR/Deputy: supports Program Review decisions and provides cost/program evaluation inputs that can materially shape both the POM and the PB [25, 28].

Program Decision Memorandum (PDM). Records SECWAR program decisions at the end of Program Review.

OPNAV Integration of Capabilities and Resources (N8). Navy resource integrator (“checkbook”): builds a balanced Navy POM and aligns affordability, risk, and trade space across portfolios [25].

OPNAV Programming Division (N80). Runs the Navy POM build and integration, adjudicating sponsor submissions into a single, fiscally constrained program [25].

OPNAV Assessment Division (N81). Provides analytic assessments and force-structure trade studies that inform POM decisions [25].

OPNAV Fiscal Management Division (N82). Navy budget formulation and execution authority (FMB): translates the POM into the annual budget and manages execution controls and allocation flows [25, 29].

OPNAV Campaign Analysis and Future Force Design (N83). Campaign analysis and future force design division: sponsors wargames, mission-level analysis, and scenario design that ground POM options in force-level outcomes [25].

OPNAV Capability Integration and Resources (N84). Capability integration and resources division: brokers cross-portfolio offsets, aligns PEO execution phasing to POM decisions, and keeps two-pass/seven-gate products synchronized with budget marks [25].

OPNAV Advanced Concepts and Wargaming (N89). Advanced concepts and wargaming division: future force design cross-checks, wargaming, and architecture integration to keep sponsor asks coherent with POM toplines [25].

OPNAV Warfare Systems (N9). Navy requirements sponsor (“demand signal”): validates and advocates warfare requirements and resourcing by mission area [25].

Warfare Integration Directorate (N91). Warfare integration directorate; sometimes shown as Integrated Warfare Directorate (N9I); cross-domain mission integration and architecture; orchestrates mission-level POM issues.

> **Warning**
>
> Office designators can shift; verify current subcodes week-of.

Expeditionary Warfare Directorate (N95). Expeditionary warfare requirements/resourcing sponsor within N9 [25].

Surface Warfare Directorate (N96). Surface warfare requirements/resourcing sponsor within N9 [25].

Undersea Warfare Directorate (N97). Undersea warfare requirements/resourcing sponsor within N9 [25].

Air Warfare Directorate (N98). Air warfare requirements/resourcing sponsor within N9 [25].

Unmanned Warfare Systems (N99). Unmanned warfare systems sponsor: integrates unmanned air/surface/undersea capabilities into Fleet architectures and defends the investment profile in PPBE trades [25].

Resources and Requirements Review Board (R3B)/Naval Capabilities Board (NCB). Navy governance gates for resourcing and requirements decisions that feed the POM process [25].

2.1.5 Programming v.s. Budgeting

Table VIII compares key differences between Programming and Budgeting. One way to think about this: (1) Budgeting is for the money for this year and (2) Programming is for the 5-year FYDP.

TABLE VIII: PROGRAMMING V.S. BUDGETING

	Programming	Budgeting
Purpose	Build a balanced force across FYDP	Price/validate an executable BY request
Lead	N8 with supporting codes/PEOs/SYSCOMs	FMB/Office of the Secretary of War (Comptroller)/Office of Management and Budget
Key products	POM, PDM	Budget estimate, reclama, President's Budget

Source: *3.1.1. PPBE (Planning, Programming, Budgeting, and Execution)*, 2025 [25]

Figure 2.4 from the Platinum Card captures how the Service-level Program/Budget reviews run in parallel after Planning guidance issues, keeping POM trades and the Comptroller's reclamas synchronized.

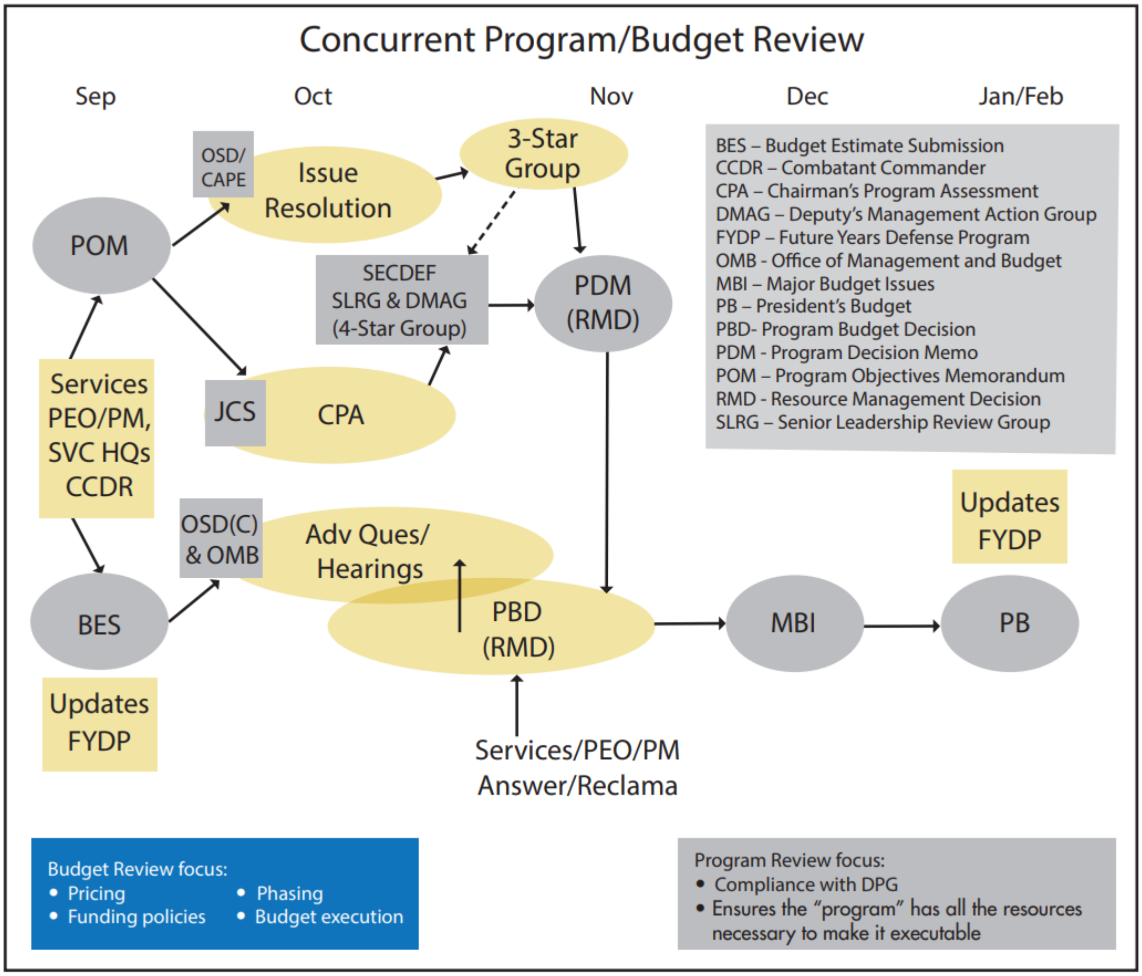


Figure 2.4. Platinum Card view of the concurrent Program and Budget reviews. Source: 3.1.1. PPBE (Planning, Programming, Budgeting, and Execution), *2025 [25].*

2.1.6 PPBE Programming

3-Star v.s. 4-Star Reviews: During OSD's Program Review, issues raised on the Services' POMs move first to a 3-Star Programmers Panel (staff-level, chaired by Director, CAPE (DCAPE)) that vets issue papers, builds options (with offsets), and forwards recommendations. Unresolved or strategy-level trades go to the 4-Star forum, the Deputy's Management Action Group (DMAG) co-led by the Deputy SECWAR (with the Vice-Chairman of the Joint Chiefs of Staff (VCJCS)), for senior adjudication. Decisions at this stage are documented as a PDM or (in recent cycles) a programmatic Resource Management Decision (RMD), which updates the FYDP and hands off to the Comptroller's Budget Review that culminates in budgetary RMDs [25].

2.1.7 Marks and reclamas (pre-enactment)

"Mark" (internal). During Program and Budget Review, OSD and Office of Management and Budget (OMB) can add/cut/realign a Component submission; decisions are captured in products such as PDMs and RMDs (and in OMB "passback") [25].

Reclama. The Component's formal, time-phased appeal of an internal mark: data-driven justification, feasible alternatives/offsets, and senior adjudication before the PB is finalized [25].

Board cue. Distinguish internal marks/reclamas (budget formulation) from congressional marks (committee markup) and from reprogramming after enactment [25, 26].

2.1.8 Appropriations Life-cycle (Colors of Money)

Figure 2.5 visualizes when each major appropriation category is Current, Expired, or Cancelled. Pair this with the obligation windows discussion so you can sketch the chart quickly at the board.

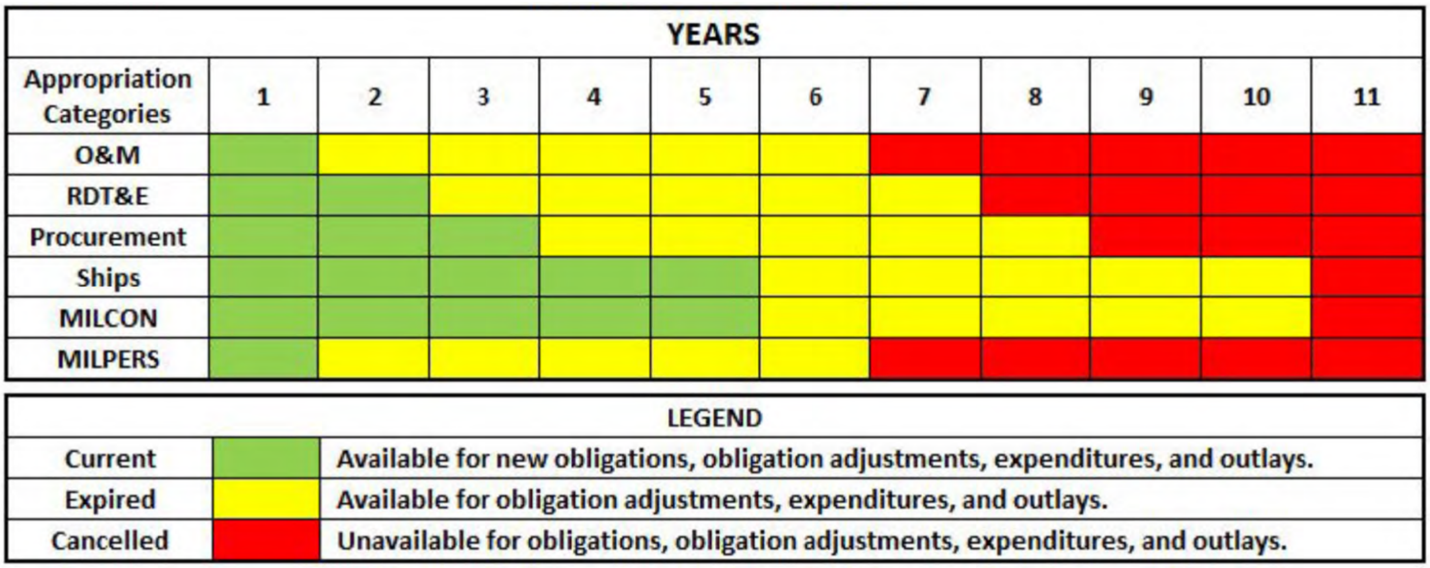


Figure 2.5. *Colors of Money Timeline. Source:* 3.1.3. Program Funding, *2025 [30].*

2.1.9 IGT (Intragovernmental Transactions)

Definition. Orders and collections between federal entities using Treasury's G-Invoicing: 7600A (order), 7600B (agreement/performance), Intra-Governmental Payment and Collection (IPAC) for collections [31].

Why it matters. Intragovernmental Transaction (IGT) acceptance *is an obligation* on the customer side and the *revenue recognition driver* for the performing Working Capital Fund (WCF) activity.

2.1.10 Common Ordering Instruments

Work Order. Internal directive used within a command to control scope/cost/schedule.

Project Order. Statutory order for a *definite, specific, and entire* project; obligational on acceptance; performs like a contract between DoW activities. (41 U.S.C. § 6307) [32]

Military Interdepartmental Purchase Request (MIPR) (DD Form 448). Economy Act order between federal entities; obligational when accepted; performed by the servicing agency. (31 U.S.C. § 1535) [31, 33]

2.1.11 Fiscal Law (Know These Cold)

Purpose Statute ("Misappropriation"). Funds must be used only for their appropriated purpose (31 U.S.C. § 1301(a)) [34].

Bona Fide Needs Rule. Use current-year appropriations only for legitimate needs of that fiscal year (31 U.S.C. § 1502(a)) [35].

Anti-Deficiency Act. Do not obligate/expend in excess of available amounts or before funds are available (31 U.S.C. § 1341, § 1517) [36].

2.1.12 Reprogramming: Moving Resources *After* Enactment

Four mechanisms (high level).

1. **Congressional Prior Approval Reprogramming (PA Reprog)**: Actions above statutory/committee thresholds or affecting congressional special-interest items or new starts. Requires approval from all four defense committees; timing driven by committee cycles [27].
2. **Internal Reprogramming (IR)**: Realignments within an appropriation that do not cross thresholds or trigger congressional interest; used for proper purpose alignment without changing totals. Notice sent to committees [27].
3. **Below-Threshold Reprogramming (BTR)**: Component-level authority to move funds below set dollar/percent limits within the same appropriation/fiscal year; limits reset annually by appropriations acts/committee guidance [27].
4. **Letter Transfer (LT)**: Treasury non-expenditure transfer moving budgetary resources between appropriations/treasury symbols when authorized (e.g., statute or enacted language) [27].

Hint

General rules of thumb—cannot change color-of-money or extend time; must complete before funds' obligation period ends; expired-year accounts limited to valid upward/downward adjustments only [27].

2.1.13 From Enactment to Field Authority

1. Treasury *warrants* the appropriation; OMB *apportions* on Standard Form 132 (Categories A/B/C) [37, 38].
2. OSD (Comptroller) issues allocations to components such as the DON.
3. OPNAV (Financial Management Budget (OPNAV) (FMB) / N82) allocates to SYSCOM Budget Submitting Offices (BSOs) (Echelon II), which suballocate to Warfare Centers (Echelon III), followed by *allowances/allotments* to the Enterprise Resource Planning (ERP) system [27].
4. Commands record commitments, obligations, and expenses, and request outlays via Defense Finance and Accounting Service (DFAS).

Category A/B/C (board cue). OMB apportions budgetary resources by time (A), by activity/project (B), or for multi-year/no-year future use (C) to prevent agency obligations at a rate that would require deficiency/ supplemental funding [37, 38].

Allotments. After apportionment, Components issue internal allotments/suballotments; obligating beyond an allotment/apportionment is an Anti-Deficiency Act (ADA) issue, even if the appropriation still has unobligated balance [27, 36].

Standard Form 133. The Report on Budget Execution and Budgetary Resources compares actual budget execution to the approved apportionment to spot potential violations early and support mid-year adjustments [38].

2.2 NWCF Essentials: Rates, Results, and Execution

2.2.1 Key Definitions (Execution)

Commitment. Administrative reservation of funds in anticipation of an obligation; reduces available authority but is not yet a legal liability [27] (e.g., National Reconnaissance Office (NRO)'s Request for Contract Action (RCA) process).

Obligation. Legal liability incurred that encumbers funds [27] (e.g., award of a contract, acceptance of an IGT order).

Expenditure/Expense. Recording of cost when goods/services are received/accepted (accrual basis) [27] (e.g., sending check to pay an invoice).

Outlay/Disbursement. Treasury cash payment to a vendor or performing activity [27] (e.g., payment check is cashed in).

2.2.2 Flow: PMO Purchase Request to Treasury Payment

1. PMO *commits* funds (approved in the ERP system).
2. Contract award or IGT order acceptance creates the *obligation* [27, 31].
3. Performance/acceptance posts *expense* (accrual).
4. DFAS schedules the invoice for payment; Treasury disburses (*outlay*) via IPAC (for IGT) or commercial Electronic Funds Transfer (EFT) [31].

2.2.3 NWCF Metrics and Equations

NOR (annual profit/loss).

$$\text{NOR} = \text{Revenue} - \text{Total Expenses}$$

Target performance remains near break-even over time (small NOR) [27].

Accumulated Operating Result (AOR) (retained earnings/equity over time).

$$\text{AOR}_t = \text{AOR}_{t-1} + \text{NOR}_t \pm \text{Other Adjustments}$$

Represents the *corpus* (net position) of the fund [27].

2.2.4 NWCF Corpus v.s. Appropriations

AOR (working capital). Revolving cash and retained earnings authorized under 10 U.S.C. § 2208 [39]; remains available without fiscal year limitation to finance operations until recovered through stabilized rates [40].

Customer Appropriations. Mission-funded (e.g., Operation and Maintenance, Navy (OMN), RDT&E, Shipbuilding and Conversion, Navy (SCN)) dollars obligated by the customer when a project order or inter/intra-governmental order is accepted; purpose, time, and amount statutes still apply to the customer [27].

Capital Investment Program (CIP). Financed by NWCF corpus but budgeted in the capital budget exhibit; used for plant/equipment modernization and amortized back through rates, not through a separate appropriation [40].

No Augmentation. NWCF activities cannot augment customer appropriations; corpus only bridges cash timing between expense recognition and reimbursement [40].

2.2.5 Stabilized Rates and SLR

Stabilized Rates. Customer prices set in the budget build to recover expected full costs (labor, material, overhead, depreciation/C CIP) with a near-zero NOR goal [27].

SLR. Published labor $/hr for a shop/code; recovers direct labor + fringe + overhead + G&A + capital depreciation recovery [27]. The rate is:

$$\text{SLR} = \frac{\text{Direct Labor} + \text{Fringe} + \text{OH} + \text{G\&A} + \text{Depreciation Recovery}}{\text{Direct Labor Hours}}$$

Adjustment Battle Rhythm. Rates are established two years ahead during the PPBE budget build and held constant throughout budget year execution; mid-year changes require Office of the Under Secretary of War (Comptroller) (OUSW(C)) approval when earned rates diverge materially from planned costs, and Navy BSOs typically review SLR accuracy monthly/quarterly to recommend any out-of-cycle adjustments [40].

2.2.6 NWCF v.s. Mission-Funded (Appropriation) Commands

Which billets at NWCF are mission-funded. Military administration and leadership billets (CO/XO/Admin) are funded by **Military Personnel (MILPERS)** and treated as mission-funded within NWCF; certain command/HQ oversight billets may also be OMN funded by policy, so verify locally. These costs are not recovered in stabilized rates. Others that are assigned to NWCF will bill hours towards work and the NWCF will reimburse MILPERS [27].

2.2.7 Standing Up or Modifying a NWCF Business Area

1. **Business Case Development**: Sponsor (e.g., ASN(RD&A) or SYSCOM) prepares analytical justification showing workload, demand signal, and ability to operate on a revolving basis without violating purpose/time/amount [40].

2. **DON Approval Chain**: Secretary of the Navy (SECNAV) (delegated to Assistant Secretary of the Navy (Financial Management and Comptroller) (ASN(FM&C))) endorses the concept and forwards to OUSW(C) while coordinating with Navy Comptroller to align PPBE exhibits [40].

3. **OUSW(C)/OMB Review**: DoW Chief Financial Officer (CFO) validates cash requirements, rate methodology, and capital plan, then seeks OMB alignment for the PB [40].

4. **Congressional Notification**: Congress must be notified (and, when required, explicitly authorize in appropriations or authorizations) before execution; 10 U.S.C. § 2208 [39] restricts creation of new WCFs without legislative awareness [40].

5. **Implementation**: Once approved, the new business area is issued a NWCF business unit code, begins PPBE rate build two years out, and transitions legacy appropriated accounts via opening balance adjustments [40].

2.2.8 NWCF Cycle of Operations (Order to Cash)

1. **Customer order.** IGT 7600A or project order is received and *accepted*, creating the customer's obligation.

2. **Work in process.** Labor and material are applied; costs accumulate; billing events are scheduled per percent complete or delivery.

3. **Revenue recognition and billing.** The NWCF recognizes revenue and bills via IPAC (IGT) or a commercial invoice if authorized.

4. **Collection (cash).** Treasury IPAC/electronic funds transfer posts; NWCF cash increases; NOR/AOR update through the period close [27, 31].

2.2.9 Operating During a Continuing Resolution

Customer Funding Limits. Appropriated customers remain bound by prior-year obligation rates and anti-deficiency constraints; NWCF orders cannot exceed apportioned CR amounts until an appropriations act is passed [41].

Cash Cushion. Existing NWCF corpus allows Warfare Centers to keep executing accepted orders (labor continues, suppliers paid) even if reimbursements lag, provided cash balances stay within the Financial Management Regulation (FMR)'s upper/lower operating limits [40].

No New Starts. CR guidance prohibits new start projects, major capital investments, or rate changes absent explicit exception; NWCF managers defer new workloads that would obligate customer funds beyond CR allowances [41].

Rate Discipline. Stabilized rates remain frozen; only emergency OUSW(C)-approved rate adjustments may occur, so BSOs focus on expense control to avoid large NOR swings during the CR period [40].

2.2.10 Why Use a Warfare Center (Organic)

Things PMOs/industry *can't* do. Inherently governmental TA warrants, certification authority, certain safety releases; highly classified or nuclear workspaces.

Things they *shouldn't* do. Independent test/assessment, spec adjudication, tech baseline control, blue-&-gold separation to avoid Organizational Conflict of Interest (OCI).

Things they *won't* do. Sustainment engineering at scale, depot-level organic repair, fleet distance support.

2.3 Congressional Enactment

2.3.1 Foundations and core definitions

Power of the purse. U.S. Constitution, Article I.

Section 8.

"The Congress shall have power to... provide for the Common Defense... and general welfare..."

Section 9.

"No money shall be drawn from Treasury, but in consequence of Appropriations made by law..."

Key terms. Budget Authority (BA) (legal authority to incur obligations), Total Obligational Authority (TOA), obligations, outlays, Legislative Proposal (LEGPROP).

Authorization v.s. Appropriation. National Defense Authorization Act (NDAA) authorizes programs/policy; appropriations provide BA.

2.3.2 Who drafts what

Authorization. House Armed Services Committee (HASC) /Senate Armed Services Committee (SASC).

Appropriations. House Appropriations Committee (HAC) /Senate Appropriations Committee (SAC).

Budget Resolution. House Budget Committee (HBC) /Senate Budget Committee (SBC) (sets topline aggregates and 302 allocations; not a law).

Independent analysis/oversight. Congressional Budget Office (CBO) (scoring), Congressional Research Service (CRS) (research), Government Accountability Office (GAO) (audits/oversight).

2.3.3 Committee markup and "marks" (congressional)

What a mark is. A committee change (add/cut/plus-up, fence, reporting language) to the PB request during subcommittee/full-committee markup and conference before the bill is enacted [26].

When it matters most. The best opportunities to fix a problematic mark are before full committee and during conference/joint explanatory statement, where House/Senate differences are resolved [26].

How programs respond. Program offices work issues through official channels (chain of command and Service Office of Legislative Affairs (OLA)); provide fact-based impacts to cost/schedule/performance and executable alternatives. After enactment, relief typically requires use of formal reprogramming authorities (and thresholds) rather than informal advocacy [26, 27].

Board cue. Distinguish congressional marks (committee action) from internal marks/reclamas (budget formulation) and from reprogramming after enactment [25, 26].

2.3.4 Regular-order timeline

Figure 2.6 shows the timeline from the PB → Appropriation Bill with approximate timelines assuming no CR.

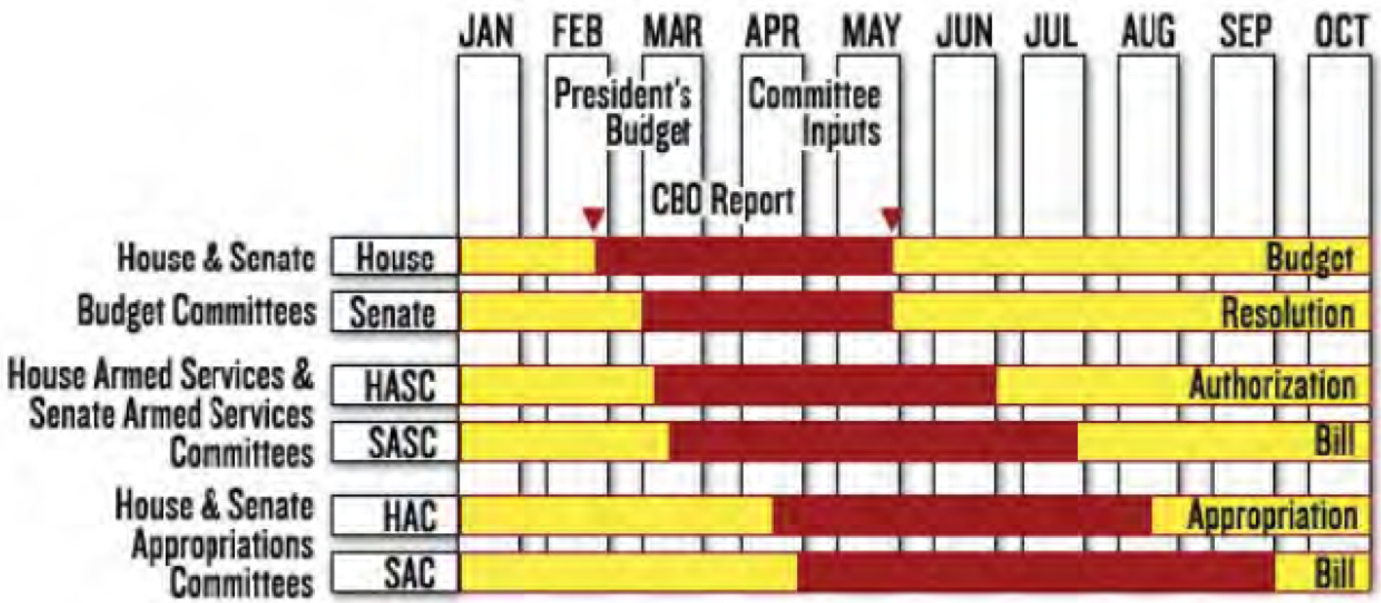


Figure 2.6. Congressional Enactment Timeline. Source: 3.1.2. Congressional Enactment*, 2025 [26].*

2.3.5 The 12 regular appropriations bills

Defense. Department of War appropriations.

Military Construction and Veterans Affairs. Military construction, housing, and Department of Veterans Affairs programs.

Energy and Water. Department of Energy, Army Corps of Engineers, and related infrastructure programs.

Homeland Security. Department of Homeland Security operations and components.

Interior and Environment. Department of the Interior and Environmental Protection Agency activities.

Labor, Health and Human Services, Education. Departments of Labor, Health and Human Services, and Education, plus related agencies.

Legislative Branch. U.S. Congress and supporting legislative branch agencies.

Financial Services and General Government. Treasury, judiciary, Small Business Administration, and other independent agencies.

Transportation and Housing and Urban Development. Departments of Transportation and Housing and Urban Development.

Commerce, Justice, Science. Departments of Commerce and Justice plus science agencies such as National Aeronautics and Space Administration (NASA) and the National Science Foundation.

State and Foreign Operations. Department of State, U.S. Agency for International Development, and foreign assistance programs.

Agriculture and Food and Drug Administration (FDA). Department of Agriculture and Food and Drug Administration activities.

Note

The bills that EDO's care about the most are Defense, Military Construction (MILCON) & Department of Veterans Affairs (VA) and Energy & Water. This because those bills directly affect funding for our acquisitions.

Hint

These bills are sometimes packaged into minibuses or a single omnibus appropriations act.

2.3.6 Continuing Resolution

Continuing resolution rules (board one-liner). *No new starts*; rate/quantity changes generally constrained unless specified; execution limited by OMB apportionment, CR anomalies, and any authorities Congress adds in the joint explanatory statement [41].

2.3.7 Sequestration

Automatic, across-the-board reductions if statutory caps/triggers are breached; applied by account unless modified by law.

2.4 Program Funding and Execution

2.4.1 BA, commitment, obligation, expenditure, outlay

1. *BA* available (legal authority).
2. *Commitment* (administrative reservation).
3. *Obligation* (legal liability via contract/order).

4. *Expenditure* (payment recorded).

5. *Outlay* (cash disbursed/Treasury).

   Source references [27, 31].

2.4.2 Execution tracking (program manager view)

Time-phased plan. Track commitments/obligations/expenditures against a time-phased spend plan by WBS/work package; execution is constrained by OMB apportionment and internal allotments, not just total appropriated BA [27, 29, 38].

Contract performance. Use EVM to determine whether cost/schedule performance supports the funding plan: Cost Performance Index (CPI)/Schedule Performance Index (SPI), Cost Variance (CV)/Schedule Variance (SV), and Estimate at Completion (EAC) trends; reconcile contractor Integrated Program Manager's Data Analysis Report (IPMDAR) data with finance reports (e.g., Contract Funds Status Report (CSFR)) [42, 43].

Board cue. A program can be "green" on obligations while still trending to an overrun if EAC rises or schedule slips; pair budget execution with EVM and technical risk indicators [43].

2.4.3 Appropriation categories, scope, obligation windows

Table IX shows the Colors of Money with their spending categories and obligation windows. Figure 2.5 from § 2.1.8 visually shows when each category is Current, Expired, or Cancelled.

Table IX: Appropriation categories, scope, obligation windows

Category	Typical Scope	Obligation Window
RDT&E	Science & technology; development; test & evaluation	2 years
OPN[a]	End items/modernization and spares	3 years
OMN	Operations, maintenance, training, minor mods	1 year
MILCON	Facilities construction and real property	5 years
MILPERS	Personnel pay	1 year

[a] SCN is a specific procurement with a 5-year obligation window.

Source: *DoW Financial Management Regulation 7000.14—R, Volume 3*, 2025 [27]

2.4.4 Funding policies

Annual OMN/MILPERS. Fund only bona fide needs of the fiscal year.

RDT&E. Fund work as it occurs by fiscal year.

Full-funding. (Procurement/MILCON). Fund the total usable end-item at award.[1]

Common techniques/exceptions: Advance Procurement (long-lead items), Multiyear Procurement, Economic Order Quantity, Cost-to-Complete when provided in law.

Exceptions to the Full Funding Policy

Below is the list of the key exceptions to the Full Funding Policy, what they allow, and the must-know rules/approvals (Table X shows full details).

Table X: Key exceptions to the Full Funding Policy: what they allow and what to remember

Exception	Typical Appropriation	What it allows	Must-know rules / approvals
MYP	APN, OPN, SCN	Single contract covering multiple fiscal years of buys to achieve predictable demand and unit-cost savings; may include EOQ purchases common across years.	Requires specific congressional authorization; stable requirements/design; credible savings; low technical risk; term generally up to 5 years; plan/budget any cancellation liability.
Advance Procurement (incl. LLTM/EOQ)	APN, OPN, SCN	Limited *early* funding (often 1–2 years before the main "buy" year) for long-lead components or EOQ lots to protect schedule or achieve price breaks.	Narrow and part-specific; must be justified in budget docs; EOQ commonly tied to MYP (or explicit block authority); does *not* equal full funding of the end item.
BB	Primarily SCN	A single contract buying multiple ships (or end items) across fiscal years to gain tooling/learning-curve efficiencies.	Requires explicit congressional authorization; funds still obligated year-by-year; not under the MYP statute, so oversight mechanics differ.
CTC	SCN	Additional funds in a later FY to finish an item (e.g., ship) when matured estimates exceed prior appropriations.	Completes original approved scope (not added capability); appears as a distinct request/justification; common in long-duration shipbuilding.
RCOH for CVN	SCN (with prior Advance Procurement)	Mid-life refueling and major overhaul funded incrementally over multiple years; early procurement of nuclear fuel/critical components.	Recognized exception due to size/complexity; incremental SCN with advance procurement years ahead of the principal execution.

Note:
(i) Economic Order Quantity (EOQ) purchases are commonly approved within Multiyear Procurement (MYP) or other explicit multi-year constructs;
(ii) Block Buy resembles MYP in intent (savings) but uses program-specific authorization;
(iii) Plan for cancellation liability under MYP when required.
Source: *DoW Financial Management Regulation 7000.14—R, Volume 3*, *10 U.S.C. § 3501*, *DFARS 217.172: Multiyear Contracts*, 2025, 2023, 2025 [27, 44, 45]

[1] Exceptions to Full funding are select SCN programs (e.g., CVNs, SSBNs, and Landing Helicopter Docks (LHDs)/Landing Helicopter Assaults (LHAs)) with Congressional approval.

2.4.5 Navy Sustainment Funding: TYCOM v.s. Fleet (O&S)

Who holds O&M,N. Type Commander (TYCOM) leaders (*TYCOMs*; e.g., Commander, Naval Surface Forces (COMNAVSURFOR), Commander, Naval Air Forces (COMNAVAIRFOR), Commander, Submarine Forces (COMSUBFOR)) are the primary executors of OMN for unit readiness: flying hours, steaming days, intermediate-level maintenance, training, and afloat/ashore support.

Fleet Commanders. U.S. Pacific Fleet (PACFLT) and U.S. Fleet Forces (USFF) set operational priorities and readiness goals, adjudicate contingencies, and may centrally manage certain Fleet-wide initiatives; they influence TYCOM allocations but most execution occurs at the TYCOM.

Program Office (SYSCOM/PEO). Funds in-service engineering, tech data, Diminishing Manufacturing Sources, and modernization using Other Procurement, Navy (OPN)/Aircraft Procurement, Navy (APN) (end items/kits) and RDT&E/OMN (installation, trials) as appropriate; manages sustainment Indefinite-Delivery, Indefinite-Quantitys (IDIQs)/Performance-Based Logistics (PBL) with color-of-money compliance.

Who pays what. OPN/APN: hardware/software upgrade kits, initial spares; OMN: installs, fleet introduction/training, minor mods, repair parts consumption; RDT&E: development/test of engineering changes. Naval Supply Systems Command (NAVSUP)/NWCF activities recover costs via rates for supply/repair.

Board cue. In operations and support, TYCOMs execute most OMN; Program Managers fund modernization and in-service support from OPN/APN/RDT&E. Align funding lines with the WBS and the Colors-of-Money rules to avoid Anti-Deficiency Act risk.

Quick information regarding exceptions to the full funding policy:

MYP. Up to 5 years; stable design; verifiable savings; cancellation liability planned.

Advance Procurement Long-Lead Time Material (LLTM)/EOQ. Focused early buys for schedule/savings; not full funding.

Block Buy (BB). Congressionally authorized multi-ship buys; obligate by Fiscal Year (FY); not the MYP statute.

Cost-to-Complete (CTC). Complete original scope when estimates mature upward; separate SCN request.

Refueling and Complex Overhaul (RCOH). Planned midlife overhaul; incremental SCN with advance procurement of cores/critical material.

2.4.6 National and Military Intelligence Program funding (National Reconnaissance Office context)

Two budget programs, two authorities. The intelligence budget is split between the National Intelligence Program (NIP) (strategic, cross-Intelligence Community (IC) priorities) and the Military Intelligence Program (MIP) (defense operational and tactical requirements); the Director of National Intelligence (DNI) manages the NIP and the Under Secretary of Defense for Intelligence and Security (USD(I&S)) manages the MIP under distinct authorities and review chains.[46]

NRO and peers straddle both. The NRO (along with Defense Intelligence Agency (DIA) and National Geospatial-Intelligence Agency (NGA)) executes both NIP and MIP dollars; their directors serve as NIP program managers and MIP component managers, driving dual oversight and reporting requirements.[46]

Governing guidance. NIP budget formulation/execution follows Office of the Director of National Intelligence (ODNI) policy in *ICD 104* and the IC planning, programming, budgeting, and evaluation system, while MIP policy follows *DoDD 5205.12* and DoW FMR execution rules; expect separate guidance, exhibits, and reviews.[27, 47, 48]

Reprogramming and oversight. NIP adjustments route through ODNI processes and intelligence committee review paths; MIP reprogramming follows DoW FMR thresholds and DoW congressional notifications, so integration work must reconcile paperwork, actors, and timelines.[27, 47]

Recent toplines (context). FY 2025 appropriated toplines: 73.3$BforNIPand$27.8B for MIP; FY 2025 requested toplines: 73.4$B(NIP)and$28.2B (MIP). Current as of 2025-12-28.[46, 49, 50]

2.4.7 Which money when (by acquisition phase)

Table XI shows the nominal types of appropriation used for Materiel Solution Analysis (MSA), Technology Maturation and Risk Reduction (TMRR), Engineering and Manufacturing Development (EMD), Production and Deployment Phase (P&D), and Operations and Support Phase (O&S)

Table XI: Appropriation types per phase

Phase	Typical appropriation(s)
MSA	RDT&E (analyses, prototyping); small OMN for studies
TMRR	RDT&E (risk reduction, development, test)
EMD	RDT&E (develop/build/test/qualify); selected long-lead/AP/EOQ in Procurement when authorized
PD	Procurement (e.g., OPN); limited RDT&E for fixes
OS	OMN sustainment; spares via Procurement as applicable

2.4.8 MILCON v.s. O&M sustainment v.s. OPN modernization

Rule of thumb: Real property construction ⇒ MILCON; keeping systems running (repair/overhaul, services) ⇒ OMN; new capability/end-item upgrades ⇒ Procurement (e.g., OPN).

2.4.9 Field Activity Financial Management (NWCF quick hits)

Goal. NWCF activities price to *break even over time*; in-year result is **NOR**, cumulative is **AOR**.

SLR. Recover direct labor + overhead + G&A; shocks flow to NOR/AOR and are managed over time.

Carryover. Unfilled orders at FY end; subject to limits/management.

IGT via G-Invoicing. Treasury-mandated platform for intra-gov orders/settlement; apportionment/availability rules still govern customer funds.

2.5 Cost Elements, Estimates, and Learning Curves

2.5.1 Direct v.s. Indirect Costs

Direct. Traceable to a final cost objective (e.g., technician hours, end-item material).

Indirect. Pooled and allocated (overhead, General and Administrative (G&A), facilities, depreciation). See the Federal Acquisition Regulation (FAR) cost principles [51].

2.5.2 Program Cost Terms (Know the Stack)

Flyaway Cost. Recurring production cost of an aircraft/weapon delivered (excludes non-recurring tooling, spares, support).

Weapon System Cost. Flyaway plus support equipment, training, data, and initial spares.

Procurement Cost. Weapon system cost plus peculiar support and MILCON as applicable.

Life-Cycle Cost (LCC). All costs from concept through disposal (RDT&E, procurement, O&S, disposal).

Total Ownership Cost (TOC). LCC plus broader infrastructure/indirect enterprise costs.

Note

Use DON/DoW cost handbook definitions; be consistent across documents. [3, 52].

Figure 2.7 shows the LCC from above and how they relate to one another. Note how each cost builds to the next as different types of expenditures are added.

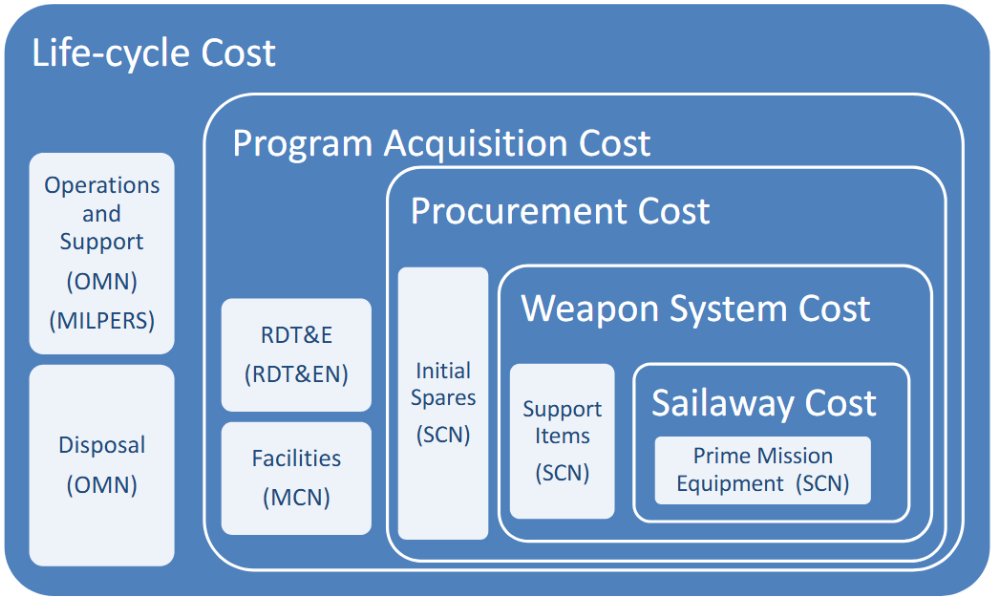


Figure 2.7. *Composition of Life Cycle Costs. Source:* 3.1.6. Cost Estimating, *2024 [53].*

2.5.3 Why the LCCE Matters

Establishes the affordability baseline, informs Acquisition Program Baseline (APB), drives color-of-money phasing, and underpins "will-cost / should-cost" targets [52]. Figure 2.8 shows the different types of funding that will be used in different phases of the MCA. Most of the LCC will be from the O&S phase.

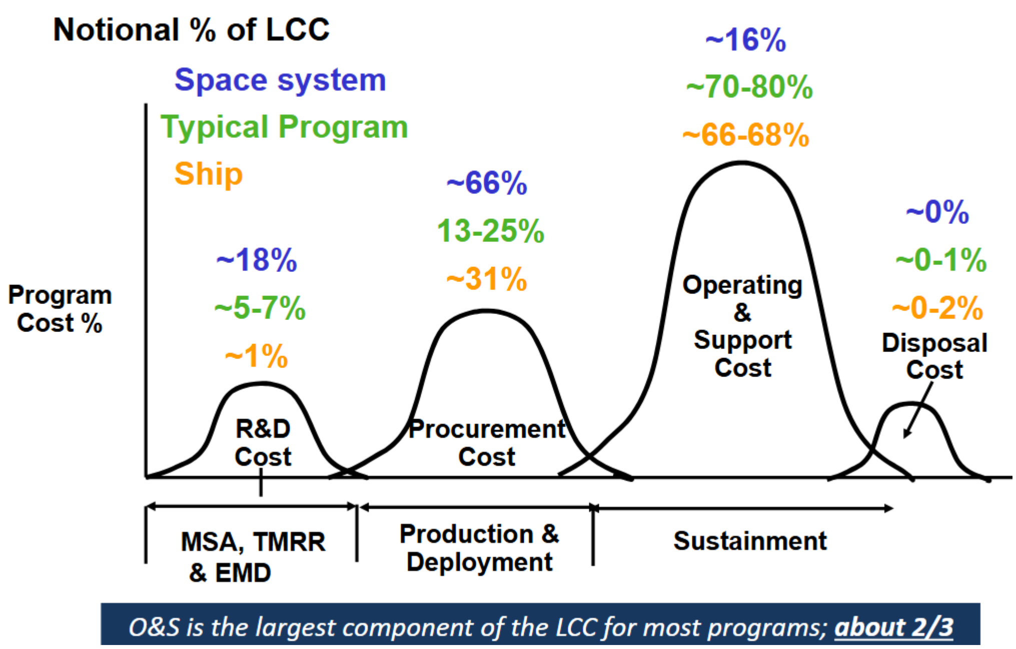


Figure 2.8. *LCC over time. The colors of money used are shown for each phase including percent of total costs. Source:* 3.1.6. Cost Estimating, *2024 [53]*

2.5.4 Estimating Methods (Types, When, and Why)

Analogy (top-down). *When:* Earliest stages (ICD/Materiel Development Decision (MDD), pre- Analysis of Alternatives (AoA)) when design is immature and only analogous programs or components exist. *Why:* Fast, defensible scoping by scaling from a close comparator using adjustment factors (size, weight, speed, complexity). Good for bounding affordability and informing trades before detailed data exists.

Parametric (Cost Estimating Relationship (CER)-based). *When:* AoA through TMRR when historical data supports CERs (e.g., weight, Source Lines of Codes (SLOCs), thrust, kW). *Why:* Enables rapid sensitivity analysis and trade-space exploration with statistical confidence intervals; supports independent cross-checks and early budget phasing.

Engineering (bottom-up). *When:* EMD and production planning when Bill of Materials (BoMs), routing, labor standards, and quotes are available. *Why:* Most traceable and auditable; ties to technical baseline and schedule networks; supports Integrated Baseline Reviews (IBRs) and negotiation objectives.

Actuals / Extrapolation. *When:* From late TMRR into production when pilots/Low-Rate Initial Production (LRIP) actuals exist and learning effects are visible. *Why:* Highest fidelity; captures learning curves and process improvements; anchors estimates to realized performance and Cost and Software Data (CSDR) actuals [54].

Table XII: Comparison of Estimating Methods

Type	Description	When Used	Why Used
Analogy	Uses cost of a similar past system, adjusted for differences	Early concept/refinement (Pre-Milestone A/B)	Quick estimate when data limited
Parametric	Uses statistical cost estimating relationships (CERs) like cost vs. weight or power	Milestone B/C; early design	Captures cost drivers using quantifiable parameters
Engineering (Bottom-Up)	Builds from detailed WBS—materials, labor, overhead	Post-CDR, Low-Rate Initial Production	High accuracy for mature designs
Actual Cost / Extrapolation	Uses historical/production actuals (learning curves)	Full-Rate Production / O&S	Most accurate; supports should-cost and sustainment budgeting

2.5.5 Products and Players

Cost Analysis Requirements Description (CARD). Technical/schedule baseline for independent estimates.

Independent Cost Estimate (ICE). Component Acquisition Executive (CAE) or OSD CAPE for ACAT I; Program Office Estimate (POE).

CSDR. Contractor cost & software data reporting for actuals/CERs.

APB/Selected Acquisition Report (SAR)/Defense Acquisition Executive Summary (DAES). Baselines and reporting [52].

2.5.6 Learning Curve (Production Efficiency)

If unit time/cost follows an s% slope, doubling quantity reduces unit cost by s%. Typical aerospace slopes: 85–90% early, flattening as processes mature. Plan lots and spares buys accordingly [54].

2.5.7 Cost Escalation (Inflation Indices)

Escalate base-year costs to then-year using approved DoW inflation indices; separate real growth from price growth; align phasing to obligation windows [27].

2.5.8 Will-Cost / Should-Cost

Will-Cost. Conservative reference estimate used for budgeting and margin planning.

Should-Cost. Management targets that drive efficiencies (e.g., lean events, value engineering, rate negotiations) [52].

3 Contracting and Solicitation

3.1 Intro to Contracting Fundamentals

3.1.1 Why the Navy Uses Contracts

- Contracts create a legally enforceable relationship between the Government and industry partners, defining rights and responsibilities for each party.
- Written terms provide structure for changes via agreed conditions, protecting both the Navy and the Contractor throughout acquisition actions.
- Binding agreements ensure accountability for cost, schedule, and performance outcomes that cannot be achieved through informal arrangements.

3.1.2 Essential Elements of a Binding Contract

Mutual Assent. Requires an offer and acceptance with a meeting of the minds on all material terms such as scope, price, quantity, and delivery.

Consideration. Both sides exchange value (e.g., payment, performance, schedule relief) that courts will evaluate for adequacy in Government contracting.

Capacity. Parties must be legally competent to enter the agreement; lack of capacity (e.g., incapacity, lack of authority) undermines enforceability.

Lawful Purpose. The contract must pursue a legal objective; agreements for unlawful acts are void.

3.1.3 Government Relationship to Contractors

Transactional. Each party enters with distinct incentives (contractor profit versus government outcomes); avoid unauthorized commitments.

Professional. Both teams must understand the contract and collaborate to meet requirements.

Collaborative. Contractors bring technical expertise while programmatic decision-making remains inherently governmental.

Constrained. Interactions are bound by ethics rules, conflicts-of-interest standards, and contract clauses.

3.1.4 Role of the Office of Management and Budget

The OMB assists the President in overseeing the preparation of the federal budget and supervising its administration in Executive Branch agencies. In contracting, OMB's Office of Federal Procurement Policy (Office of Federal Procurement Policy (OFPP)) shapes the acquisition landscape.

Budget Development & Execution. OMB reviews agency budget requests, formulates the President's Budget, and apportions appropriated funds to agencies for execution.

Procurement Policy. Through OFPP, OMB provides overall direction for government-wide procurement policies, regulations, and procedures, promoting economy and efficiency.

Regulatory Review. OMB's Office of Information and Regulatory Affairs (Office of Information and Regulatory Affairs (OIRA)) reviews significant regulations (including FAR changes) to ensure consistency with Presidential priorities.

Legislative Clearance. OMB coordinates the agency's legislative recommendations and testimony to ensure alignment with Administration policy.

3.1.5 FAR System and NAVSEA Overlays

FAR. Government-wide acquisition regulation establishing policy, procedures, and contract clauses [51].

Defense Federal Acquisition Regulation Supplement (DFARS). DoW-level supplement tailoring FAR provisions for defense-unique requirements, including competition, source selection, and industrial base policy [55].

Navy Marine Corps Acquisition Regulation Supplement (NMCARS). DON supplement that adds Navy-specific directives such as approval thresholds, peer reviews, and templates [56].

SUPSHIP Operation Manual (SOM) Chapter 3. NAVSEA guidance for SUPSHIP on contract formation, modification, and surveillance *(Appendix C.2: RDML Jonathan J. Jett-Parmer leads SUPSHIP as NAVSEA Deputy Commander, with RDML Robert J. Dodson (United States Navy Reserve (USNR)) as reserve deputy)*[57].

NAVSEA Source Selection Guide. Standardizes competitive source selection procedures, roles, and documentation for SYSCOM procurements [58].

NAVSEA Contracts Handbook. Practical desk reference covering policy interpretations, clause usage, and best practices for NAVSEA contracting professionals [59].

3.1.6 Types of Contracting Officers (Know All Three)

Procuring Contracting Officer (PCO). Leads acquisition planning through award; signs contracts and bilateral modifications on behalf of the Government.

Administrative Contracting Officer (ACO). Oversees post-award administration, surveillance, and contractor performance; typically assigned via Defense Contract Management Agency (DCMA) or NAVSEA field offices.

Termination Contracting Officer (TCO). Manages partial or complete contract terminations, settlement proposals, and equitable adjustments.

All contracting officers must hold a warrant that delineates dollar and authority limits; only a warranted Contracting Officer (KO) can obligate the United States [51].

3.1.7 PM and KO Partnership

PM. Accountable to the Milestone Decision Authority (MDA) for cost, schedule, and performance; integrates technical authority, requirements, and budget execution across the lifecycle.

KO. Provides acquisition strategy execution expertise, ensures compliance with statute/regulation, and is responsible for contract integrity and enforceability.

Board Cue. PMs lead program outcomes; KOs safeguard the contracting instrument. Neither can assume the other's authorities, so coordination before solicitations, negotiations, or modifications is mandatory.

3.1.8 Competition in Contracting Act Requirements

Full and Open Competition. Default posture: all responsible sources may compete under the FAR [51, 60].

Full and Open After Exclusion of Sources. Permits set-asides (e.g., small business, 8(a)) or alternate-source strategies when justified by the FAR [51].

Approval for Other than Full and Open. Requires documented justification and senior approval per FAR Subpart 6.3 [51, Subpart 6.3].

Seven Exceptions to Full and Open Competition (Memorize)

1. Only one responsible source will satisfy agency requirements (see FAR [51, § 6.302-1]).
2. Unusual and compelling urgency (see FAR [51, § 6.302-2]).
3. Industrial mobilization; engineering, developmental, or research capability (see FAR [51, § 6.302-3]).
4. International agreement (see FAR [51, § 6.302-4]).
5. Authorized or required by statute (see FAR [51, § 6.302-5]).
6. National security (see FAR [51, § 6.302-6]).
7. Public interest (see FAR [51, § 6.302-7]).

Hint

Be ready to cite an example scenario for each exception and the approval level required.

3.1.9 Responsiveness, Responsibility, and Key Determinations

Responsiveness (sealed bidding). Bid must conform to all material terms of the Invitation for Bids; nonconforming bids are rejected without discussion per the FAR [51].

Responsibility. Prospective contractor must possess adequate resources, schedule compliance, performance record, integrity, and necessary systems to receive award (FAR [51, § 9.104]).

3.1.10 J&A v.s. D&F

Justification and Approval (J&A). Documents the rationale for other-than-full-and-open competition, identifies the chosen statutory exception, and records approval by the appropriate official. Must be posted to `https://sam.gov` after award with required redactions [51].

Determination and Findings (D&F). Formal determination that specific conditions are satisfied before taking an action (e.g., use of special contract types, multiyear contracting); states the findings that support the determination [51].

Note

At the NRO, documents and rationale are posted on the low and high-side Acquisition Research Center (ARC). We are required to post other-full-and-open competition for five days on the ARC to allow opportunity for other contractors to bid.

3.1.11 Who Signs D&Fs (and When)

General Rule. FAR [51, § 1.704] requires the contracting officer to sign D&Fs when the action is within their delegated authority, unless a higher approval level is specified elsewhere in the regulation or delegation memo.

Head Contracting Activity (HCA). FAR [51, §§ 16.603-3, 16.504(c)(1)(ii)(D)] reserve approval for actions such as issuing a letter contract or awarding a single-award task/delivery IDIQ expected to exceed $100M; DON HCAs may redelegate no lower than a flag/Senior Executive [56, § 5201.707].

SAE. Multiyear contracting, extraordinary contractual relief, or other actions identified in FAR [51, § 17.105-1] and DFARS [55, § 217.172] require a D&F signed by the Service SAE (ASN(RD&A) for the Navy, who is also the DON Senior Procurement Executive) or a specifically delegated official.

Document Content. Every D&F must cite the specific statutory/regulatory authority, describe supporting facts, and state the determination in clear language; expiration dates and any required follow-on reviews must also be included per FAR [51, § 1.707] and NMCARS [56, § 5201.707] guidance.

3.2 Solicitation Preparation

3.2.1 Summary

Uniform Contract Format (UCF) scaffolding. Tailor the UCF (Sections A through M) so Section C defines the requirement, Section L tells offerors how to respond, and Section M mirrors evaluation factors and the stated basis of award per the FAR [51, 61].

Release readiness. Do not release the Request for Proposal (RFP) until the acquisition plan/strategy is approved, funds are certified, legal review is complete, synopsis rules are satisfied, and the source-selection teams are chartered per FAR/NMCARS rules [51, 56, 61].

Communication discipline. Request for Informations (RFIs), draft RFPs, industry days, and Questions and Answers (Q&A) all flow through the KO; once released, clarifications must be shared with all offerors through written amendments or controlled exchanges in accordance with the FAR [51, 61].

3.2.2 Practitioner Steps

1. Finalize the requirement package (spec /Statement of Work (SOW), Contract Data Requirements Lists (CDRLs), Independent Government Cost Estimate (IGCE), market research report) with IPT concurrence and Small Business coordination documentation per the FAR [51, 61].

2. Structure Section L instructions: volumes, page limits, submission medium, proposal due date, classification controls, and FAR [51, § 5.203] synopsis timing.

3. Align Section M evaluation factors, subfactors, and relative importance statements with the basis of award (Lowest Price Technically Acceptable (LPTA) v.s. trade-off) and rating methodology approved by the Source Selection Authority (SSA), KO, legal, and cost/price community consistent with the FAR [51, 58].

4. Verify pre-release approvals and postings: acquisition plan or strategy, Determination of Acquisition Strategy (as applicable), certified funds, and `https://www.SAM.gov` (System for Award Management (SAM)) synopsis [56, 61].

5. Manage amendments and records after release: capture bidder Q&A, issue conformed RFPs, adjust milestones when requirements shift, and file every action in the official contract record per the FAR and activity guidance [51, 59].

3.2.3 Checklist

- Acquisition plan/strategy signed; conformed requirements package appended per NMCARS [56].
- Funding document (Purchase Request (PR), project order, or MIPR) signed and funds certified for the planned obligation per the FAR [51].
- Market research report and Small Business Coordination Record approved per the FAR [51].

- Source Selection Plan endorsed by SSA, legal, and cost/price; nondisclosure/OCI statements executed for SSA, Source Selection Advisory Council (SSAC), Source Selection Evaluation Board (SSEB), and advisors [58].
- Section L instructions mirror Section M factors, include proposal structure, late proposal policy, and submission portal guidance per the FAR [51].
- `https://www.SAM.gov` synopsis posted (15 days before closing for most supplies/services; 30 days for Research and Development (R&D) unless justified) in accordance with the FAR [51].
- Amendment template/change log prepared; routing matrix established for expedited approvals per NAVSEA guidance [59].

3.2.4 Navy Overlays and Tailoring

NMCARS overlays. Follow [56, § 5205.303/5205.301] for synopsis exceptions and NAVSEA contracting notices for Contract Requirement Package checklists and release approvals [56, 59].

Technical data. Coordinate with TA on [55, § 252.227] clauses, distribution statements, and CDRL tailoring to protect Navy equities.

Shipbuilding tailoring. For availabilities, modernization, or planning efforts, ensure Section C and the WBS matches the avail sequence and integrates warfare-center technical direction and installation responsibilities [61].

3.2.5 Industry Engagement & Communications

Pre-release. RFIs, sources-sought notices, draft RFPs, and industry days are vetted through the KO; responses shared with all potential offerors to avoid unequal access [51].

Post-release. Clarifications and Q&A must be issued as amendments for all offerors; exchanges stay within FAR [51, § 15.201] boundaries until competitive range is established [59].

Small business focus. Coordinate with the Small Business Professional and Competition Advocate on set-aside decisions, subcontracting plan reviews, and mitigation of potential limitations on subcontracting per the FAR [51].

3.2.6 Documentation & Approvals to Capture

- Signed Source Selection Plan and SSA/SSAC/SSEB appointment letters with confidentiality and OCI certifications [58].
- Competition Advocate memoranda where narrowed sources or synopsis deviations are requested under the FAR/NMCARS rules [51, 56].
- Legal sufficiency memorandum for the conformed solicitation (Sections C, H, L, and M) retained in the contract file per NAVSEA contracting guidance [59].
- KO release memorandum noting solicitation number, issue date, amendment log, and proposal due date/time [61].

3.2.7 Competitive Method Selection: Sealed Bidding v.s. Negotiated Procurement

Sealed bidding (FAR Part 14). Preferred when requirements are well-defined, award will be based on price alone, discussions are not needed, and the Government expects to receive more than one responsive bid. It uses public opening, responsiveness determinations, and fixed-price awards [51, Part 14].

Negotiated procurement (FAR Part 15). Used when the Government needs to evaluate technical approach or past performance, intends to conduct discussions, or when requirements or pricing demand flexibility. Negotiated acquisitions support best-value trade-offs and cost-reimbursement vehicles [51, Part 15].

When to choose. If the team can describe performance in terms of precise specifications with minimal risk and expects no need for exchanges, sealed bidding streamlines award. When innovation, risk, or integration complexity requires subjective evaluation, or when schedule/budget changes are likely, negotiated procurement is mandatory under the FAR [51, 58].

3.2.8 Contract Type Landscape and Risk Allocation

Fixed-price family. Contractor accepts the cost risk; Government locks price once requirements are stable. Includes:

- **Firm-Fixed-Price (FFP)**: Used for mature, low-risk requirements, often in Production and Deployment [51, 52].
- **Fixed-Price Incentive (Firm Target) (FPIF)**: Shares cost variance through a negotiated share ratio; encourages cost control while keeping final price bounded by a ceiling. Used during EMD or LRIP when cost risk remains but production discipline is needed [51, 52].
- **Fixed-Price with Economic Price Adjustment (FPEPA)**: Adds indexed adjustments for volatile commodities or long-lead items [51].
- **Fixed-Price Redetermination (FPR)**: Re-prices after defined milestones when initial uncertainty is expected to retire [51].

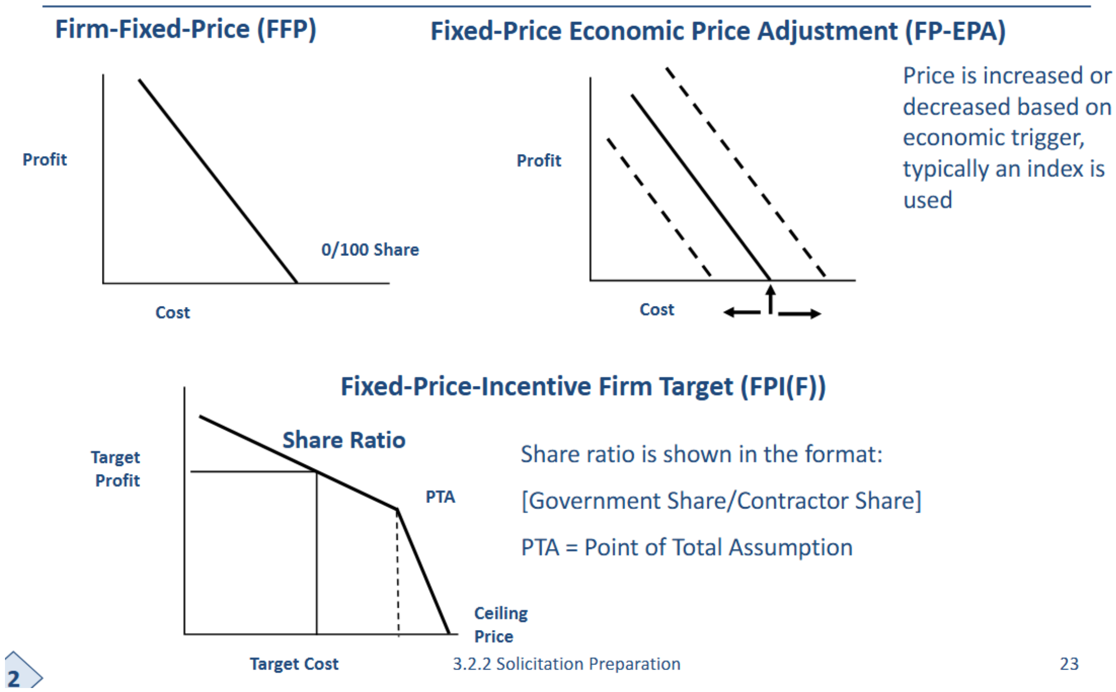


Figure 3.1. *Fixed-price incentive geometry (cost, profit, PTA).*

Cost-reimbursement family. Government carries more cost risk; contractor must deliver the best effort. Includes:

- **Cost-Plus-Fixed-Fee (CPFF)**: Used in early R&D or prototyping (MSA/TMRR) when effort is exploratory [51, 52].
- **Cost-Plus-Incentive-Fee (CPIF)**: Shares cost variance similar to FPIF but without a price ceiling; encourages efficiency while acknowledging uncertain cost baseline [51].
- **Cost-Plus-Award-Fee (CPAF)**: Adds subjective award-fee evaluation for mission effectiveness or management performance that cannot be captured in objective metrics [51].

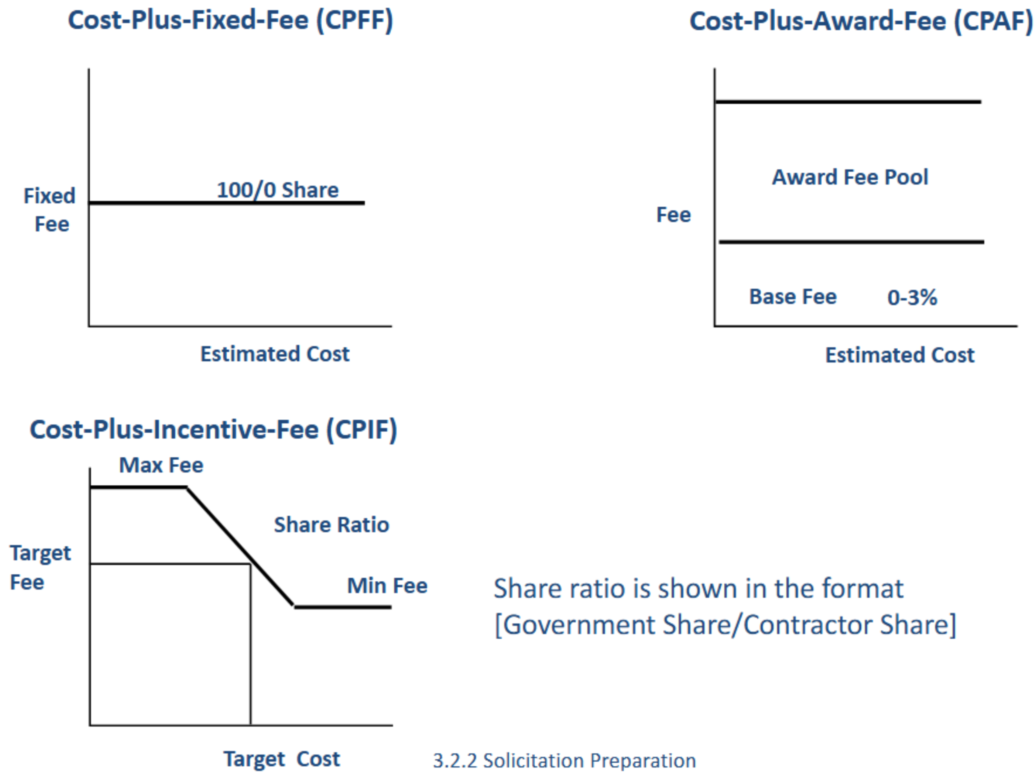


Figure 3.2. *Cost-reimbursement contract risk/fee characteristics. Source:* 3.2.2. Solicitation Preparation, *2025 [61].*

Award v.s. incentive fees. Incentive fees (cost or performance) are calculated from predetermined formulas tied to measurable outcomes (cost, schedule, technical) under the FAR; award fees are earned through periodic board evaluations against tailored factors and may be unearned if performance is merely satisfactory [51, 59].

Award-fee governance. The Award Fee Determining Official (often the PM or SSA designee) chairs an Award Fee Board, uses the approved plan, and issues a determination memo. Fees are paid after each period and must be commensurate with value delivered; scores below the threshold earn zero dollars for that segment [51, 58].

MCA phase alignment. MSA and TMRR generally use CPFF or CPAF because of design uncertainty; EMD can transition to CPIF or FPIF as risk retires; Production and Deployment favors FFP/FPIF; Operations and Support relies on FFP or FFP Level of Effort (LOE) and can incorporate sustainment IDIQs for depot work [52].

MSA. Analytical trades and early prototypes benefit from CPFF flexibility as requirement scope evolves.

TMRR. CPIF/CPAF maintain incentives while the Government still absorbs major technical risk during competitive prototyping.

EMD. FPIF or CPIF balance contractor motivation and cost control once the design stabilizes, with ceilings to protect the Government.

P&D. FFP/FPIF dominate because production baselines and learning curves are known; contractor should own execution risk.

O&S. Sustainment leverages FFP, FFP LOE, or IDIQs task orders to manage repeatable workloads and retain competition for modernization.

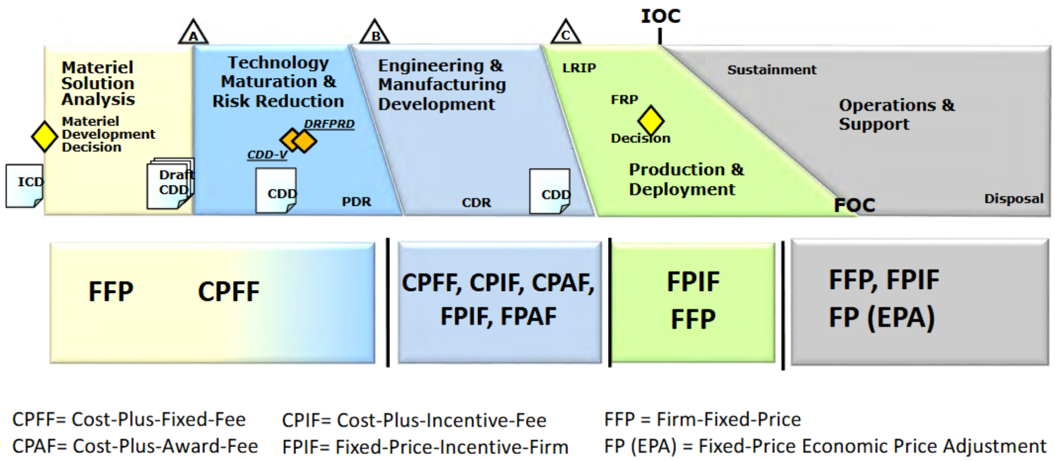


Figure 3.3. *Contract-type emphasis across the Major Capability Acquisition phases.*

3.2.9 Fixed-Price Incentive (Firm Target) Mechanics

Share ratio. Defines how cost overruns or underruns are split between Government and contractor (e.g., 70/30 means Government absorbs 70% of variance, contractor 30%) per the FAR [51].

Point of Total Assumption (PTA). The cost point beyond which every additional dollar of overrun is borne entirely by the contractor:

$$PTA = \frac{\text{Ceiling Price} - \text{Target Cost}}{\text{Government Share Ratio}} + \text{Target Cost}.$$

Example. Target cost \$100M, target profit \$10M (target price \$110M), ceiling price \$120M, share ratio 70/30. If actual cost is \$112M:

$$\text{Contract price} = \$110\text{M} + 0.3 \times (\$100\text{M} - \$112\text{M}) = \$106.4\text{M}.$$

The PTA occurs at $\$113.\overline{3}\,\text{M}$; beyond that, profit erodes dollar-for-dollar until exhausted [51, 59].

3.2.10 Indefinite & Flexible Contract Forms

IDIQ. Indefinite-Delivery, Indefinite-Quantity contracts provide flexibility in quantity and timing with task/delivery orders; suited for recurring installations, sustainment, or services [51].

Time-and-Materials (T&M)/Labor-Hour (LH). Hybrid vehicles when it is not possible to estimate effort precisely; require ceiling price, surveillance, and justification because Government assumes cost risk on labor hours [51].

Undefinitized Contract Actions (UCAs). Letter contracts or other actions authorized before all terms are settled (e.g., urgent ship repair). Must be definitized within the regulatory timeline, with strict management for obligation limits and profit [51, 55, 56].

3.2.11 Debriefings and Protests (Forums and Timelines)

Debriefings. Post-award (and certain pre-award) debriefings are conducted per the FAR [51, § 15.506]. For DoW enhanced debriefings, the question period extends the debrief conclusion for protest timeliness purposes; verify current Class Deviation guidance.

Protest forums. (i) Agency-level to the contracting activity (KO/Head of Contracting Activity) [51, § 33.103]; (ii) GAO bid protests [51, § 33.104]; (iii) United States (U.S.) Court of Federal Claims (Court of Federal Claims (COFC)) under 28 U.S.C. § 1491(b).

Who handles them. Agency: procuring activity and counsel; GAO: GAO decides, agency produces the record and implements any stay; COFC: Department of Justice litigates, COFC judge adjudicates and may grant injunctive relief.

Timelines (common rules). Pre-award challenges to solicitation terms must be filed before the proposal due date. For post-award protests at GAO: file within 10 calendar days of when the basis is known, or within 5 business days after debriefing conclusion to obtain an automatic stay of performance under Competition in Contracting Act (CICA). Agency-level protest timeliness follows similar 10-day knowledge rules in [51, § 33.103]. COFC has no fixed day-limit but relief is equitable; prompt filing is expected.

3.2.12 Why Cost-Plus After Low-Rate Initial Production? (PD and O&S Exceptions)

High residual technical risk. Post-LRIP, major engineering change proposals, tech refresh, or obsolescence-driven redesign can reintroduce uncertainty better handled under CPIF/CPAF than FFP.

Unstable scope or emergent defects. Corrective actions discovered during Initial Operational Test and Evaluation (IOT&E)/early fielding may require iterative investigation and rework where a best-effort *cost-reimbursable* vehicle protects schedule and access to talent.

Sustainment stand-up. Initial depot activation, organic transition, or complex PBL standing up new measurements and data flows can warrant short-duration CPFF/CPAF with incentives while processes stabilize.

Data rights/technical baseline gaps. Where the Government lacks detailed technical data to price FFP confidently, a temporary cost-plus arrangement with aggressive CDRL deliverables can bridge to FFP.

Governance. Treat as exceptions with a plan to transition to FFP/FPIF once uncertainty retires; document rationale in the Acquisition Strategy and obtain approvals consistent with [51, 55].

3.2.13 Simplified Acquisition and Small Business Programs

Simplified Acquisition Threshold (SAT). Generally $250,000 (higher in contingency or overseas circumstances). Benefits include streamlined competition, quotation-based awards, and mandatory small-business reservations above the micro-purchase threshold [51].

Small Business Administration (SBA) programs. FAR Part 19 [51, Part 19] implements 8(a), Historically Underutilized Business Zone (HUBZone), Service-Disabled Veteran-Owned Small Business (SDVOSB), and Women-Owned Small Business (WOSB) programs. The contracting officer must consider set-asides, sole-source thresholds, and subcontracting plans to meet DON goals.

3.2.14 Alternative Acquisition Authorities

Other Transaction (OT) agreements. Authorized by 10 U.S.C. § 4021 (prototype) and § 4022 (follow-on production) to rapidly engage nontraditional performers, provided cost share or significant participation criteria are met [62, 63].

Small Business Innovation Research (SBIR). 15 U.S.C. § 638 establishes Phase I (feasibility), Phase II (prototype), and Phase III (commercialization/production). SBIR awards leverage R&D funds to deliver Navy-unique technology with data rights protections for five years [64].

3.2.15 Technical Data Rights Overview

Unlimited rights. Government may use, disclose, and authorize others without restriction (typically for data developed exclusively at Government expense) per the DFARS [55].

Government purpose rights. Government may use or authorize contractors for Government purposes for five years, after which rights convert to unlimited. Applies to mixed-funding developments under the DFARS [55].

Limited/Restricted rights. Contractor retains control; Government use is constrained to internal purposes (limited for noncommercial technical data, restricted for commercial computer software). Negotiated licenses or specifically negotiated licenses can expand access per the DFARS [55].

3.2.16 Uniform Contract Format Essentials

Sections A through H (contract). Cover Standard Form 33, supplies/services, packaging, inspection, delivery, special contract requirements, clauses, and attachments per the FAR UCF [51].

Sections I through K (legal/representations). Incorporate FAR/DFARS clauses, representations, certifications, and instructions for completion [51, 55].

Sections L and M (solicitation/Evaluation). Instructions to offerors, proposal structure, page limits, and evaluation criteria that must remain synchronized to any change to factors or relative order requires an amendment under the FAR [51, 58].

3.3 Cost and Price Evaluation

3.3.1 Summary

KO accountability. The KO must determine that every negotiated price is fair and reasonable, integrating price, cost, technical, field-pricing, and risk analysis inputs before award per FAR 15.404-1(a)(1)–(5) [51, § 15.404-1(a)].

Cost v.s. price lens. Use price analysis when certified cost or pricing data are not required, and escalate to cost analysis when element-by-element scrutiny is needed to support the prenegotiation objective under the FAR [51, §§ 15.404-1(b), 15.404-1(c)].

Rate discipline. Forward pricing rates and structured profit analysis keep the Government prenegotiation objective current; the objective is always proposed cost plus profit or fee per FAR policy [51, §§ 15.401, 15.404–4, 15.407–3].

Financial health cue. Profitability ratios (Return on Sales (ROS), Return on Assets (ROA)) and cash-flow checks expose responsibility risk and focus Defense Contract Audit Agency (DCAA)/ DCMA field-pricing support requested by the KO [51, § 9.104-1] [65].

3.3.2 Practitioner Steps

1. Baseline the offeror's proposal: confirm certified cost or pricing data requirements, request data other than certified cost or pricing data when warranted, and map contractual requirement traces per the FAR [51, §§ 15.403-1, 15.404-1(a)].

2. Execute price analysis first—compare competition results, historical buys, catalog/commercial data, and independent estimates; pivot to cost analysis when price analysis alone cannot demonstrate reasonableness per the FAR [51, §§ 15.404-1(b), 15.404-1(c)].

3. Decompose direct labor, material, and Other Direct Costs (ODCs); verify indirect pools and allocation bases; and apply the five allowability tests before accepting proposed cost elements [51, §§ 31.201-2, 31.202, 31.203].

4. Establish the profit/fee objective using the agency's structured approach and update forward pricing rates or agreements as necessary to avoid stale factors in negotiations under the FAR [51, §§ 15.404-4, 15.407–3].

5. Pull in field-pricing support early: blend DCAA audit results, DCMA production assessments, and program team technical evaluations into the prenegotiation briefing and contract file [51, § 15.404-1(a)(5)] [51, § 42.101(b)] [55, § 242.302(a)(13)(A)].

3.3.3 Proposed Price, Cost, Profit, and Fee

The breakdown of the proposed price is showing in Figure 3.4 and described below.

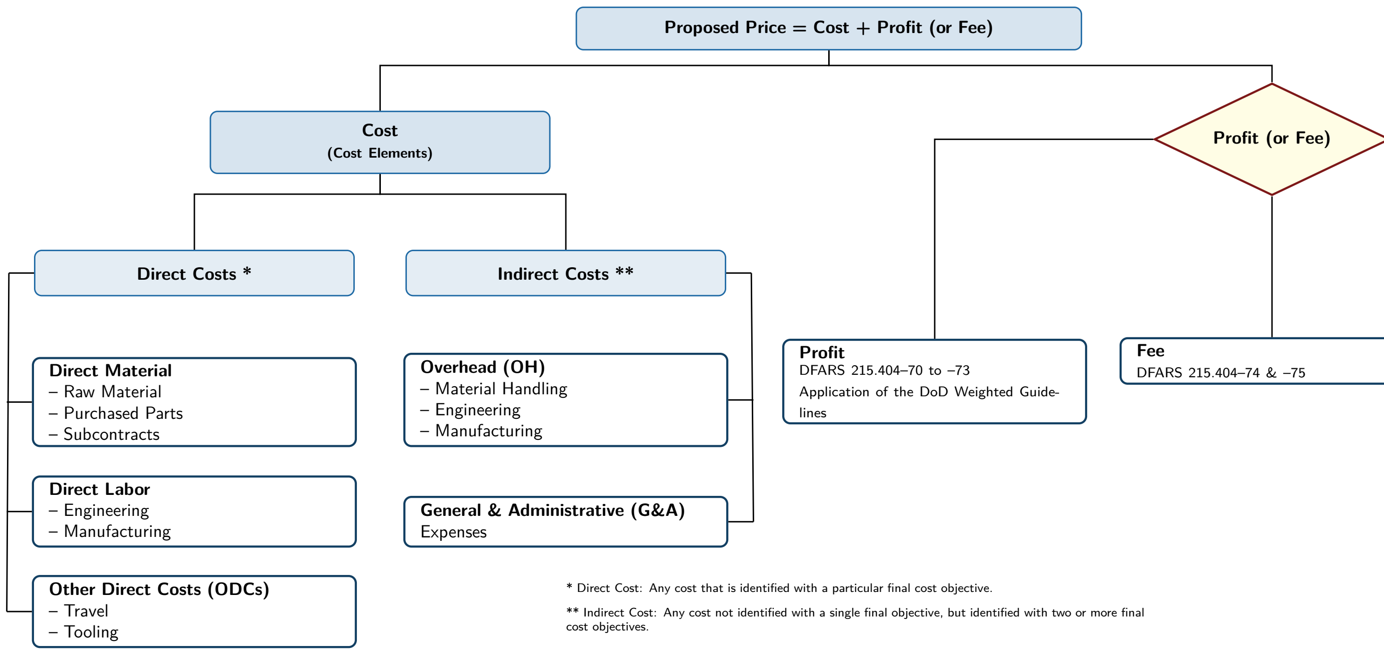


Figure 3.4. *Mapping direct v.s. indirect contractor cost elements.Source:* 3.2.4. Cost and Price Evaluation, *2024 [65]*

Proposed price. The offeror's total price equals estimated cost plus any profit or fee applicable to the contract type per the FAR [51, § 15.401].

Cost. Sum of allowable direct and indirect costs required to deliver the contract—all subject to reasonableness, allocability, and cost-principle limits under the FAR [51, §§ 31.201-2, 31.202, 31.203].

Profit. Negotiated incentive element above allowable cost for fixed-price or incentive contracts, set through structured analysis per the FAR [51, § 15.404-4(a)].

Fee. Fixed remuneration on cost-reimbursement vehicles that does not vary with actual cost (e.g., CPFF fee) [51, § 16.306(a)].

3.3.4 Direct and Indirect Cost Structure

Direct cost pools. Direct labor, materials, and ODCs traceable to a single final cost objective; charged through project-specific accounts backed by bills of material, timecards, travel authorizations, or subcontract quotes [51, § 31.202] [65].

Indirect cost pools. Overhead (manufacturing, engineering) and G&A accrued across multiple objectives, then allocated using consistent bases that reflect benefits received [51, § 31.203].

3.3.5 Other Direct Costs and Cost Allowability

ODCs. Travel, subcontract services, specialized tooling, and similar charges that can be singled out for the contract even though they are not labor or bulk material categories [51, § 31.202] [65].

Cost allowability. A cost is billable only if it is reasonable, allocable, consistent with applicable Cost Accounting Standards (CAS) / Generally Accepted Accounting Principles (GAAP), compliant with contract terms, and within any Subpart 31.2 limitations [51, § 31.201-2].

Note

Slides may refer to "cash allowability"; FAR terminology is *cost* allowability.

3.3.6 Cost and Price Analysis

Price analysis. Examination of the total proposed price without breaking out cost elements using competition, historical buys, or market indicators per the FAR [51, § 15.404-1(b)].

Cost analysis. Evaluation of individual cost elements (direct, indirect, and profit) when price analysis alone cannot demonstrate reasonableness under the FAR [51, § 15.404-1(c)].

Integrated view. Technical analysis, field-pricing support, and risk assessments feed both techniques to substantiate the final fair and reasonable determination per the FAR [51, § 15.404-1(a)(5)].

3.3.7 Forward Pricing Rates and Fully Burdened Labor

Forward Pricing Rate Proposal (FPRP). Contractor-submitted forward pricing rate proposal that lays out projected indirect rates and factors for the pricing period [51, § 15.407-3(a)].

Forward Pricing Rate Agreement (FPRA). Negotiated agreement-often executed by DCMA-that locks indirect rates for one to three years, reducing repetitive audits and speeding negotiations [51, § 15.407-3(c)] [65].

Fully burdened rate. Applies direct labor, indirect burdens, and any negotiated profit/fee to a common labor-hour baseline so skill mixes can be compared on an apples-to-apples basis [65].

3.3.8 Indirect Rate Math and Profitability Checks

Apply the standard relationships from the cost principles and the coursebook when validating proposals; they are board favorites for quick-turn computations covering indirect pools, Total Cost Input (TCI), G&A, and profitability metrics such as ROS and ROA [51, §§ 31.201-2, 31.202, 31.203] [65].

$$\text{Indirect Cost Rate} = \frac{\text{Indirect Cost Pool}}{\text{Allocation Base}} \tag{3.1}$$

$$\text{TCI} = \text{Direct Cost} + \text{Overhead Cost} \tag{3.2}$$

$$\text{G\&A Cost} = \text{G\&A Rate} \times \text{TCI} \tag{3.3}$$

$$\text{Fully Burdened Labor Rate} = \frac{\text{Direct Labor Cost} + \text{Indirect Costs} + \text{Profit/Fee}}{\text{Direct Labor Hours}} \tag{3.4}$$

$$\text{ROS} = \frac{\text{Operating Profit}}{\text{Net Sales}} \tag{3.5}$$

$$\text{ROA} = \frac{\text{Net Income}}{\text{Total Assets}} \tag{3.6}$$

Track ROS and ROA against industry benchmarks, cash-on-hand, and debt loads to detect responsibility concerns or unsustainable buy-ins before negotiations [65].

3.3.9 Oversight and Field Pricing Support

DCMA production insight. Delegated contract administration offices deliver manufacturing surveillance, schedule risk analysis, and forward-pricing coordination; DFARS highlights that payment authority stays with the buying command even when DCMA supports rates [55, § 242.302(a)(13)(A)] [65].

DCAA audit coverage. DCAA is the responsible Government audit agency for most contractors, providing proposal adequacy reviews, incurred cost audits, and financial capability analyses [51, § 42.101(b)].

Integrated negotiation record. Document how DCMA, DCAA, technical, and program inputs influenced the prenegotiation objective and the final price reasonableness determination [51, § 15.404-1(a)(5)] [65].

3.3.10 Checklist

- File both price and cost analysis results (including technical inputs) showing how the fair-and-reasonable price was determined per the FAR [51, § 15.404-1].
- Verify each significant cost element passes the five allowability tests and that allocation bases match current practice under the FAR [51, §§ 31.201-2, 31.203].
- Confirm the current FPRA/FPRP is in the file or document why legacy rates remain valid [51, § 15.407-3].
- Capture DCAA audit opinions, DCMA rate guidance, and profitability ratio trends in the prenegotiation briefing to evidence responsibility diligence [51, §§ 9.104-1, 42.101(b)] [65].

3.4 Earned Value Management (EVM)

3.4.1 What EVM is

EVM integrates scope, schedule, and cost to quantify performance by comparing planned work to completed work and associated costs; it is widely used in government and defense programs to keep delivery on time and within budget.

3.4.2 Key Components of EVM

Core cost and schedule metrics establish the earned value performance baseline [42, 43].

Planned Value (PV) / Budgeted Cost of Work Scheduled (BCWS): The budgeted cost for the work scheduled to be completed by a specific date.

Earned Value (EV) / Budgeted Cost of Work Performed (BCWP): The budgeted cost for the work actually completed by a specific date.

Actual Cost (AC) / Actual Cost of Work Performed (ACWP): The actual cost incurred for the work completed by a specific date.

Budget at Completion (BAC): The total budget allocated for the project.

EAC: The forecasted total cost of the project based on current performance.

CPI: A measure of cost efficiency, calculated as $\mathrm{CPI} = \mathrm{EV}/\mathrm{AC}$. A CPI less than 1 indicates a cost overrun.

SPI: A measure of schedule efficiency, calculated as $\mathrm{SPI} = \mathrm{EV}/\mathrm{PV}$. A SPI less than 1 indicates a schedule delay.

SV: The difference between the earned value and the planned value, calculated as $\mathrm{SV} = \mathrm{EV} - \mathrm{PV}$. A negative SV indicates a schedule delay.

CV: The difference between the earned value and the actual cost, calculated as $\mathrm{CV} = \mathrm{EV} - \mathrm{AC}$. A negative CV indicates a cost overrun.

Info

The first three are equivalent terms; this guide uses the PV, EV, AC forms.

3.4.3 When to Use EVM

EVM is mandated on cost or incentive contracts (and applicable subcontracts) when the program meets the policy thresholds and the effort is discretely measurable [42, 55]. Key decision points for the PM include:

$100 million and above: Implement the full American National Standards Institute (ANSI)/Electronic Industries Alliance (EIA)-748 standard and ensure the contractor's system is formally validated by DCMA.

$20 million to $99.9 million: Implement ANSI/EIA-748 guidelines, with validation required when performance indicates risk.

Additional policy considerations follow [42, 52].

- Applicability determinations must confirm the work scope can be discretely measured before mandating EVM.
- The Milestone Decision Authority may approve EVM on FFP, time-and-materials, or LOE contracts, but such use is discouraged absent clear benefit.
- For efforts below $20 million, the PM conducts a risk-based cost-benefit analysis to decide whether EVM adds value.
- Contracts that require EVM must also deliver IPMDAR data and complete an IBR within six months of award.

3.4.4 EVM Compliance

EVM clauses are flowed in the solicitation and award package (DFARS 252.234-7001 and 252.234-7002) alongside the Contractor Business Systems clause (DFARS 252.242-7005) to anchor validation and surveillance requirements [55]. Compliance expectations follow [42, 52].

- The program office integrates EVM planning into the WBS, maintains the Performance Measurement Baseline, and ensures timely receipt and analysis of IPMDAR submissions.
- DCMA leads EVM system acceptance, validation, and ongoing surveillance against ANSI/EIA-748 and the DoW EVM Implementation Guide; DCAA supports with accounting system audits.
- Service focal points such as Acquisition Data and Analytics and the DON Center for EVM adjudicate applicability determinations and coordinate policy updates.
- SUPSHIP can execute validation and surveillance responsibilities for shipbuilding programs on behalf of DCMA *(Appendix C.2: RDML Jonathan J. Jett-Parmer leads SUPSHIP; RDML Robert J. Dodson (USNR) is the reserve deputy)*.

3.4.5 EVM Principal Players

Programmatic, contracting, and oversight roles synchronize to sustain disciplined performance management [42].

PMO: Implements EVM on the contract, ensures solicitations and awards contain the required clauses, and works with the contracting activity to tailor reporting and Integrated Master Schedule (IMS) requirements.

DCMA: Serves as the DoW EVM Executive Agent by validating, accepting, and surveilling contractor systems, maintaining the official acceptance roster, and analyzing IPMDAR and schedule submissions (including the 14-point IMS review) to focus government attention on emerging issues.

DCAA: Conducts accounting system audits, corroborates cost data, and supports DCMA during surveillance events and reviews.

Acquisition Data and Analytics (ADA): Acts as the department-wide focal point for EVM policy, guidance, and competency management; issues interpretations to maintain consistent application across programs.

DON Center for EVM: Coordinates applicability determinations with Deputy Assistant Secretary of the Navy (Acquisition Policy and Budget) (DASN (APB)) and OSD, and serves as the Navy's central point of contact for EVM matters.

SUPSHIP: Performs many DCMA/DCAA surveillance roles for shipbuilding contracts when delegated by the cognizant program office. *(Appendix C.2: See Appendix C.2 for current SUPSHIP flag billet holders)*

3.4.6 PMB

Figure 3.5 shows the characteristic S-curve depicting cumulative PV/BCWS, while Figure 3.6 summarizes the progression from defining the SOW to final PMB adjustments. PMBs are:

- Scoped, scheduled, and budgeted plans for the authorized work
- Time-phased budgets that align to the master schedule
- The basis for cost and schedule performance management
- Effectively the PV for the entire project

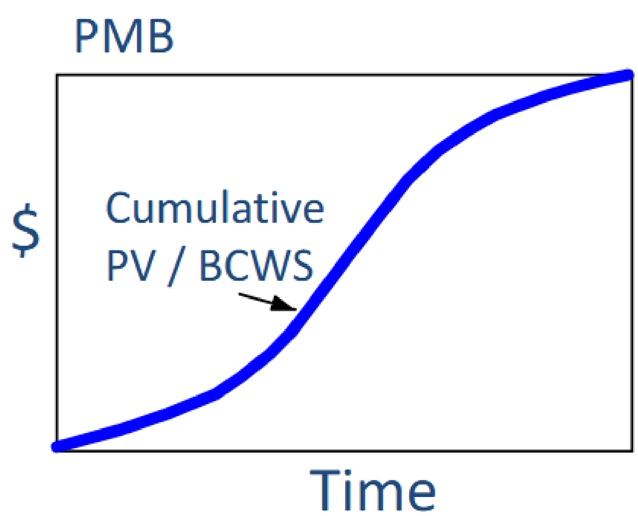


Figure 3.5. *PMB cost v.s. time chart showing the "S"-curve. Source:* 3.6.1. Introduction to EVM, *2025 [42].*

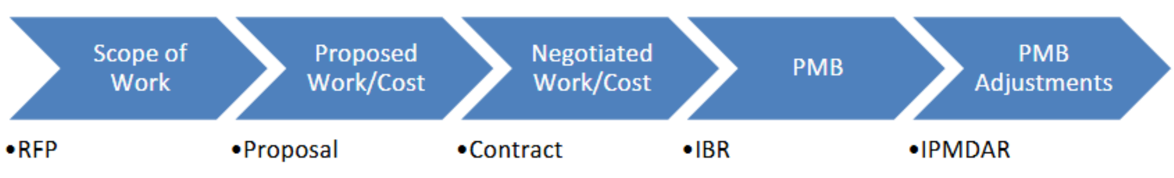


Figure 3.6. *The development flow for PMB. Source:* 3.6.1. Introduction to EVM, *2025 [42].*

PMB Development

Step 1: *Define all work*. Decompose the effort using the WBS and align it with the organizational structure. Control accounts are the natural management points where Control Account Managers (CAMs) are assigned, and each control account contains work packages and (if needed) planning packages.

Step 2: *Schedule the work*. Develop the integrated schedule (often visualized with Gantt charts) that sequences the WBS elements, includes key milestones, and forms the backbone of the time-phased budget.

Step 3: *Budget the work*. Assign time-phased budgets to each work package, establish management reserve for in-scope known unknowns, and confirm that the sum of the control accounts equals the contract budget base.

Changes to the PMB must be formally controlled and documented. Reasons for changes include:

- Contract changes
- Internal replanning
- Formal reprogramming

3.4.7 EVM Reviews and Reports

Formal reviews validate the integrity of EVM data products before and after system acceptance [42].

Post-Acceptance Review

- Objective: ensure performance data remain accurate after system acceptance and identify any corrective actions required to reaffirm compliance.
- Timing: conducted as needed following system acceptance and prior to the next IBR.
- Led by the Review Director (typically DCMA) with membership tailored to the purpose of the review.
- Culminates in a formal report prepared by the Review Director.

Initial Compliance Evaluation

- Objective: validate that the Contractor's EVM system description matches actual practice and satisfies the Earned Value Management System (EVMS) criteria.
- Timing: executed prior to the initial IBR (and subsequently only if needed).
- Led by the DCMA Review Director with cross-functional support as required.
- Results documented in a report signed by the Review Director.

Integrated Baseline Review

- Participants: the contractor, PM and deputy, DCMA/DCAA/SUPSHIP personnel, and relevant technical staff.
- Purpose: joint assessment to verify the realism and accuracy of the PMB, confirm the entire technical scope is captured, and ensure resources and schedules are achievable.
- Timing: conducted within six months of contract award (or major replanning event) in accordance with the DoW EVM Implementation Guide and service policy [42].

Integrated Program Management Data Analysis Report

This contractually required report delivers cost, schedule, and technical status derived from the Contractor's EVM system, enabling the PMO to identify performance problem areas and emerging risks. It is required on every contract that mandates EVM and must be delivered at least monthly. The data set is tailored for each contract based on risk, size, and integration considerations, but the canonical formats are:

1. WBS (most common)
2. Organizational categories
3. Program Management Baseline
4. Staffing
5. Explanation and Problem Analysis
6. IMS

A CSFR complements the IPMDAR by providing funding (price) information that reconciles to the cost-focused earned value data [42].

Problems

When using EVM, the following are indications that a problem exists [43].

- Use of management reserves
- Significant revisions to the PMB
- Zero variance
- Sudden change in trends
- Unreasonable Total Cost Performance Index (TCPI)

3.4.8 EVM Data Analysis

Figure 3.7 shows the S-curve representation of the EVM variables; the gaps between the curves are the cost and schedule variances that drive performance assessments [43].

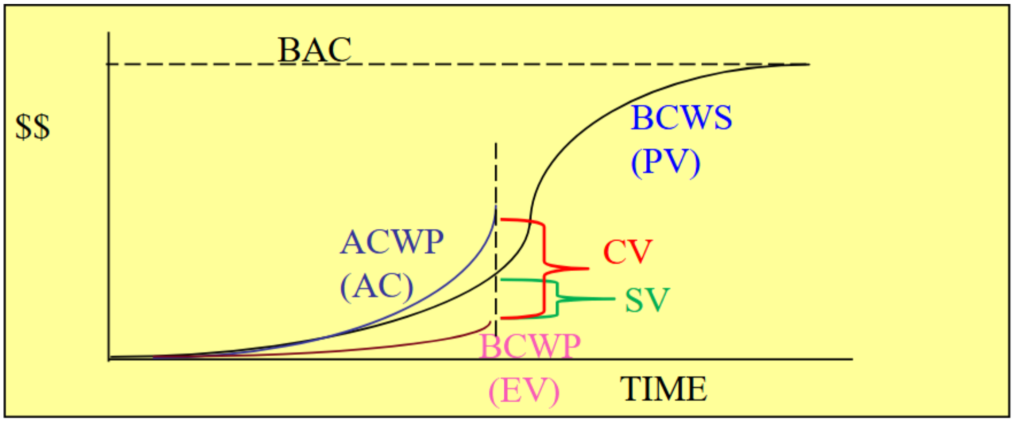


Figure 3.7. *EVM chart showing the independent variables and variances. Source:* 3.6.2. EVM Data Analysis, *2025 [43]*

Data Analysis Steps

1. **Get current status**: capture the latest cost and schedule performance using CPI, SPI, and narrative explanations.
2. **Identify trends**: analyze indices and variance trajectories over time to highlight emerging risks and opportunities.
3. **Predict completion**: develop independent EAC and Variance at Complete (VAC) assessments to forecast final outcomes.
4. **Determine management actions**: decide where to apply resources, contract changes, or risk mitigations to recover performance [43].

Performance Measurement Techniques

Percent Complete: Applies to long-duration tasks lacking interim milestones; progress is reported as the earned value fraction of total budget.

Weighted Milestones: Uses milestone weights for long-duration tasks with clear waypoints, crediting earned value as milestones are completed.

Percent Start/Percent Finish: Suitable for short-duration tasks (less than two reporting periods) with credit apportioned at start and finish.

0/100 Method: Provides earned value only when the task is complete, offering a conservative measure for discrete, short tasks [43].

Equations

Standard cost and schedule performance calculations include [43].

$$
\begin{aligned}
\mathrm{CV} &= \mathrm{EV} - \mathrm{AC} \\
\mathrm{SV} &= \mathrm{EV} - \mathrm{PV} \\
\mathrm{CPI} &= \frac{\mathrm{EV}}{\mathrm{AC}} \\
\mathrm{SPI} &= \frac{\mathrm{EV}}{\mathrm{PV}} \\
\mathrm{CV\%} &= \left(\frac{\mathrm{CV}}{\mathrm{EV}}\right) \times 100 \\
\mathrm{SV\%} &= \left(\frac{\mathrm{SV}}{\mathrm{PV}}\right) \times 100 \\
\%\ \text{Complete} &= \left(\frac{\mathrm{EV}}{\mathrm{BAC}}\right) \times 100 \\
\%\ \text{Spent} &= \left(\frac{\mathrm{AC}}{\mathrm{BAC}}\right) \times 100 \\
\mathrm{EAC} &= \frac{\mathrm{BAC}}{\mathrm{CPI}} \\
\mathrm{EAC} &= \mathrm{AC} + \frac{\mathrm{BAC} - \mathrm{EV}}{\left(\frac{\mathrm{EV}}{\mathrm{AC}}\right)_{\text{3 months}}} \\
\mathrm{EAC} &= \mathrm{AC} + \frac{\mathrm{BAC} - \mathrm{EV}}{\mathrm{CPI} \times \mathrm{SPI}} \\
\mathrm{EAC} &= \mathrm{AC} + \frac{\mathrm{BAC} - \mathrm{EV}}{0.8\,\mathrm{CPI} + 0.2\,\mathrm{SPI}} \\
\mathrm{VAC} &= \mathrm{BAC} - \mathrm{EAC} \\
\mathrm{TCPI} &= \frac{\text{Work remaining}}{\text{Budget remaining}} \\
\mathrm{TCPI}_{\mathrm{EAC}} &= \frac{\mathrm{BAC} - \mathrm{EV}}{\mathrm{EAC} - \mathrm{AC}} \\
\mathrm{TCPI}_{\mathrm{BAC}} &= \frac{\mathrm{BAC} - \mathrm{EV}}{\mathrm{BAC} - \mathrm{AC}}
\end{aligned}
$$

4 Defense Acquisitions

4.1 Acquisition Policy

4.1.1 Summary

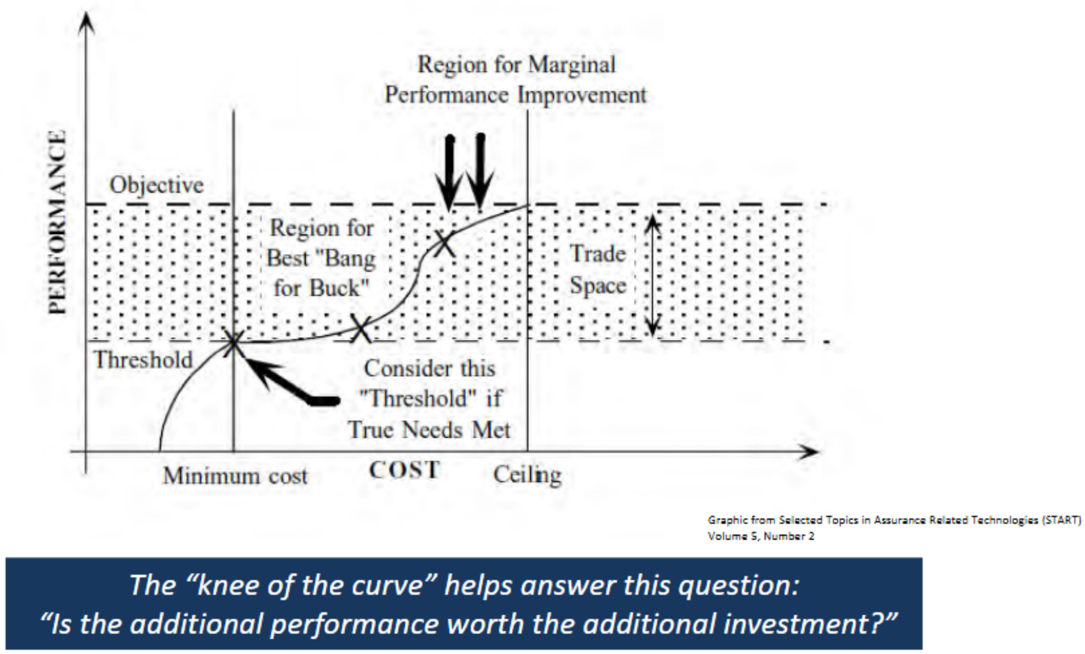
Statute drives authority. Title 10 charges the Military Departments to equip the force, while each annual NDAA refreshes acquisition authorities and reporting duties—know the latest delegation trail before advising a board [66].

Policy = DoW 5000 series. DoDD 5000.01 establishes acquisition principles and senior roles; DoDI 5000.02 operationalizes the Adaptive Acquisition Framework (AAF) pathways; DoDI 5000.85 prescribes MCA execution details [52, 67, 68].

Regulation stack. FAR > DFARS > NMCARS translates statute and policy into enforceable contracting rules; Secretary of the Navy Instruction (SECNAVINST) 5000.2G tailors the AAF for DON programs and makes ASN(RD&A) the SAE [3, 51, 55].

The main drivers that a PM must balance is cost, schedule, and performance as shown in Figure 4.1. The trade space for performance versus cost is shown in Figure 4.2, which shows the relation of performance and cost with the threshold and objective requirements.

Figure 4.1. The cost, schedule, and performance triangle that must be balanced and traded. Source: 3.3.1. Acquisition Policy and Players, *2025 [66].*



3.3.1 Acquisition Policy and Players

Figure 4.2. *Trade space of performance v.s. cost showing the range of viable options with the region of best "value." Source:* 3.3.1. Acquisition Policy and Players*, 2025 [66].*

4.1.2 Practitioner Steps (Board Prep Focus)

1. Confirm statutory authority and delegation: identify the Defense Acquisition Executive (DAE) (Under Secretary of War for Acquisition and Sustainment (USW(A&S))) or delegated MDA, the SAE, ASN(RD&A), and the resource sponsor accountable for the requirement [3, 66, 67].

2. Select the correct AAF pathway (or hybrid) and align entry/exit criteria, statutory reports, and decision forums (Milestone (MS) A/B/C, MDD, Business Case Analyses (BCAs), Congressional notices) [52, 68].

3. Map contracting rules to the pathway: FAR/DFARS clauses, competition requirements, and Tailored Acquisition Strategy approvals [51, 55, 66].

4. Crosswalk Navy overlays: SECNAVINST 5000.2G, SECNAVINST 5400.15D organization responsibilities, and PEO/DRPM charters for technical authority touchpoints [2, 3].

5. Prepare decision documentation: update the Acquisition Strategy /Systems Engineering Plan (SEP) /Test and Evaluation Master Plan (TEMP), ensure statutory certifications (Clinger-Cohen, PPBE affordability caps) are current, and pre-brief the chain (Program Office → PEO → ASN(RD&A)) [52, 66].

4.1.3 Policy Stack and Authorities

Title 10, U.S.C. Establishes Service responsibilities (Subtitle C for DON organization, Subtitle A Part V for acquisition management) and empowers annual NDAAs to adjust acquisition thresholds or pilot authorities [66].

DoW Directives/Instructions. DoDD 5000.01 sets acquisition principles, governance forums, and senior leader responsibilities; DoDI 5000.02 implements the AAF with six scalable pathways; DoDI 5000.85 gives MCA-specific statutory requirements (Joint Requirements Oversight Council (JROC), cost reporting, baseline control) [52, 67, 68].

Regulations. FAR and DFARS governs all federal contracting; DFARS adds DoW-specific clauses (e.g., earned value, data rights, cybersecurity); NMCARS adds DON policy and ASN(RD&A) approval levels [51, 55, 56].

Service overlays. SECNAVINST 5000.2G tailors milestone documentation, Naval SYSCOM oversight, and Naval Accelerated Acquisition; SECNAVINST 5400.15D assigns PEO and SYSCOM responsibilities for acquisition program execution [2, 3].

Advisory guidance. The Defense Acquisition Guidebook (DAG) captures best practices—it is not directive authority, but boards expect you to know how it informs planning reviews and tailoring memoranda [69].

Info

Boards expect you to quote the controlling document *and* state who owns the decision. Memorize the policy ladder: Statute → Directive → Instruction → Regulation → Service overlay.

4.1.4 Acquisition Players and Decision Forums

DAE. USW(A&S) chairs the Defense Acquisition Board (DAB), is MDA for ACAT ID/Major Automated Information System (legacy ACAT I equivalent) (IAM) programs unless delegated, and approves key acquisition policies [66, 67].

SAE. ASN(RD&A) serves as the DON SAE; appoints PEOs, assigns MDA for ACAT II and below, and ensures Navy acquisition compliance with DoW policy [2, 3].

Chief of Naval Operations/Resource Sponsor. Validates requirements and PPBE resourcing, provides integrated warfare/community priorities to ASN(RD&A) and the PEOs [66].

PEO/DRPM. Executes programs within delegated authorities; maintains acquisition baseline control, briefs ASN(RD&A) and DAB-level forums, and ensures SYSCOM technical authority integration [2, 52].

PM. Accountable for cost, schedule, performance; leads the IPT, maintains statutory certifications, and readies milestone documentation [52, 66].

Governance forums. Milestone Decision Reviews, the DAB, Configuration Steering Boards, Overarching IPTs, and Navy Program Executive Council reviews provide structured oversight and risk adjudication [52, 66, 67].

Note

Edge case: Rapid acquisition authorities (e.g., UCA) compress governance. Ensure delegation letters document any waived statutory certifications before you recommend skipping a DAB or ASN(RD&A) review [68].

4.1.5 Adaptive Acquisition Framework Pathways

The AAF can be visualized in Figure 4.3. The figure shows all six pathways with their purposes and general flow of the acquisition for each.

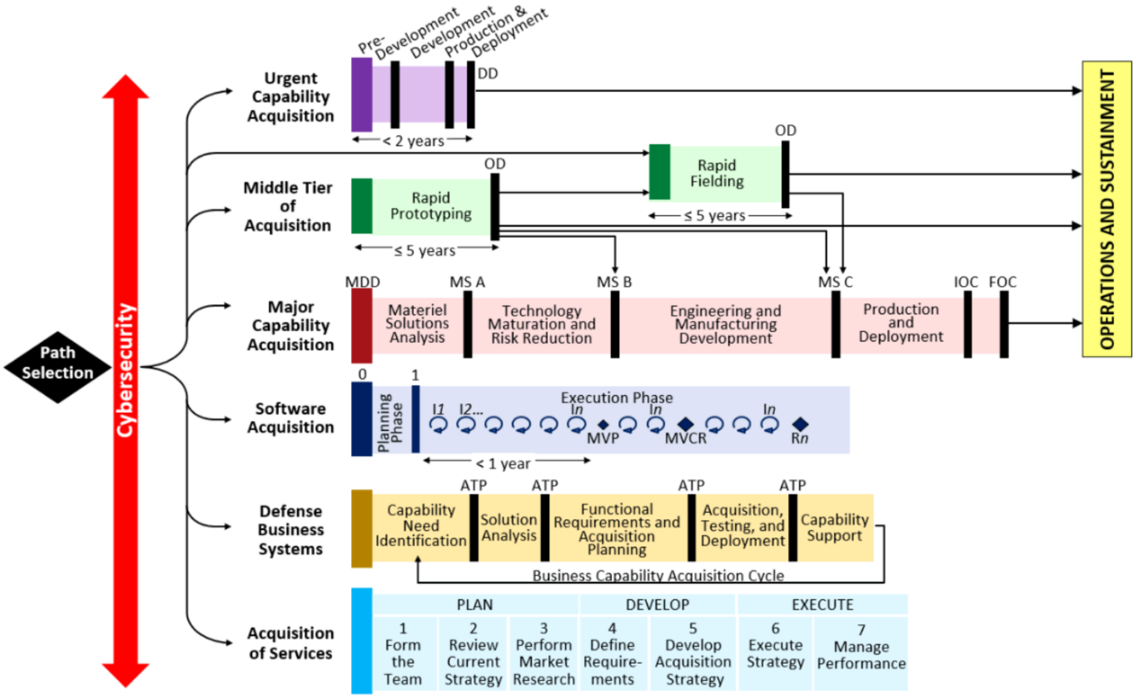


Figure 4.3. Adaptive Acquisition Pathways. Source: Adaptive Acquisition Framework, *2025 [54].*

The MCA is the main acquisition pathway that EDOs will operate in. As shown in Figure 4.4, it can be broken down into five phases, four major decision points and three milestones.

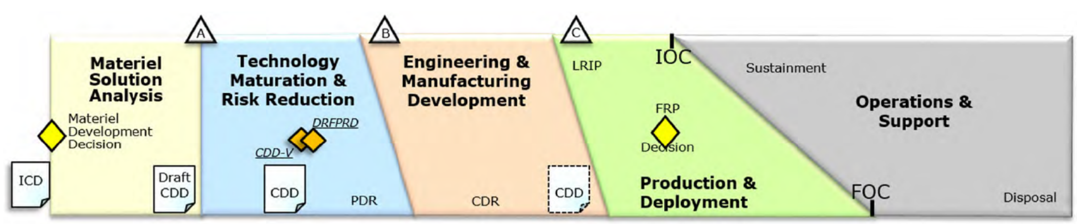


Figure 4.4. Major Capability Acquisition with phases, milestones, and decisions points. Source: 3.3.1. Acquisition Policy and Players, *2025 [66]*

Pathway Playbooks

Major Capability Acquisition.

Entry. Default when a validated ICD, PPBE affordability certification, and Title 10 Major Defense Acquisition Program (MDAP) thresholds are triggered; the DAE (USW(A&S)) or delegated MDA authorizes the Material Development Decision [52, 68, 70]. *Major events.* Material Solution Analysis, TMRR, EMD, Production & Deployment, and Operations

& Support framed by MS A/B/C, Production Readiness Reviews, Configuration Steering Boards, and Full-Rate Production Decisions [52]. *Outputs.* Acquisition Strategy, SEP, Life-Cycle Sustainment Plan (LCSP), TEMP, SAR/DAES, and when applicable Live Fire and survivability certifications [52, 71, 72]. *What happens.* The Program Manager leads integrated product teams, controls the Acquisition Program Baseline, and coordinates statutory certifications before each milestone package is staffed through the Navy Program Executive Council and the DAB [52]. *Pros.* Maximum statutory confidence, predictable decision forums, and the ability to tailor documentation while retaining congressional transparency [52]. *Cons.* Heavier documentation, longer lead times, and concurrency risks if requirements or funding discipline erodes between milestones [52].

Middle Tier of Acquisition.

Entry. Warfighter sponsor certifies the capability can be prototyped or fielded within 5 years, and the MDA approves a rapid prototyping or rapid fielding strategy in lieu of JCIDS milestone paperwork [73]. *Major events.* Competitive prototyping sprints, operational demonstrations, and limited fielding increments documented in the Middle Tier of Acquisition (MTA) Strategy and semiannual execution reports to Under Secretary of War for Research and Engineering (USW(R&E))/USW(A&S) [73]. *Outputs.* MTA Acquisition Strategy, Test Strategy (tailored TEMP), transition plan to either MCA or sustainment funding, and congressional notifications when cost or schedule deviates [73]. *What happens.* Programs lean on OTs, digital engineering, and combined Developmental Test (DT)/Operational Test (OT) events; logistics and cybersecurity artifacts are right-sized but must be ready before transition to production [73]. *Pros.* Compresses documentation, accelerates learning, and protects prototyping budgets from the full milestone stack. *Cons.* Sustainment planning and PPBE wedges still have to be established before transition; poor exit planning strands prototypes without a production home [73].

Software Acquisition Pathway.

Entry. Capability owner commits to iterative Development, Security, and Operations (DevSecOps) delivery, establishes a software factory or enterprise pipeline, and designates a product manager vice a traditional milestone PM [74]. *Major events.* Plan, Execute, and Deliver phases with quarterly (or faster) capability drops, authority-to-operate reciprocity reviews, and portfolio synchronization reviews in lieu of MS A/B/C [74]. *Outputs.* Software Acquisition Strategy, Product Roadmap, backlog burn-down metrics, continuous test/accreditation evidence, and sustainment metrics for cloud hosting costs [74]. *What happens.* The government product owner jointly manages the pipeline with contractors, employs automated testing, and updates requirements via configuration-controlled roadmaps instead of static CDDs [74]. *Pros.* Continuous user feedback, cyber resiliency baked into each sprint, and the ability to budget in capability releases instead of monolithic blocks [74]. *Cons.* Requires mature pipelines, disciplined product management, and close PPBE coordination to realign color-of-money profiles from hardware-centric norms [74].

Urgent Capability Acquisition.

Entry. A validated Joint Urgent Operational Need (JUON)/Joint Emergent Operational Need (JEON)/Urgent Operational Need (UON) identifies a combatant-commander risk that cannot wait two years; ASN(RD&A) or USW(A&S) approves entry with an affordability determination and sustainment exit criteria [75, 76]. *Major events.* Rapid assessment within days, solution selection inside 30 days, fielding no later than 24 months, and close-out/transition reviews that assign residual sustainment ownership [76]. *Outputs.* Urgent Capability Acquisition Strategy, tailored test/safety releases, congressional notification packages, and a signed disposition plan for residual hardware [76]. *What happens.* The Rapid Acquisition Cell coordinates funding reprogramming, waivers, and senior integration group decisions while the program office executes minimal-but-sufficient contracting and test [76]. *Pros.* Weeks-to-months fielding, ability to leverage existing production lines or Commercial Off-the-Shelf (COTS) solutions. *Cons.* Limited longevity (normally five years or less), intense oversight on waivers, and explicit requirement to transition to another pathway or divest [76].

Defense Business Systems.

Entry. A portfolio exceeds the statutory Defense Business System (DBS) cost thresholds or is designated critical; requires Business Process Reengineering certification, a Functional Sponsor, and alignment to the Business Enterprise Architecture before funds may be released [77]. *Major events.* Capability Needs Statement approval, Functional Requirements Board adjudication, incremental capability releases, and annual Investment Review Board certification to Congress [77]. *Outputs.* Business Case, Functional Strategies, Capability Support Plan, data standards, and interoperability determinations to satisfy Net-Ready Key Performance Parameter (KPP) expectations [77]. *What happens.* Functional leads co-chair the IPT with the Program Manager, enforcing process improvements before automating legacy workflows [77]. *Pros.* Forces portfolio-level prioritization, ties funding to measurable process change. *Cons.* Extra governance (Investment Review Board (IRB), Defense Business Council) and Clinger-Cohen certifications can slow schedules if stakeholders are not aligned early [77].

Defense Acquisition of Services.

Entry. Non-IT services exceeding simplified acquisition thresholds, or any effort designated special interest, must use DoDI 5000.74 governance and appoint a Services Category Manager before solicitation [78]. *Major events.* Service Requirement Review Board, Acquisition Strategy Panel, independent cost estimate for Category I/II services, and annual review of contractor performance metrics [78]. *Outputs.* Services Acquisition Plan, Quality Assurance Surveillance Plan, Performance Work Statement, and documented affordability/cost realism analyses [78]. *What happens.* Multifunctional teams (requirements owner, contracting, legal, comptroller) collaborate early, determine best-fit contract type, and tailor competitive ranges for recurring availabilities or modernization services [78]. *Pros.* Aligns recurring services with category management, encourages outcome-based metrics. *Cons.* Additional reviews for Category I/II services can add lead time if the requirement was scoped as a product effort originally [78].

Info

Pathways can be combined (e.g., prototype under Middle-Tier, transition to MCA, then adopt the Software Pathway for continuous upgrades). Boards expect you to articulate which statutory obligations remain even when tailoring [68].

4.1.6 Warfighting Acquisition System Imperatives (2025)

Purpose.

SECWAR redesignated the DAS as the Warfighting Acquisition System (WAS) on 7 Nov 2025 to treat acquisition as a warfighting function, make speed-to-field the primary metric, and push decisions to the accountable portfolio leaders [79].

Portfolio accountability.

Portfolio Acquisition Executives (PAEs) chair Capability Trade Councils, own integrated portfolio scorecards, and are tasked to deliver minimum viable combat capability in months—not years—while continuously iterating increments; they are expected to waive non-statutory documents, reallocate within portfolio controls, and document trades in decision memoranda [79, 80].

Mission engineering and experimentation.

The Mission Engineering and Integration Activity (MEIA) replaces lengthy requirements staffing with rapid kill-chain analysis, cross-Service experiments, and industry demos, so requirement adjustments are anchored in real data before the DAE or PAE commits resources [79, 81].

Requirements–budget coupling.

Key Operational Problems (KOPs) and the Requirements and Resourcing Alignment Board (RRAB) ensure every joint problem statement is paired with funding guidance (including Joint Acceleration Reserve (JAR) taps) inside the same PPBE cycle, eliminating the historical gap between validated requirements and appropriated dollars [81].

Implications for EDOs.

Expect Capability Trade Council taskers in lieu of Configuration Steering Boards, portfolio scorecards that track cycle time from need to Initial Operational Capability, and requests for mission-thread data packages to feed MEIA campaigns—ships, systems, and industrial-base briefs must now quantify time-based performance alongside cost/schedule [79, 80].

Info

Board cue: explain how you would tailor milestone documentation under the WAS memo while preserving statutory certifications (e.g., Clinger-Cohen, cybersecurity) and how you will capture trade rationales now that Capability Trade Councils replace legacy Configuration Steering Boards [79].

4.1.7 Urgent Operational Needs Governance

JUON.

Combatant commanders submit JUONs to the Joint Rapid Acquisition Cell when an unforeseen operational gap threatens lives or mission success within the current fight; the Joint Staff validates within days, assigns a lead Service, and requires fielding in ≤24 months with a plan to transition or terminate after the contingency [75].

JEON.

A JEON covers emerging threats with a slightly longer horizon (2–5 years) but still demands joint prioritization, Functional Capabilities Board staffing, and explicit affordability bounds before the Joint Requirements Oversight Council renders a decision [82].

UON.

Fleet, Type, and numbered Force Commanders elevate UONs per Office of the Chief of Naval Operations Instruction (OPNAVINST) 5000.42E; requests must include the operational vignette, Doctrine, Organization, Training, Materiel, Leadership and Education, Personnel, Facilities, and Policy (DOTMLPF-P) analysis, recommended sponsor, and incremental funding estimate before Gate 1 of the two-pass/seven-gate process [83].

Request content.

Every urgent package documents the risk of not acting, desired operational effect, timeframe to theater, sustainment/retrograde plans, cybersecurity implications, and classification handling; incomplete data delays Joint Staff or OPNAV acceptance [75, 83].

Acquisition handoff.

Validated packages flow to the UON Management Board and then to ASN(RD&A) for alignment with the Urgent Capability Acquisition (UCA) pathway, rapid contracting, and source selection; the MDA records any tailored documentation and must still close on lifecycle ownership before the temporary solution sunsets [76].

4.1.8 Security Cooperation and Capital Alignment (2025 Reforms)

Arms-transfer realignment.

SECWAR shifted the Defense Security Cooperation Agency (DSCA) and the Defense Technology Security Administration (DTSA) under USW(A&S) to unify Foreign Military Sales, Direct Commercial Sales, technology protection, and industrial-base production decisions under one accountable enterprise that can plan exportability features early and keep allied demand synced with U.S. production lines [84].

National Security Capital Forum.

Section 1092 of the FY 2025 NDAA and subsequent SECWAR direction established the National Security Capital Forum (NSCF), chaired by the Office of Strategic Capital, to convene federal and non-federal financiers, monitor capital flows that affect defense production, and flag transactions that threaten supply-chain resilience [85].

EDO relevance.

Program briefs now need to show how exportability, co-production, and investment incentives are built into acquisition strategies because DSCA/DTSA decisions and NSCF findings will increasingly shape schedule risk (supplier solvency, capital availability) and board questions on allied burden-sharing [84, 85].

4.1.9 Acquisition Categories and Delegated Decision Authority

USW(A&S) serves as the DAE; SECNAV designates ASN(RD&A) as both the Navy SAE and CAE, while SECNAVINST 5000.2G and the Warfighting Acquisition Strategy empower PAEs to manage portfolios and request tailored delegation [3, 67, 80].

ACAT I (MDAP).

Programs that meet the MDAP thresholds (≥ $525M RDT&E or ≥ $3.6B procurement in FY 2020 constant dollars) default to DAE oversight; ACAT ID programs (e.g., *Columbia*-class SSBN) keep USW(A&S) as MDA, whereas ACAT IC efforts (e.g., DDG(X) combat systems refresh) are normally delegated to ASN(RD&A) with reporting to Congress via SAR/DAES [52, 66, 70].

ACAT IA.

Major Automated Information Systems exceeding $360 M life-cycle cost or designated by Under Secretary of War (Comptroller) (USW(C))/CAE; the Navy SAE is the default MDA and may assign PEO Digital or the relevant PAE to execute under the Software or Business Systems pathways (examples: Navy ERP, Maritime Maintenance IT portfolio) [68, 77].

ACAT II.

Programs below MDAP thresholds but above $200 M RDT&E or $920 M procurement (FY 2020 constant dollars) are typically delegated from ASN(RD&A) to the cognizant PAE/PEO; examples include shipboard radars, launch-and-recovery systems, or complex weapon upgrades where the PEO retains the MDA [52, 66].

ACAT III.

Efforts below ACAT II thresholds (but still acquisition programs) normally have the PAE, SYSCOM commander, or PM acting as MDA; documentation is tailored, yet APBs, TEMPs, and configuration control still apply [3, 52].

ACAT IV (DON unique).

SECNAVINST 5000.2G divides ACAT IV into IVT (test) and IVM (monitor) for sustainment, modernization, and In-Service Engineering efforts; NAVSEA/NAVAIR/NAVWAR commanders often serve as MDA, delegating narrowly scoped efforts to PMs while ensuring technical authority concurrence [3].

Delegation chain. The DAE retains ACAT ID decisions unless a signed memo delegates to ASN(RD&A); the Navy SAE typically keeps ACAT IC/IA programs and can further delegate ACAT II/III/IV to PAEs. SECNAVINST 5000.2G requires each delegation letter to specify statutory certifications retained by ASN(RD&A), while the Warfighting Acquisition Strategy directs PAEs to chair Capability Trade Councils and reassign funds across their portfolios when trades stay within cost/schedule thresholds [3, 80].

4.1.10 Project Overmatch Status

Project Overmatch is the Navy's operational architecture initiative to link platforms, weapons, sensors, and data fabrics so that Fleet Commanders can execute CJADC2 and Joint Fires at decision speed [22, 23]. VADM Seiko Okano (EDO) serves as the direct reporting program manager (DRPM) and steering committee chair while serving as PMILDEP; she sponsors Five Eyes data-sharing arrangements, Maven Smart Systems overlays, and Open DAGIR integrations that move capability from concept to fielded software in weeks [22, 23].

The DRPM title matters: a DRPM reports directly to ASN(RD&A), can tailor milestone documentation, and retains signature authority for integration decisions that cut across multiple PEOs. Under the Oct 2024 Five Eyes Project Arrangement, Capt. Remil Capili (PM), Shayna Bond (international deputy), and David Flowers (international lead) are embedding Cooperative Project Personnel from Australia, Canada, New Zealand, and the United Kingdom in San Diego, building a coalition lab, and preparing RIMPAC 26 demonstrations following Trident Warrior 24 and Talisman Sabre 25 events [22]. Capili, as PM, drives day-to-day execution and reports to Okano; Okano retains DRPM/steering-chair authority even after moving from NAVWAR to PMILDEP, preserving direct reporting to ASN(RD&A) and cross-PEO tailoring authority [22].

As PMILDEP, Okano now chairs Program Decision Meetings when delegated, synchronizes PAEs across portfolios, and uses Open DAGIR/Maven to inform requirements trades and PPBE issue papers; her FY26–27 focus is expanding the coalition network, institutionalizing rapid AI/ML onboarding, and ensuring Overmatch digital threads stay visible in milestone packages so new ship, submarine, and aircraft programs inherit the data standards by default [3, 22, 23].

4.1.11 Navy Overlays and Touchpoints

SECNAVINST 5000.2G. Implements ASN(RD&A) Program Decision Meetings, requires risk-based tailoring plans, and codifies DON-level documentation (Acquisition Plan, LCSP, Plan of Action and Milestones (POA&M)) [3].

SECNAVINST 5400.15D. Aligns PEOs and SYSCOMs, clarifies technical authority (NAVSEA 05, NAVWAR 5.0) integration, and designates Deputy ASN(RD&A) portfolio responsibilities [2].

Resource Sponsor coordination. OPNAV N9, N4, or equivalent sponsor must endorse requirements and PPBE positions before ASN(RD&A) decisions [66].

Note

When SECNAV policy conflicts with older DoW memoranda, defer to the most current directive from the higher authority unless ASN(RD&A) grants written tailoring. Document the rationale in the Acquisition Strategy.

4.2 Milestones and Phases

4.2.1 Defense Acquisition System Overview

The DAS is the governance framework that aligns requirements, resourcing, and acquisition execution to deliver warfighter capability. It is policy-driven by DoDD 5000.01 and process-enabled through DoDI 5000.02, using the AAF to tailor pathways while preserving statutory oversight and Navy-unique controls [67, 68].

4.2.2 Adaptive Acquisition Framework Pathways (Board Snapshot)

MCA. Default for complex weapon systems; uses phased milestones with mandatory statutory certifications; emphasizes affordability trades, test rigor, and Navy two-pass / seven-gate touchpoints [52].

MTA. Rapid prototyping or rapid fielding to deliver an operational capability within five years; minimizes documentation, but demands robust transition planning to MCA or sustainment funding [73].

UCA. Accelerated path to field urgent capability within 2 years or provide an interim solution in less than 5; leverages tailored test, streamlined acquisition decision memoranda, and dedicated senior oversight [76].

Software Acquisition Pathway. Iterative, DevSecOps-focused delivery of software increments with continuous integration, enterprise cloud alignment, and recurring user feedback cadences [74].

Acquisition of Services Pathway. Structures service requirements, performance metrics, and contracting strategies to assure mission outcomes and oversight thresholds (e.g., Multi-Functional IPT requirements review) [78].

Defense Business Systems Pathway. Rationalizes business processes, data standards, and Business Capability Acquisition Cycle (BCAC) to unlock enterprise interoperability before investing in solutions [77].

Note

Pathways are not exclusive—an EDO may begin in MTA for prototyping, then transition to MCA for production once requirements and sustainment plans stabilize. Programs can also leverage UCA for an initial bridge while planning a transition to MCA.

4.2.3 Baseline Discipline: Acquisition Program Baselines, Deviations, and Nunn–McCurdy

APB. The cost-schedule-performance contract between the MDA, resource sponsor, and PEO; defines objective and threshold values that bound trade space [52].

Program Deviations. Triggered when performance, schedule, or cost thresholds breach; the Program Manager must notify the MDA within 30 days and submit a corrective action plan within 60 [52].

Nunn–McCurdy Act. Statutory reporting when Program Acquisition Unit Cost (PAUC) or Average Procurement Unit Cost (APUC) breaches significant (≥15% current / 30% original) or critical (≥25% current / 50% original) thresholds; critical breaches require certification to Congress within 60 days or program termination [86].

4.2.4 JCIDS Flow into the Defense Acquisition System

ICD. Supports the MDD with validated capability needs and initial attributes [82].

CDD. Provides KPPs and Key System Attributes (KSAs) to anchor Milestone B and the Technical Requirements Baseline [82].

Capability Production Document (CPD). Updates KPP/KSA values for production decisions and sustainment planning at Milestone C [82].

Note

Maintain requirements traceability matrices linking KPP thresholds to test plans and APB values for board discussions.

4.2.5 KPPs v.s. KSAs (Know the Difference)

KPP. A performance attribute of a system critical to mission success. Each KPP has a threshold (minimally acceptable) and objective (desired) value. Failure to meet a *threshold* normally triggers program reassessment or revalidation of the requirement. *Why it matters:* Drives design trades, tempers requirement creep, and is reportable in baselines and decision reviews [82].

KSA. Important performance attribute not critical enough to be a KPP. KSAs are managed and traded to balance cost/schedule/performance without jeopardizing mission effectiveness. *Why it matters:* Provides disciplined trade-space for the Program Manager and requirements sponsor [82].

Difference. KPPs are mission-critical and tightly controlled; KSAs are important but offer more flexibility. Both appear in the CDD/CPD with threshold/objective values, but KPPs carry higher visibility and risk if breached.

4.2.6 Acquisition Strategy Essentials

Purpose. Documents tailored acquisition approach, contract strategy, intellectual property posture, and affordability constraints for MDA approval [52].

Key Contents. Pathway selection, phasing plan, incentive structure, test and evaluation strategy, risk burn-down, sustainment concept, cybersecurity, international cooperation, and business case for software/data management [52, 67].

Approval Event. MDA concurrence at Milestone A (initial) with updates prior to Milestone B and significant re-baselines; Navy 2-pass/7-gate integrates resource validation before signature [83].

4.2.7 International Acquisition Objectives

Foreign Military Sales (Foreign Military Sales (FMS)). Strengthen partner capability while leveraging U.S. economies of scale; requires disclosure review and Letter of Offer and Acceptance alignment [52].

Direct Commercial Sales (Direct Commercial Sales (DCS)). Contractor-led exports; Program Office ensures configuration control and technology-security compliance [52].

Cooperative Development. Shared investment with allies (e.g., Memoranda of Understanding) to reduce life-cycle cost and enhance interoperability [52].

Armaments Cooperation / International Programs Security. Integrates disclosure, export licensing, and technology protection plans into the Acquisition Strategy [52].

4.2.8 Navy Two-Pass / Seven-Gate Review

Pass 1 – Capability. Led by OPNAV N9 to mature the requirement: Gate 1 validates the capability gap, Gate 2 refines solutions and mission analysis, Gate 3 assesses affordability and alignment with Navy enterprises, and Gate 4 authorizes program initiation (supports MDD) [83].

Pass 2 – Resourcing. Led by OPNAV N8 to align budgeting: Gate 5 verifies contracting/test readiness before Milestone B (often synchronized with the Development RFP Release Decision), Gate 6 confirms production and sustainment resources prior to Milestone C, and Gate 7 evaluates sustainment performance and modernization plans [83].

Note

For board prep, highlight where SECNAV Gate Reviews or Defense Acquisition Board touchpoints overlay Navy gates to demonstrate cross-tier synchronization.

4.2.9 Technology Readiness Levels

Table XIII summarizes Technology Readiness Level (TRL) definitions with representative naval examples.

Table XIII: Technology Readiness Levels with representative naval examples.

TRL	Definition	Example (EDO context)
TRL 1	Basic principles observed	Novel corrosion-resistant alloy proven in laboratory
TRL 2	Technology concept formulated	Hypothesis for integrating alloy into hull structures
TRL 3	Analytical and experimental proof-of-concept	Coupon tests validating stress performance in relevant environment
TRL 4	Component validation in laboratory	Scale panel fabrication and testing under controlled loads
TRL 5	Component validation in relevant environment	Module integrated into land-based test site with salt-fog exposure
TRL 6	System/subsystem prototype in relevant environment	Full-scale hull section installed on test article or surrogate vessel
TRL 7	System prototype in operational environment	Prototype hull segment installed on operational ship during limited sea trials
TRL 8	Actual system completed and qualified	Final production drawings released, inspections passed for lead ship
TRL 9	Actual system proven in mission operations	Sustained operational deployment meeting performance metrics

Source: *Technology Readiness Assessment Guide*, 2020 [87]

Hint: Match TRLs to TMRR exit criteria: critical technologies should be ≥ TRL 6 before Milestone B.

Note

Operational environment testing typically begins at **TRL 7** (system prototype in an operational environment). **TRL 8** means the actual system is completed and qualified (e.g., qualification/acceptance testing, certifications). **TRL 9** means the actual system is *proven in mission operations* (e.g., successful fleet employment meeting mission metrics).

4.2.10 T&E Organizations and Roles

OSD Director, Operational Test and Evaluation (DOT&E). Independent oversight of operational test adequacy, approves/concurs in TEMP for covered programs, and reports to Congress on operational suitability and effectiveness.

Service Operational Test Agencies. Navy *Commander, Operational Test and Evaluation Force (COMOPTEVFOR) (COTF)*, *Air Force Operational Test and Evaluation Center (AFOTEC)*, *U.S. Army Test and Evaluation Command (ATEC)*, *Marine Corps Operational Test and Evaluation Activity (MCOTEA)*. Plan/execute IOT&E and report independent findings to the Service and DOT&E.

DT. Program Offices with SYSCOM Warfare Centers (e.g., Naval Air Warfare Center Aircraft Division (NAWCAD), Naval Air Warfare Center Weapons Division (NAWCWD), NSWC, NUWC) plan and run DT to verify requirements, qualify designs, and feed Operational Test Readiness Review (OTRR).

Interoperability/Joint. *Joint Interoperability Test Command (JITC)* assesses joint interoperability/Net-Ready KPP compliance where applicable.

Ranges/Land-based sites. Service ranges and land-based engineering/test sites provide representative and operational environments to support DT/OT.

Artifacts. TEMP aligns DT/OT scope and phases; OTRR certifies readiness for IOT&E; SAR / DAES capture results for governance [52].

4.2.11 Major Capability Acquisition Lifecycle Overview

MSA. Shape the trade space and select a preferred materiel solution; conclude with an AoA recommendation and technology maturation plan [52].

TMRR. Mature technologies, refine requirements, and reduce integration risk to enter EMD; manage competitive prototyping where feasible [52].

EMD. Complete detailed design, verify system performance, and prepare for production and deployment decisions [52].

P&D. Produce capability, conduct IOT&E, field initial units, and decide on Full-Rate Production (FRP) [52].

O&S. Sustain readiness, execute modernization, and plan for disposal while feeding affordability back into PPBE cycles [52].

Figure 4.5 Shows the major phases, review and decision points within MCAs. The following are the reviews that occur during the phases:

Alternative Systems Review (ASR). Validates the preferred materiel solution and technology maturation approach emerging from the AoA; inputs include threat assessments, cost-schedule trades, draft system concepts, and updated Technology Development Strategy / SEP outlines, and the primary output is an ASR report that authorizes drafting authoritative system-level requirements for the upcoming System Requirements Review (SRR) [52, 69].

SRR. Confirms that system-level performance, interface, cyber, and logistics requirements trace to the validated ICD / CDD and are testable; requires a draft System Specification, mission threads, interface control documentation, and technical performance measures, and produces a baselined system requirements set and updated verification strategy feeding System Functional Review (SFR) planning [52, 69].

SFR. Demonstrates that the functional architecture and allocated requirements can satisfy the mission needs within cost/schedule constraints; leverages functional allocation matrices, preliminary hazard analyses, and draft reliability growth plans, and yields an approved functional baseline plus a risk/opportunity burn-down plan ahead of Preliminary Design Review (PDR) [52, 69].

PDR. Assesses whether the allocated baseline and preliminary design (drawings, models, manufacturability assessments, support concepts) are mature enough to enter detailed design; the review outputs include an approved allocated baseline, configuration-controlled action list, and direction for completing qualification hardware prior to Critical Design Review (CDR) [52, 69].

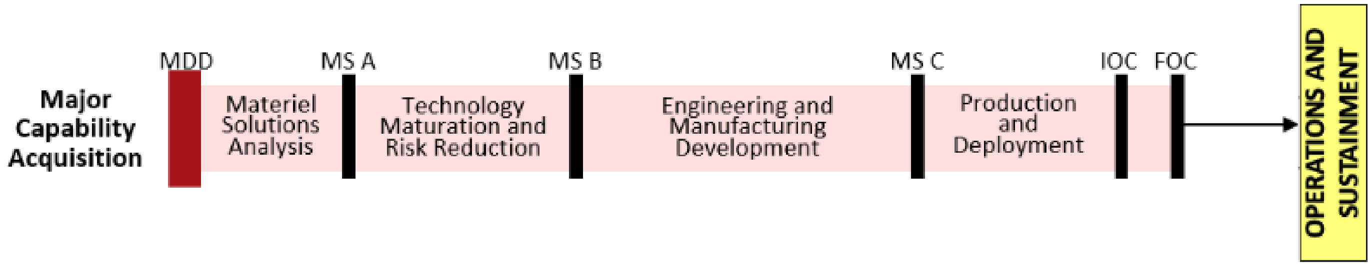
CDR. Confirms the detailed design meets all requirements with acceptable technical risk; inputs include product baseline technical data packages, qualification/analysis results, producibility assessments, and cybersecurity artifacts, and the decision produces an approved product baseline plus direction to release build-to packages and procure long-lead items [52, 69].

Test Readiness Review (TRR). Evaluates readiness to execute a specific test event by reviewing test articles, instrumentation, facilities, procedures, safety releases, and deficiency dispositions; a successful TRR outputs an authorization to start test with documented entrance/exit criteria and risk mitigations [52, 69].

System Verification Review (SVR). Verifies the integrated system meets the product baseline and is ready for production decision reviews; it uses completed verification cross-reference matrices, configuration audits, and qualification / DT results to produce a verified system baseline, deficiency adjudication list, and recommendation for Full-Rate Production Decision Review (FRPDR) preparation [52, 69].

OTRR. Certifies that production-representative assets, training, logistics products, safety artifacts, and test ranges are ready to support initial operational test; the review outputs the operational test readiness certification and any restrictions or corrective actions before DOT&E concurrence [52, 69].

Production Readiness Review (PRR). Determines whether manufacturing processes, supply chain, quality systems, and sustainment enablers can support LRIP / FRP quantities; requires control plans, process capability data, staffing/budget burn rates, and yields a decision to proceed into production along with corrective actions for any shortfalls [52, 69].



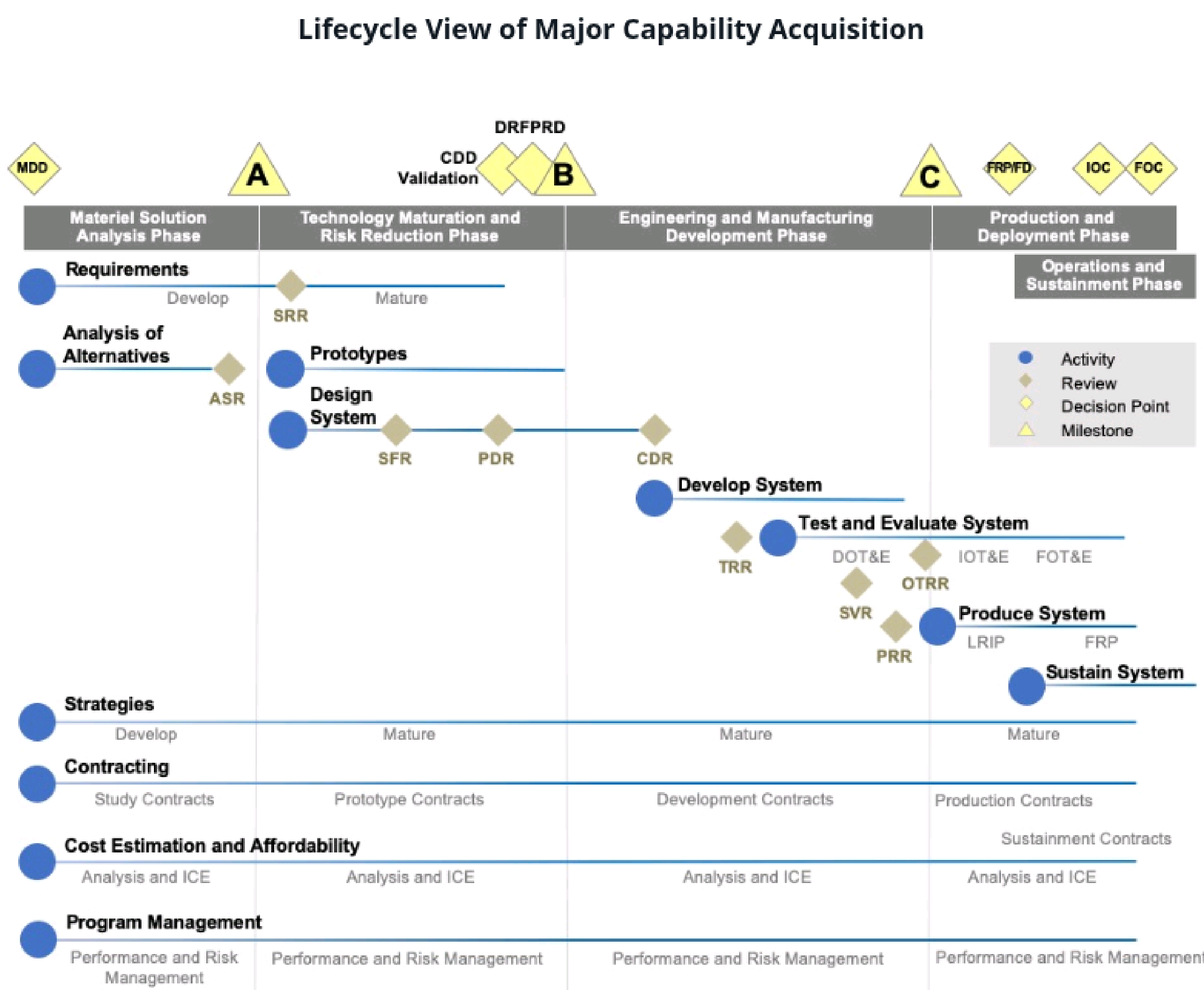


Figure 4.5. Lifecycle view for a MCA showing all functions and activities for each phase.Source: DoDI 5000.85, *2020 [52]*

4.2.12 Materiel Development Decision

Inputs. Validated ICD, initial affordability analysis, preliminary acquisition strategy, and resource sponsor endorsement [68].

Decision. MDA authorizes entry into MSA (or an alternate pathway) and designates the lead PEO/PM [52].

Outputs. Acquisition Decision Memorandum (ADM), competitive AoA guidance, and direction to form the IPT structure [52].

4.2.13 Materiel Solution Analysis Phase Essentials

Purpose. Analyze alternatives, define the preferred materiel solution, and scope technology maturity and risk reduction tasks [52].

Key Activities. Conduct AoA, establish initial system performance metrics, develop early test strategies, and plan technology maturation for critical subsystems [52].

Entrance Criteria. Approved MDD, resourced AoA study guidance, and draft acquisition strategy [68].

Exit Criteria. Approved TMRR acquisition strategy update, technology risk assessments (i.e., TRL ≥ 3–4), and draft CDD inputs for Joint staffing [52].

4.2.14 Milestone A Focus Areas

Purpose. Authorize entry into TMRR and approve the initial Acquisition Strategy and APB [52].

Board Topics. AoA results, technology maturation plans, affordability caps, cost estimates, and early test strategies [52].

Outputs. ADM, approved initial APB, direction for competitive prototyping, and signed Technology Development Strategy if required [52].

4.2.15 Technology Maturation and Risk Reduction Phase Essentials

Purpose. Reduce integration risk, mature the Technical Baseline, and prepare for Engineering & Manufacturing Development source selection [52].

Key Activities. Competitive prototyping (if directed), systems engineering trade studies, reliability growth, cyber-resilience planning, and Capabilities Development Document Validation (CDD-V) [52].

Entrance Criteria. Approved Milestone A ADM, funded risk-reduction plan, and established IMS aligned to Navy Gate 5 timeline [52, 83].

Exit Criteria. Technology Readiness Assessments showing TRL ≥ 6, draft Milestone B documentation (e.g., SEP, TEMP), and configured product baseline ready for the Development Request for Proposal Release Decision (DRFPRD) [52].

4.2.16 Milestone B Focus Areas

Purpose. Authorize entry into EMD and establish the program as a major defense acquisition program with formal reporting [52].

Board Topics. Affordability caps, independent cost estimates, TEMP approval, LRIP planning, and production-representative prototype evidence [52].

Outputs. Updated APB, approved Acquisition Strategy revision, and contractual authority to release EMD RFPs [52].

4.2.17 Engineering and Manufacturing Development Phase Essentials

Purpose. Complete detailed design, build and test engineering development models, and validate manufacturing processes [52].

Key Activities. CDR, system verification, developmental test events, cybersecurity accreditations, and sustainment baseline development [52].

Entrance Criteria. Approved Milestone B ADM with resourced contracts, finalized technical baseline, and product-support planning aligned to Navy Gate 6 [52].

Exit Criteria. Successful IOT&E readiness, manufacturing readiness ≥ TRL 8, and validated sustainment strategies supporting Milestone C [52].

4.2.18 Milestone C Focus Areas

Purpose. Authorize production and deployment following demonstration of operational effectiveness and suitability [52].

Board Topics. IOT&E results, sustainment and supply-chain readiness, affordability assessments, and updated manpower/training plans [52].

Outputs. FRPDR plan, updated APB, and authority to obligate procurement funding for production lots [52].

4.2.19 Production and Deployment Phase Essentials

Purpose. Produce and deliver the system while validating operational performance and support concepts [52].

Key Activities. LRIP execution, IOT&E, Configuration Steering Board (CSB) governance, and ramp to Full-Rate Production [52].

Exit Criteria. FRP approval, attainment of Initial Operational Capability (IOC), and sustainment plans ready for fleet release [52].

4.2.20 Operations and Support Phase Essentials

Purpose. Sustain mission effectiveness, execute modernization, and manage total ownership cost until disposal [52].

Key Activities. Performance-Based Logistics execution, Depot maintenance planning, Post-Deployment reviews, and CSBs for major updates [52].

Milestones. Full Operational Capability (FOC), disposal decision documentation, and updates to capability requirements feeding future planning cycles [52].

5 Requirements

5.1 Joint Capabilities Integration & Development System

5.1.1 Requirements Demand Versus Acquisition Supply

JCIDS. Establishes the joint demand signal by translating strategic guidance into validated capability requirements, risk statements, and DOTMLPF-P change recommendations that scope non-materiel and materiel options [88, 89].

DAS. Converts validated requirements into executable acquisition strategies, cost/schedule baselines, and lifecycle sustainment plans under Department of War Directive (DoDD) 5000.01 and Department of War Instruction (DoDI) 5000.85 [52, 67].

> **Note**
>
> Board cue: be ready to explain where a program sits in both systems; Gate 2 packages are dead on arrival if the JCIDS paperwork is stale even when the DAS strategy is current [89].

5.1.2 Capability Requirements and the Capability-Based Assessment Cadence

Capability requirements are mission-task statements (task, condition, standard) that are solution-agnostic yet measurable enough to drive trades, and Capability-Based Assessments (CBAs) provide the disciplined analysis to justify them [88]. Each CBA follows repeatable phases:

Study initiation. The sponsoring component issues the study notice, nominates a lead Functional Capabilities Board (FCB), scopes timelines, and aligns analytical support and governance expectations [88, Annex B].

Operational context & mission threads. Analysts decompose national / Combatant Command (CCMD) guidance into joint concepts, mission threads, and integrated architectures to keep the study "concept-based and threat-informed" [88, Encl. A].

Gap and risk assessment. Existing and programmed capabilities are mapped against required mission effects to quantify gaps, assess risk, and prioritize by warfighting impact and threat timelines [88].

Solution analysis. DOTMLPF-P are explored first; only when risk remains unacceptable are materiel approaches carried forward, together with preliminary affordability bounds and schedule realism [88].

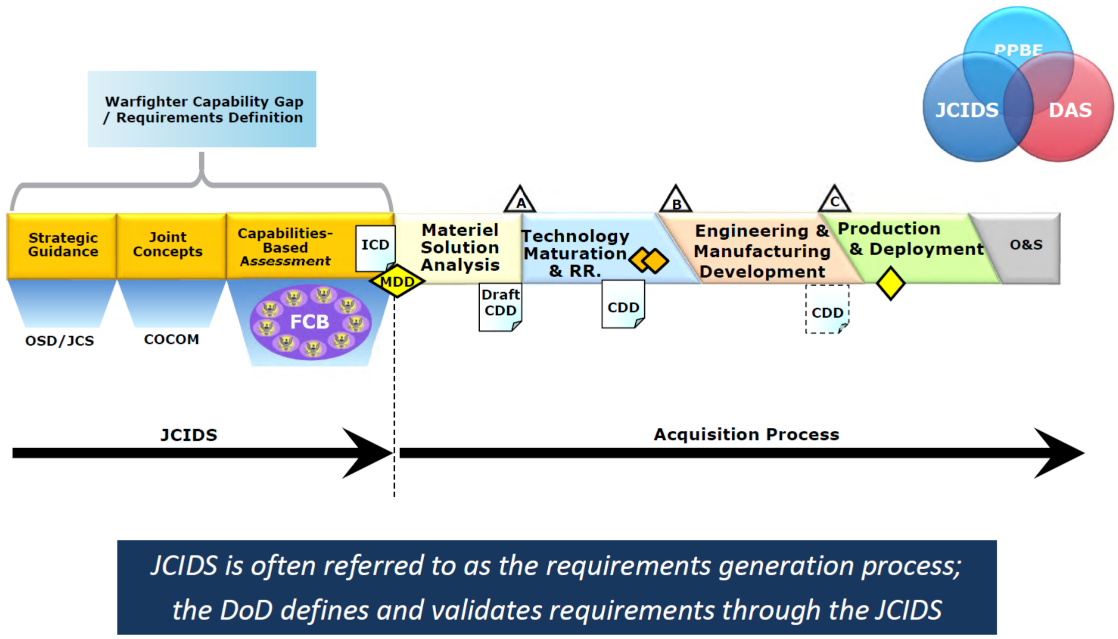


Figure 5.1. *Relationship between the Joint Capabilities Integration and Development System and the Defense Acquisition System. Source:* 3.3.3. Joint Capabilities Integration and Development System, *2025 [89]*

Documentation & validation path. The resulting evidence package (mission context, gaps, DOTMLPF-P recommendations, risk, and proposed documents) feeds the Gatekeeper, FCB, and validation authority decision on which JCIDS artifact (ICD, DOTMLPF-P Change Recommendation (DCR), update) to staff [88, Encl. A].

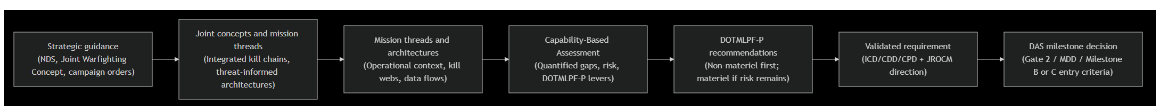


Figure 5.2. *Concept-to-requirement flow that must stay synchronized with acquisition execution. Source:* 3.3.3. Joint Capabilities Integration and Development System, *2025 [89].*

5.1.3 Strategy-to-Requirement Synchronization

Joint demand starts with strategic direction (National Defense Strategy, Joint Warfighting Concept, campaign orders) and is progressively tailored until a validated requirement enters the DAS [88, 89]. Key touchpoints:

Strategic guidance. SECWAR and CJCS translate national objectives into force design problems and assign lead sponsors; the Joint Staff J-7 houses concept development and readiness assessments that set the analytical agenda [88].

Joint concepts & mission threads. Mission engineers decompose guidance into mission areas, integrated kill chains, and threat-informed architectures so that requirements stay "concept-based and threat-informed" before solutioneering [88, Encl. A].

CBAs. Sponsors conduct the structured CBA described above to document measurable capability requirements, prioritize risks, and recommend DOTMLPF-P levers with quantitative evidence [88].

Gatekeeping & staffing. The Joint Staff Gatekeeper assigns a Joint Staffing Designator (JSD), routes packages through Knowledge Management and Decision Support (KM/DS), and tasks the appropriate FCB for certification prep; timelines and suspense assignments live inside KM/DS so Navy leads can monitor in real time [88].

Validation. Depending on the JSD, the JROC, Joint Capabilities Board (JCB), or delegated Service/CCMD authority validates requirements, issues memoranda Joint Requirements Oversight Council Memorandums (JROCMs), and records mandatory actions (e.g., certifications, follow-on DCRs) that acquisition teams must honor [88, 90].

Acquisition handoff. Validated artifacts, with their JROCM direction and Capability Trade Council (CTC) or CSB trade expectations, become entry criteria for Gate 2, the MDD, and milestone reviews, ensuring the PAE and PEO communities inherit coherent demand signals [89].

5.1.4 Major Capability Acquisition Requirement Set

ICD. Captures the operational context, quantified gaps, and risk that justify materiel and non-materiel approaches; it is required to support the Materiel Development Decision and is deliberately solution-agnostic [88, Appendix B-A].

CDD. Establishes system-level performance attributes (KPPs, KSAs, Additional Performance Attributes (APAs)) that anchor TMRR planning and must be validated before the Development RFP release and Milestone B; it becomes the authoritative source for the Technical Requirements Baseline [52, 88].

CDD updates. Refine thresholds/objectives when an EMD phase spans multiple lots or when KPPs trades are needed before Milestone C; updates keep production-representative values authoritative without re-opening earlier portions of the document [88].

CPD. Translates the validated CDD into production-ready parameters, sustainment metrics, and deployment considerations for Milestone C and Full-Rate Production/Fielding decisions [52, 88].

5.1.5 How KPPs Stay Integrated with DAS

Validated KPPs define the trade space that Milestone Decision Authorities use to approve baselines, certifications, and contracts. DoDI 5000.85 requires KPP ownership to be traceable from the CDD into the System Requirements Review, Developmental Test plans, and Acquisition Program Baseline; breaches trigger Gatekeeper tripwire reviews and Configuration Steering Boards [52, 88]. Navy PMs must therefore align CDD/CPD updates with SEA 05 warrant packages and ASN(RD&A) Gate 6 readiness to keep statutory certifications (Net-Ready (NR)-KPP, cybersecurity, safety) synchronized [83, 89].

5.1.6 Prioritizing Materiel and Non-Materiel Solutions

JCIDS policy directs sponsors to work through every DOTMLPF-P lever before defaulting to expensive materiel approaches, both to reduce risk and to speed delivery [88]. Practical applications include:

Doctrine. Update joint/combined warfighting concepts or tactics to mitigate gaps quickly (e.g., change Anti-Submarine Warfare (ASW) Concept of Operations (CONOPS) before buying sensors).

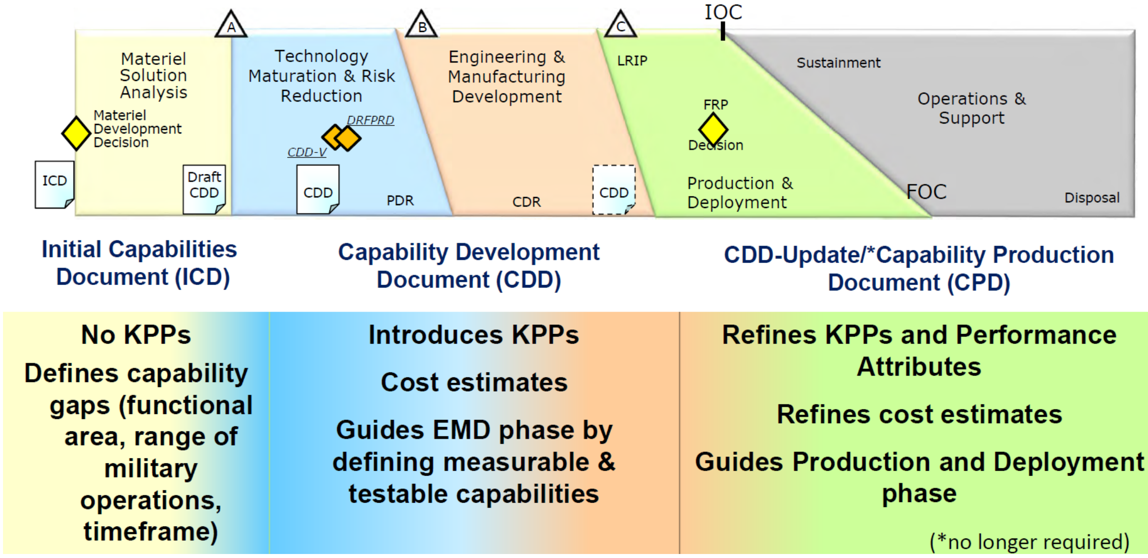


Figure 5.3. *Placement of Joint Capabilities Integration and Development System artifacts inside the phased Major Capability Acquisition pathway. Source:* 3.3.3. Joint Capabilities Integration and Development System, *2025 [89].*

Organization. Re-align Fleet/NAVWAR command relationships or mission tailoring to close gaps without new hardware.

Training. Modify Fleet Readiness Training Plan events or simulator syllabi so crews can exploit existing systems better.

Materiel. Only after non-materiel levers fail should new materiel increments be scoped, with clear linkage to DAS milestones.

Leadership. Insert leadership development or acquisition workforce upskilling when governance bottlenecks (e.g., JROC literacy) create the gap.

Personnel. Adjust billets, Navy Enlisted Classifications, or civilian skill codes to align manpower with capability employment.

Facilities. Invest in pier-side infrastructure, cyber ranges, or test venues necessary to realize capability performance.

Policy. Seek changes to SECNAV/OPNAV directives or coalition agreements when authorities constrain capability delivery.

5.1.7 Governance Roles and Validation Chain

JROC. Executes Title 10 § 181 duties—prioritizing capability gaps, validating KPPs, delegating authority, and issuing JROCM guidance that binds both requirements and acquisition communities [88, 90].

Joint Staff Gatekeeper & JCB. The J-8 Gatekeeper screens every submission, assigns Joint Staff suspense, and recommends whether the JROC or JCB (chaired by the VCJCS) is the right validation venue; the JCB focuses on issues that are joint but lower risk, accelerating queue time for most Navy programs [88].

FCBs. Functional Capabilities Boards (e.g., C4/Cyber, Protection, Logistics) chair working groups, broker certifications (Net-Ready, Force Protection, System Survivability), monitor implementation of DCRs tasks, and elevate tripwires to the JCB/JROC [88].

Service validation authorities. OPNAV N9/N9I act as the Navy Gatekeeper, ensuring Gate 0/1 packages satisfy OPNAVINST 5000.42E expectations before forwarding to the Joint Staff, and they remain responsible for Service-specific "Joint Integration" decisions even when joint validation is delegated [83].

Requirements are therefore validated either by the JROC, the JCB (if delegated), or by the cognizant Service/CCMD authority when the Joint Staff assigns a Joint Staffing Designator of "Joint Integration/Information" [88].

5.1.8 IT, C4I, and Coalition Interoperability Implications

Net-Ready. Every Information System (IS) ICD, IS CDD, and Capability Drop version of the CDD must secure Net-Ready certification from the C4/Cyber FCB chair, demonstrating compliance with enterprise architecture, data standards, and mission thread-level interoperability [88].

Architecture and reuse. Sponsors must deliver Department of War Architecture Framework (DoDAF) views (All Viewpoint (AV), Operational Viewpoint (OV), Systems Viewpoint (SV), Capability Viewpoint (CV)) that trace requirements to reusable services, Application Programming Interfaces (APIs), and data models, enabling software pathways and rapid iteration [68, 88].

C4I interoperability. DoDD 5000.01 compels all acquisition programs to plan for cross-Service and allied interoperability; JCIDS enforces this by requiring coalition releasability considerations and Combined/Joint Mission Threads inside every requirement [67].

Allied impacts. Capability sponsors must document how U.S. and allied systems exchange data, share bandwidth, and remain cyber survivable before validation; failure to do so can halt Gate 2 or Net-Ready KPPs endorsements [88, 89].

5.1.9 Warfighter Acquisition System: Replacing JCIDS in Requirements and Acquisition

In 2025, the Department of War undertook a sweeping reform to speed up capability development by disestablishing the JCIDS and replacing it with a new streamlined framework integrated with a WAS [91, 92]. This new system shifts the focus from lengthy, centralized requirements bureaucracy to a problem-centric, warfighter-oriented approach. It is backed by high-level policy changes, including executive orders and official memoranda, that prioritize urgency, accountability, and flexibility in defense acquisition. The subsections below summarize the governing policies, explain the operational flow of the new system (phases, decision points, and key documents), map inputs/outputs at each stage, and compare the new framework to the legacy JCIDS process in terms of authority, timeline, and flexibility.

Governing Policies and Official Guidance

The replacement of JCIDS and the transition to the WAS are codified in recent DoW directives and memoranda. A key catalyst was Executive Order 14265, "Modernizing Defense Acquisitions and Spurring Innovation in the Defense Industrial Base," which directed a comprehensive review of JCIDS with the goal of streamlining and accelerating acquisitions [93]. Following this review, the Secretary of War (with the Deputy Secretary) issued memoranda in August and November 2025 that formally terminated JCIDS and established the new processes. Notably, a memo titled "Reforming the Joint Requirements Process to Accelerate Fielding of Warfighting Capabilities" directs the disestablishment of JCIDS and rescission of its governing instructions within 120 days [81]. It instructs the JROC to cease validating Service-level requirements documents (except where legally required), effectively returning most requirements authority back to the Military Services [81]. Several new governance structures and policies were introduced: Key Operational Problems (KOPs): Instead of JCIDS' exhaustive capability documents, the JROC now annually produces a concise list of the top-priority joint warfighting problems–referred to as KOPs–derived from the National Defense Strategy and combatant commanders' needs [81, 92]. These KOPs define the outcomes the Joint Force urgently needs to achieve, focusing the requirements community on solving real operational challenges rather than writing lengthy documents. Requirements and Resourcing Alignment Board (RRAB): A new high-level board co-chaired by the Deputy Secretary of War and the Vice Chairman of the Joint Chiefs was chartered to bridge requirements and budgeting [81]. The RRAB convenes each budget cycle to analyze the top KOPs and recommend programming guidance and funding assignments, including use of a special fund called the Joint Acceleration Reserve (JAR) [81, 92]. This ensures that when a critical need is identified, resources are allocated in the same budget cycle to address it, a sharp break from JCIDS where validated needs often waited years for funding [92]. Joint Acceleration Reserve (JAR): The JAR is a dedicated, flexible funding mechanism established to rapidly resource solutions to high-priority problems [92]. It is essentially a centrally managed fund in the budget (maintained by CAPE and the USW(C)) that can be applied mid-cycle to promising projects that emerge from experimentation or urgent needs [81]. The JAR is designed to eliminate the infamous "valley of death" by providing quick $$ to mature prototypes or concepts so they can transition into fielded programs [92]. Mission Engineering and Integration Activity (MEIA): To replace JCIDS' analytical and document-heavy approach, the USW(R&E) and USW(A&S) stood up the MEIA as an OSD-led joint activity [81]. The MEIA's role is to bring together engineers, warfighters, and industry to perform rapid mission engineering, architecture integration, and iterative experimentation in response to each KOP [81]. Instead of lengthy JCIDS studies, the department will now run structured experiments and prototype demonstrations to refine requirements and explore solutions in real time [81]. Successful concepts proven through MEIA campaigns can be fast-tracked with funding recommendations to the RRAB and JAR [81]. Capability Portfolio Management and Portfolio Acquisition Executives (PAEs): On the acquisition side, the memo directs the reorganization of acquisition programs into portfolios led by Portfolio Acquisition Executives [79]. These PAEs (elevated from existing PEO positions) are empowered as single accountable leaders for delivering outcomes across a portfolio of related programs [79]. They have authority to make requirement trade-offs (through "capability trade councils"), adjust program content, waive non-statutory processes, and tailor documentation–all to prioritize speed of delivery over bureaucracy [79]. This aligns with policy changes directing delegation of decision authority to the lowest appropriate level and reduction of required documentation to the statutory minimum [79]. In short, the Services and their acquisition executives are given more freedom (and accountability) to accelerate programs within broad boundaries. These reforms are supported by additional guidance to adapt or create DoW issuances. For example, the Joint Staff was directed to rescind the JCIDS Manual and related Chairman's instructions (Chairman of the Joint Chiefs of Staff Instruction (CJCSI) 5123 series) and issue interim guidance reflecting the new process [93]. The USW(A&S) is tasked to update the DoW 5000-series and other acquisition regulations to incorporate the WAS changes (e.g. redefining Milestone reviews, enabling portfolio management, and embedding "wartime" urgency in program oversight) [79]. Even the Defense Acquisition University (Defense Acquisition University (DAU)) is to be transformed into a "Warfighting Acquisition University" focusing on experiential training and warfighter needs instead of compliance checklists. All these policy changes underscore that the new system is institutionalizing flexibility and speed: every stakeholder from requirements generation through budgeting and contracting is now expected to ask, "Does this step get capability to the Warfighter faster?" and eliminate it if the answer is no [91].

Operational Flow: Phases, Decision Gates, and Documentation

Under the new framework, the process from identifying a military need to delivering a capability has been reshaped to emphasize agility and iterative development. The major phases and decision points in the Warfighter Requirements and Acquisition flow are outlined below: Need Identification and Prioritization: This first phase begins with recognizing a capability gap or emerging threat. Inputs can come from Combatant Commanders (urgent operational needs), the Military Services (Service-specific modernization needs), or strategic guidance (the National Defense Strategy, Joint Warfighting Concepts). Under JCIDS, needs were documented in lengthy Initial Capabilities Documents (ICD) and funneled through a cumbersome validation chain. Now, under the new system, needs of joint importance are distilled into Key Operational Problem (KOP) statements–concise descriptions of critical warfighting challenges that require new solutions [81, 92]. The JROC's annual "top problems" list serves as the gate output of this phase: it ranks the most pressing joint problems to tackle in the coming cycle [81]. For needs that are narrower (Service-specific), each Service's own requirements process can validate them internally without JROC review, greatly shortening the cycle [93]. Decision Gates: For joint needs, a JROC session approves the KOP list (replacing the JCIDS Gate 1 of ICD approval). For Service needs, the decision authority is a Service Requirements Board or similar, now unencumbered by JROC unless a joint impact is identified. Documentation: This phase produces brief problem statements or initial mission need statements–far shorter and faster to generate than the old ICDs. Solution Analysis and Experimentation: Once a KOP is identified (or a Service need validated), the next phase explores potential solutions. Rather than producing a formal Capabilities Development Document (CDD) upfront as in JCIDS, the department now uses mission engineering and rapid prototyping to define requirements through experimentation [81]. The MEIA leads this for joint KOPs: it convenes cross-functional teams (including Service labs, industry partners, and warfighters) to brainstorm concepts and then build prototypes or conduct demonstrations targeting the KOP [81]. This iterative experimentation campaign helps refine what is technically feasible and what operational concepts work, effectively "learning by doing" instead of writing requirements on paper. For Service-specific efforts, the Services pursue similar activities–many have adopted "speed to fleet" experimentation initiatives or innovation pipelines that mirror the MEIA approach. Decision Gates: A key decision in this phase is determining which solution(s) show promise. Rather than a traditional JCIDS JROC CDD approval, the outputs here are data and insights from demonstrations, leading to a recommended solution approach. For joint efforts, the conclusion of a MEIA campaign might be a report or briefing to the RRAB and JROC, confirming that a certain capability (say, a prototype drone or a software application) meets the need. Internally, a Service could decide to proceed with an existing solution or requirement refinement based on experiment results. Documentation: Instead of CDDs, the documentation is often an experiment report, technical assessments, and a refined concept of operations. Any formal requirements are kept minimal–e.g. a short list of key performance parameters for the chosen solution–to preserve flexibility. This drastic reduction in paperwork (replacing "hundred-page requirements documents" with live experimentation outcomes) is a deliberate hallmark of the new system [92]. Resource Alignment and Decision: In parallel with solution development, the funding aspect is addressed in this phase. Under JCIDS/PPBE in the past, there was a disconnect–a requirement might be approved and sit unfunded until the next budget cycle or two. The new framework tightly couples the requirement to resourcing in real time. The Requirements & Resourcing Alignment Board (RRAB) serves as the decision gate here. Each cycle (likely tied to Planning, Programming, Budgeting, and Execution timelines), the RRAB reviews the top KOPs and the results of any associated experiments. It then makes decisions on programming resources for those solutions immediately [92]. For example, if a MEIA experiment validated a prototype for a counter-drone system that addresses a KOP, the RRAB can recommend that in the upcoming budget, funding be carved out (via the JAR or reprioritization) to start procuring that system. This means the solution is "budgeted for" almost concurrently with being defined, a major acceleration. The Deputy Secretary of War and VCJCS co-chairing ensures both civilian budgeting leadership and military operational leadership are aligned on these trade-offs [92]. Meanwhile, Service-specific needs enter the Services' budget process directly, but now with guidance that they, too, should integrate their requirements with resource plans quickly (each Service was directed to reform its requirements process to better merge with budgeting) [81, 92]. Decision Gates: The RRAB decision (approved by the Deputy Secretary) is effectively a new Milestone 0 for joint programs–a commitment that "we will fund this solution and move to acquisition." It replaces the former JROC validation as the key go/no-go, by adding the crucial element of funding. For Service needs, the gate might be a SAE's decision to launch a program of record with funding in the Service POM. Documentation: The outputs here include programming guidance or funding decisions (e.g. inclusion in the FYDP, allocation from JAR) rather than a JCIDS Capability Production Document (CPD). In essence, a combined "requirements-resolution and funding memo" is the key record. The RRAB's recommendations are documented for leadership approval and to inform Congress of new starts or rapid reallocations (ensuring transparency for oversight). Acquisition Execution (Development/Fielding): Once funding is lined up, the solution transitions to the execution phase–an acquisition program or rapid fielding project is initiated to deliver the capability to the warfighter. The WAS introduces changes here to compress timelines and delegate authority. If the solution is mature (say a prototype that just needs low-rate production), the Services can use streamlined acquisition pathways (e.g. a Middle Tier of Acquisition for rapid fielding, which bypasses some traditional milestones) [79]. The memo explicitly calls for cutting out unnecessary regulatory requirements and reviews to speed this phase, such as replacing formal Analysis of Alternatives with competitive prototype trials when appropriate [79]. Under a PAE's leadership, programs are structured in short, outcome-focused increments–delivering a minimum viable capability first, then upgrading it iteratively [79]. Decision Gates: Traditional acquisition Milestones (A, B, C) still exist in a loose sense, but many authorities for those decisions are pushed down to the SAE or even the new PAEs for agile programs [79]. For instance, a PAE might have the authority to approve low-rate production start (analogous to Milestone C) once a prototype meets a threshold, without having to seek OSD or JROC permission. In cases of larger programs, Milestone decisions will focus on confirming that the program's scope aligns with the original KOP solution and that risks are managed–not on re-validating requirements. Documentation: This phase produces the standard acquisition artifacts (e.g. an acquisition strategy, test plans, perhaps a streamlined CDD if needed for record) but the directive is to keep documentation to the minimum required by law [79]. Many bureaucratic reports and process documents have been eliminated or merged. For example, configuration steering boards are replaced by "capability trade councils" under the PAE that make real-time trades without formal reports [79]. Test and evaluation is being transformed to be continuous, so instead of a massive Test & Evaluation Master Plan, the focus is on an accredited pipeline that can rapidly test incremental updates [79]. In summary, paperwork in execution is trimmed, and the emphasis is on delivering and proving capability in the field. Fielding, Feedback, and Iteration: After initial fielding, the process does not stop. A core tenet of the new approach is continuous modernization. Feedback from the warfighters using the new capability is fed back into the requirements process quickly. If the initial solution only met part of the KOP, further increments (upgrades or additional systems) go through the pipeline perhaps in an accelerated loop. The Portfolio Acquisition Executive (PAE) monitors the portfolio's performance using new time-based metrics (e.g. how fast from need identification to Initial Operational Capability) [79]. Portfolio "scorecards" are being introduced to measure this and hold the acquisition community accountable for speed [79]. Decision Gates: In this phase, decisions might include scaling up production (if the threat demand increases), starting a next increment, or, conversely, terminating a solution that did not pan out (with residual funding quickly reallocated to alternatives via the JAR/RRAB process). Documentation: Lessons learned and metrics are captured, but again with lean reporting. The continuous nature of this cycle is a departure from the static "requirements freeze" of JCIDS–instead, requirements can be adjusted on the fly in subsequent increments if threats evolve, as long as the changes are within the portfolio's authorized scope [79]. In practice, the new system's operational flow is not strictly linear; it encourages parallel efforts and rapid iterations. For example, even as one solution is being fielded, the MEIA might already be experimenting with the next-generation approach for the same KOP. The guiding philosophy is to deliver capability to the field faster and update it often, rather than waiting for a perfect solution. This is encapsulated in the Department's wartime mentality: "move with urgency – China will not wait for our forms to finish routing" [92]. Every stage of the process is geared toward compressing timelines: identifying needs quicker, prototyping sooner, funding immediately, and delegating decisions to avoid delay. By cutting out interim approval layers and documentation, the new WAS aims to collapse the time from recognizing a threat to fielding a response, from years down to months in many cases [92].

Inputs and Outputs at Each Stage of the Process

To clearly illustrate how information and decisions flow in the new framework, Table XIV maps the inputs and outputs of each major stage from mission need to acquisition decision. This mapping shows how a raw operational need is transformed into fielded capability under the new system, highlighting the key decision points and artifacts at each step. As shown in Table XIV, one of the most significant changes is the coupling of the Requirements Decision with a Resourcing Decision at the RRAB stage. Under JCIDS, a requirements document could be approved (output of JCIDS) without any immediate input into the budgeting process–a disconnect that led to delays [92]. In the new system, the decision to pursue a solution is inherently linked to allocating money to it, ensuring that validated needs are not just shelved but promptly acted upon [92]. Another notable feature is the emphasis on experimentation outputs as a formal input to decisions. This means tangible evidence (prototypes tested, technologies demonstrated) drives the requirement definition and acquisition planning, rather than theoretical studies alone [81, 92]. Documentation at each stage has been minimized and made purpose-driven. For example, rather than an ICD and CDD, the combined output of the first two stages (Need and Experimentation) might be a short Problem Statement with a Solution Outline and an experiment report–sufficient to justify a new program start. Traditional JCIDS documentation like the Capability Production Document (CPD) that was required before full-rate production is expected to be replaced by more agile documentation (perhaps a fielding decision review by the PAE and users) focusing on whether the capability meets operational needs, not whether every requirement checkbox is met [79]. This reflects a broader shift to "outcome over process"–the measure of merit is delivering a working capability, not producing documents.

Key Differences between JCIDS and the New System

The table below summarizes major differences between the legacy JCIDS requirements process and the new Warfighter-centric requirements and acquisition framework, highlighting changes in authority, timelines, flexibility, and other dimensions: As Table XV illustrates, the transformation from JCIDS to the new system is profound. Authority has shifted from a joint committee-centric model to one empowering Service chiefs and portfolio leaders–with the Joint Staff refocusing on truly joint issues (KOPs) rather than micromanaging every requirement [92, 93]. This not only speeds things up (fewer approval stops) but also places accountability squarely on those who will execute the programs. The timeline has been compressed by design: what used to be a protracted front-end process (sometimes nearly *three years to approve an ICD/CDD*) is replaced by concurrent problem-solving and funding decisions that can be made in months [92]. In fact, one of the explicit goals is that a critical need identified in year X can start getting funded and prototyped by year X+1 (if not sooner) [92]–a pace unimaginable under the old system. Another key difference is the level of flexibility. JCIDS imposed uniform documentation and gating for all programs, whether a small software fix or a major weapons system; the new framework recognizes multiple pathways and encourages tailoring. For example, if a commercial technology can meet a need quickly, the WAS guidance pushes the use of Other Transactions or commercial-like acquisitions to bypass lengthy FAR processes [79]. Middle Tier Acquisition (Section 804) pathways are explicitly championed to deliver prototypes or field limited capabilities in under 5 years, with even those streamlined further by removing bureaucratic reports [79]. In essence, the new system says: use *any* legal acquisition method that will get the job done faster–a stark contrast to JCIDS which was indifferent to how slow or fast the acquisition went after requirement approval. The cultural mindset is also different. JCIDS fostered a compliance culture (check all the boxes, meet all requirements, avoid failure at all costs). The warfighter acquisition mindset is more entrepreneurial: take smart risks, deliver something useful quickly, then iterate. The memos even call for incentivizing Program Managers who take bold steps and penalizing those who are perpetually behind schedule [79]. The end result sought is a military acquisition system that behaves more like it would in wartime urgency, rather than in peacetime bureaucracy. To conclude, the system replacing JCIDS–anchored by the WAS and the new joint requirements paradigm–represents a major shift toward agility and warfighter focus in defense capability development. Engineering Duty Officers and acquisition professionals can expect fewer obtuse staff processes and more direct responsibility for outcomes. They will operate in an environment where requirements are shorter and problem-focused, decisions are faster and informed by real-world prototypes, and the measure of success is delivering combat capability on relevant timelines rather than producing paperwork. While the transition will require adaptation (after decades of JCIDS habits), the intent is clear: *speed, agility, and innovation are to take precedence, so that U.S. forces get the capabilities they need, when they need them* [79, 92]. The new framework's success will depend on vigorous implementation, but if realized, it promises to resolve many of the frustrations of JCIDS and restore a competitive edge in military technology development.

Integrated Resourcing Artifacts: POM and BES

Under WAS, the RRAB aligns requirements decisions directly with the PPBE process, specifically influencing the POM and Budget Estimate Submission (BES) to ensure funding is available for validated needs.

POM. A recommendation from the Service to the Secretary of Defense detailing resource allocation over a five-year period to meet strategic objectives.

- *Example:* The Navy submits a POM proposing accelerated frigate construction and increased hypersonic R&D funding, justified by the National Defense Strategy (NDS).

BES. The detailed budget request for the first two years of the POM, translating programmatic goals into priced-out line items for the President's Budget.

- *Example:* Following the POM, the BES includes specific procurement line items for two Constellation-class frigates and associated manpower costs for the upcoming fiscal year.

5.2 Test & Evaluation

5.2.1 Board Prep Summary

T&E provides the objective evidence that a system can achieve the warfighter outcomes promised in requirements documents before obligating the fleet to fielding, sustainment, and congressional reporting commitments [52, 94]. A test is the controlled activity (lab, range, digital thread, fleet event) that produces data, while an evaluation is the analytic judgment that compares that data to thresholds, suitability expectations, and risk acceptance criteria [94]. Effective T&E integrates developmental, operational, and live-fire evidence so that each MCA milestone is backed by verified design data and validated mission performance insights rather than optimistic modeling alone [71]. Programs that synchronize T&E with systems engineering and program control expose performance gaps early, recycle fixes through the technical baselines, and enter OTRR with a defendable TEMP, resource plan, and deficiency scrub.

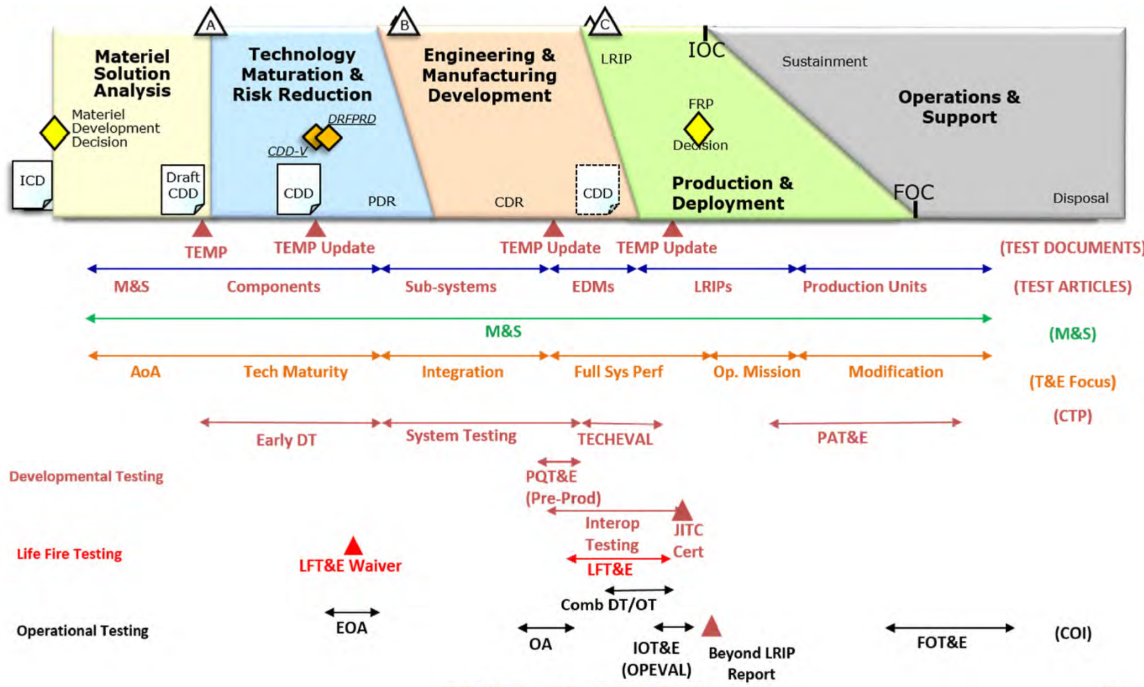


Figure 5.4. Key MCA test and evaluation touchpoints that frame DT, OT, and certification decisions. Source: 3.5.5. Test and Evaluation Part I, *2025 [94].*

5.2.2 Definition and Relationship to Verification and Validation

Developmental Test and Evaluation (DT&E) verifies "did we build the system right" by showing the technical baseline meets allocated requirements, standards, and interface controls, while Operational Test and Evaluation (OT&E) validates "did we build the right capability" by proving Sailors and Marines can accomplish missions under representative threat, environment, and sustainment conditions [68]. Verification relies on repeatable measurements (e.g., propulsion plant efficiency in a land-based engineering site) and feeds configuration control; validation uses operational scenarios, tactics, and logistics supportability to demonstrate mission effectiveness and suitability. DoDI 5000.88 further directs programs to execute frequent, iterative end-user validation of features and usability so mission thread owners can accept the delivered capability before major decisions [95]. Verification and Validation (V&V) and Verification, Validation, and Accreditation (VV&A) activities must be planned in the TEMP and SEP so that models, simulations, hardware-in-the-loop facilities, and live events share common data assumptions and confidence levels before milestone reviews [71]. *Verification* therefore proves compliance with the documented specification (analysis, inspection, test, demonstration), whereas *validation* proves the warfighter accepts the delivered capability and can close the mission threads; both sides of the V&V matrix must be satisfied before the MDA can certify readiness for production or fleet release [71].

5.2.3 Statutory Requirements and Key Roles

10 U.S.C. § 139. Establishes the DOT&E as an independent advisor to SECWAR and Congress; DOT&E approves operational test plans, monitors execution, and must issue a public report for covered programs [96].

10 U.S.C. § 2399. Prohibits a program from proceeding beyond LRIP or requesting Full-Rate Production approval until DOT&E and the lead Operational Test Agency submit their independent reports to Congress [97].

10 U.S.C. § 2366. Requires Live Fire Test and Evaluation (LFT&E) or an approved waiver for covered systems (personnel, platform, and munitions lethality/survivability) before beyond-LRIP decisions; it also directs reporting of vulnerability/lethality findings to defense committees [72].

Deputy Assistant Secretary of War for Test and Evaluation (DASW(T&E)). Serves as the OSD focal point for test resources, designates Chief Developmental Testers (CDTs) for ACAT I programs, charters Lead Developmental Test and Evaluation Organizations (LDTOs), and co-chairs Defense-level T&E Working Integrated Product Teams [71].

CDT and LDTO. The CDT orchestrates the integrated test schedule, data requirements, and deficiency resolution cadence, while the LDTO (often a warfare center or system command test directorate) furnishes instrumentation, modeling, and range execution authority for DT&E [71].

Operational Test Agencies. COMOPTEVFOR, AFOTEC, ATEC, and MCOTEA plan and execute OT&E Missile Defense Agency (MDA) record [98].

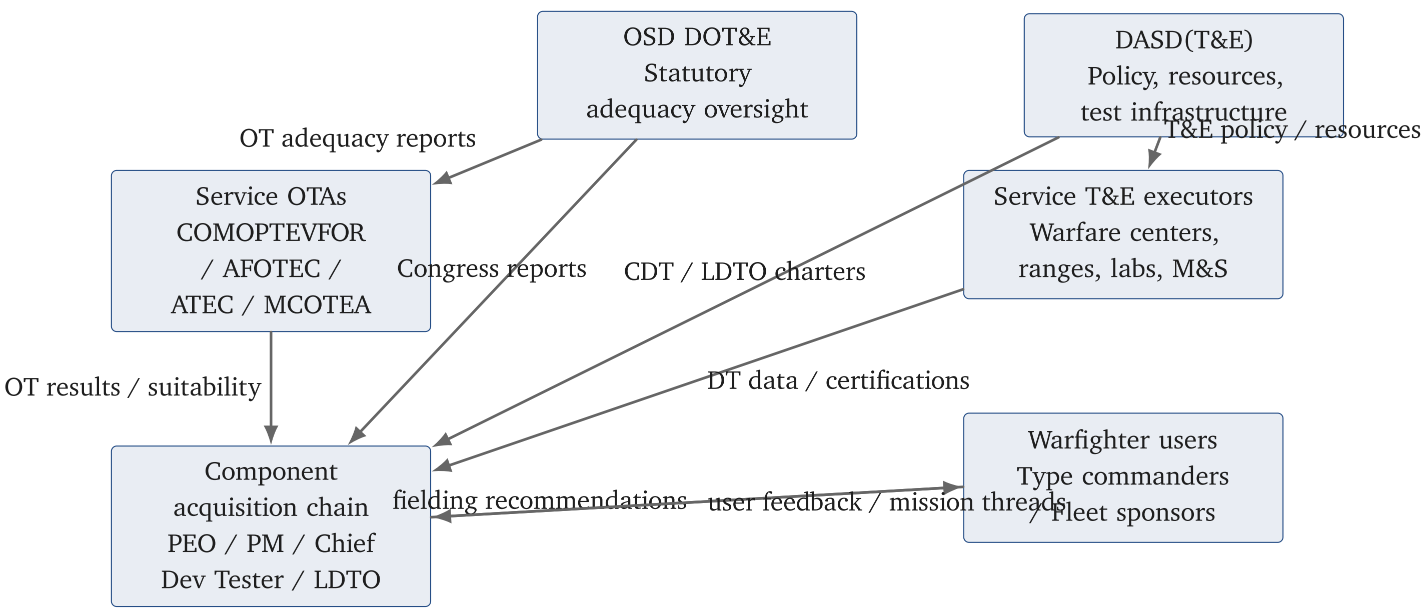


Figure 5.5. *DoW test and evaluation governance chain linking OSD oversight, Service Operational Test Agencies, warfare centers/ranges, and program-level test leadership.*

5.2.4 Developmental Test Activities

Component and qualification testing. Bench, lab, and land-based engineering site events that verify materials, software builds, cyber hardening, interoperability, and safety factors before integration; data feeds configuration audits and reliability growth curves [94].

Subsystem and integration testing. Hardware-in-the-loop, systems integration labs, and shore-based platform integration tests that expose interface defects and mission thread timing issues before at-sea trials [94].

System-level DT&E. Full-up prototypes or first-of-class hulls execute combined developmental/operational scenarios, cybersecurity assessments, and logistics demonstrations to validate readiness for TRR and Technical Evaluation (TECHEVAL) [52].

Reliability growth and maintainability demonstrations. Planned stress events and failure-mode tracking prove that Mean Time Between Operational Mission Failure (MTBOMF) and corrective maintenance timelines align with sustainment strategies before Operational Test Readiness [71].

Certification testing. Airworthiness, safety, information assurance, weapon certification, and electromagnetic environmental effects tests that deliver statutory certifications needed for fielding [94].

Note

Edge case: Software pathway efforts may collapse TECHEVAL/Operational Evaluation (OPEVAL) into continuous DevSecOps assessments, but the program must still show how combined testing and user evaluations satisfy Title 10 adequacy requirements before fielding [68].

5.2.5 Live Fire Test and Evaluation

When required.

Title 10 demands LFT&E for every new or significantly upgraded crew-occupied platform, combat vehicle, major munitions program, or missile defense component that will face threat lethality; full-up system-level shots must be realistically stressed prior to requesting Full-Rate Production approval [71, 72].

When waived.

Only the SECWAR can waive full-up LFT&E when the system is unmanned, when destructive testing would be unreasonably expensive or impractical, or when alternative testing (component shots, validated modeling, allied data) provides equivalent insights; Congress must be notified of the rationale and the alternative plan [71, 72].

Execution and use.

Programs plan vulnerability and lethality events in the TEMP, employ full-up articles plus critical subsystems and surrogates, and collect battle damage, crew survivability, and cascading-effect data to inform design hardening, tactics, and survivability equipment upgrades; the results become an annex to the DOT&E report before the FRPDR [52, 71].

5.2.6 Operational Test Phases

Early Operational Assessment (EOA). A qualitative review, often using simulations or limited field events, that informs CDD refinement and acquisition strategy decisions before Milestone B [98].

Operational Assessment (OA). Focused operational events (sometimes embedded in Fleet Battle Problems or Fleet Exercises) that characterize mission effectiveness, training load, and logistics demands prior to LRIP or dedicated IOT&E [98].

Initial Operational Test and Readiness Evaluation (IOT&RE). Navy programs often combine IOT&E and Readiness Evaluation to prove both mission performance and Fleet introduction readiness; Title 10 requires that the IOT&E portion be adequate, realistic, and reported to Congress before FRP [97].

Follow-on Operational Test and Evaluation (FOT&E). Conducted after FRP or major configuration changes to verify corrective actions, CPD thresholds, cybersecurity resilience, and software drops prior to wide deployment [71].

5.2.7 Fundamental Test and Evaluation Terms

Critical Operational Issue (COI). The mission-outcome questions ("Can the ship defend itself against a raid?") that structure OT&E plans and reports [98].

Measure of Effectiveness (MOE) & Measure of Suitability (MOS). Measures of Effectiveness and Suitability quantify warfighting value (probability of raid annihilation) and fleet readiness attributes (reliability, maintainability, training burden) tied to COIs [98].

Measure of Performance (MOP). Measures of Performance (e.g., radar detection range) show lower-level technical behaviors that roll up to MOEs/MOS [94].

Deficiency categories. Category I deficiencies are mission-critical failures that could cause death, severe injury, or major mission loss; Category II items are significant but less critical issues requiring tracking and correction [98].

Combined testing. Planned events that collect developmental and operational data simultaneously to save schedule and align datasets for evaluation teams [68].

5.2.8 Critical Technical Parameters and Technical Performance Measures

Critical Technical Parameters (CTPs) are the small subset of design attributes that must be controlled to assure a KPP or KSA remains achievable; they are derived from the ICD/CDD and baselined inside the APB [52]. Each CTP maps to one or more Technical Performance Measures (TPMs) that engineers track continuously using lab data, digital models, or DT&E results; TPMs trends highlight when technical debt threatens TEMP and SEP assumptions so leadership can adjust scope, schedule, or funding. Programs are expected to show a clean trace from COI → KPP/KSA → CTP → TPM → planned test points in the TEMP, ensuring every test objective underwrites an operational outcome and every requirement has an observable verification method [71, 94].

5.2.9 Test and Evaluation Master Plan

The TEMP captures the integrated T&E strategy, schedule, resources, and data management approach across the lifecycle; it harmonizes the SEP, LCSP, and APB so that every milestone package shows how risk will be retired [69]. Part I summarizes program context and key performance requirements, Part II defines the integrated test program (DT, OT, certification, modeling, and analysis), Part III outlines resource needs (ranges, threat surrogates, workforce, instrumentation), and annexes address data rights, stats, and cyber accreditation [71]. ACAT I(D) TEMPs require DOT&E approval; other TEMPs are approved by the Component Acquisition Executive. Updates are mandatory when requirements change, critical deficiencies emerge, or significant schedule/funding revisions alter test sequencing [52].

5.2.10 Modeling and Simulation

Modeling and simulation provide the only practical way to explore the mission design space before hardware exists, rehearse high-risk scenarios (e.g., hypersonic raid defense), and extend sparse live-test data across the operational envelope [94]. Programs must document model purpose, pedigree, uncertainty, and VV&A status in the TEMP; accredited models can bound live-test needs, size instrumentation, and support decision-quality evaluations when safety, cost, or treaties limit destructive testing [71]. Mission engineering digital threads (system-of-systems simulations, campaign models) allow combined DT/OT teams to choreograph events, align data management plans, and feed statistical confidence calculations before and after live events.

5.2.11 Navy Overlays and Integration

OPNAVINST 3960-series policy (captured in the coursebook) requires Navy programs to plan TECHEVAL (led by the engineering agent) before OPEVAL, to obtain OPNAV N9 concurrence on test objectives that bear on Fleet introduction risk, and to route OTRR packages through COMOPTEVFOR for certification that logistics, training, and safety enablers are in place [98]. First-of-class ships also execute Builder's and Acceptance Trials, Board of Inspection and Survey (INSURV) events, and dedicated combat system Ship Qualification Trials that integrate with TECHEVAL/OPEVAL data packages. Navy Commander, Operational Test Force (COTF) retains authority to recommend restricted Fleet release if Category I deficiencies remain; those actions feed the Key Roles appendix update for DASW(T&E), CDT, and COTF responsibilities.

5.2.12 Fiscal and PPBE Touchpoints

RDT&E appropriations fund most DT&E events, test articles, range time, and contractor support, while procurement colors such as SCN or Weapons Procurement, Navy buy dedicated OT assets when production-representative hardware is required for statutory adequacy [27, 94]. Operations and Maintenance funds cover Fleet participation, temporary duty, and sustainment of government test ranges. Because OT adequacy is a precondition for FRP and sustainment decisions, Program Objective Memorandum issue papers must protect test range modernization, threat-representative targets, and Modeling & Simulation accreditation costs or risk a PPBE disconnect when COMOPTEVFOR identifies unfunded test objectives.

5.2.13 Assumptions and Uncertainties

- Coursebook Modules 3.5.5 and 3.5.6 (March 2025 editions) remain the latest DON-specific T&E references.

- COMOPTEVFOR manual updates post-2024 were not available offline; Navy-specific terminology (e.g., OPEVAL taxonomy) is based on coursebook guidance.
- Upcoming NDAA changes to Title 10 T&E statutes have not been incorporated pending official publication.

Process Stage	Inputs	Outputs (Decision & Artifacts)
Need Identification	- Threat analysis, capability gap or - Urgent operational need (field reports) - Strategic guidance (NDS, Joint Concepts)	- Defined requirement or problem statement - If joint: KOP nomination for JROC [81] - If Service-specific: Service validates need internally [93]
Problem Prioritization	- KOP candidates from across DoW - Inputs from Combatant Commanders and Services	- JROC-approved list of top *Key Operational Problems* (annual) [81] - (Service needs not requiring JROC appear on Service priority lists)
Solution Exploration & Experimentation	- KOP definition (problem to solve) - Industry proposals, tech concepts - Existing prototypes or studies	- Experimentation results (data, demo performance) [81] - Refined requirements (based on what works) - Identified promising solution(s) (e.g. chosen prototype design)
Requirements & Resourcing Decision	- Prioritized KOP (with solution recommendation) - Cost, schedule, performance info from prototypes - Service budgets (baseline programs)	- RRAB decision: funding allocation to solution [92] - Joint Acceleration Reserve (JAR) funds committed if needed [92] - Designation of lead Service/agency for acquisition - If Service need: SAE approves new program start in POM
Acquisition Execution (Development & Fielding)	- Funded requirement (from RRAB/Service) - Prototype design or tech package - Acquisition strategy (tailored)	- Program of record (or rapid fielding project) initiated - Milestone decisions (delegated to PAE/Service for speed) [79] - Initial units produced and fielded (IOC achieved) - Feedback from Warfighters on capability
Iteration & Updates	- Warfighter feedback and operational data - New technology opportunities - Evolving threat (changes to problem)	- Updated requirements for next increment (if needed) - Follow-on experiments or spiral development - Adjusted portfolio plans (via PAE & possibly new RRAB input if joint)

TABLE XIV. INPUTS AND OUTPUTS THROUGH THE STAGES OF THE NEW WARFIGHTER REQUIREMENTS AND ACQUISITION PROCESS.

Aspect	JCIDS (Legacy System)	New Framework (KOPs & WAS)
Requirements Approval Authority	Centralized in JROC for most major requirements. JROC and FCB committees reviewed and validated Service proposals, often adding layers of oversight.	Decentralized to Services for component-specific needs; JROC only validates a short list of joint *Key Operational Problems* annually [92, 93]. Services now have authority to approve their own requirements without Joint Staff involvement in most cases.
Speed and Timeline	Slow, often requiring **~2+ years (800 days)** just to approve a requirement document [92]. The process was sequential and could not easily respond within a budget year.	Accelerated, with a **time-bounded process** aiming to fund solutions in the *same budget cycle* a need is identified [92]. Experimentation and development start immediately instead of waiting for paperwork [92], enabling delivery in months to a couple of years, rather than many years.
Focus and Inputs	Focused on detailed requirements documents (ICD, CDD, CPD) and exhaustive analysis. Input often a lengthy study or formal request that went through multiple staff layers. Needs were defined in isolation from resources.	Focused on solving operational **problems** with mission outcomes. Input is often a scenario or threat-based problem statement. Directly ties needs to resources ("joint problems paired with joint money" [92]) – ensuring no gap between identifying a need and funding a solution.
Resource Alignment	Requirements determination was largely separated from budgeting – validated needs could linger unfunded for years [92]. Funding followed in the normal PPBE cycle (if at all), often creating a disconnect between what JROC approved and what the budget supported.	Requirements and resourcing are **integrated**. The RRAB allocates funds to approved solutions in real-time [92]. A Joint Acceleration Reserve (new funding mechanism) is used to rapidly finance promising prototypes or urgent needs within the year [92], preventing the "valley of death" where projects died waiting for funds.
Flexibility and Adaptability	One-size-fits-all process and rigid documents. Changes required re-approvals. Emphasis on meeting every requirement in a single step, often making programs inflexible. Risk aversion was common – programs strove for full capability in one go.	Tailorable and **adaptive process**. Requirements can be adjusted as new information comes from experimentation. Programs use incremental development: deliver a minimum viable capability first, then improve iteratively [79]. Trade-offs in requirements vs. schedule are encouraged to field something useful fast [79]. The new system explicitly promotes accepting some risk and adapting on the fly to get solutions out quicker.
Documentation Burden	Very high: numerous lengthy documents (ICD, CDD, CPD, JCIDS analysis studies) and mandatory JCIDS manual formats. "Hundred-page" requirements documents were common [92]. Each Milestone review required validated documents.	**Streamlined documentation**: only minimal essential documents are produced [79]. Emphasis on factual outputs of experiments and concise statements of required capability. The culture shifts from writing reports to demonstrating results. As one commentary noted, "gone are the endless stacks of validated memos" in the new process [92].
Oversight Structure	Multiple layers (JCIDS boards, OSD oversight, etc.) often fragmented authority. No single person owned end-to-end outcome; Program Managers answered to many stakeholders, and approval authority was diffused [79].	**Streamlined governance with clear accountability**. Portfolio Acquisition Executives are empowered as the single accountable leaders for delivering warfighting capabilities in their portfolios [79]. Oversight reviews are cut down to only

6 Maintenance

6.1 Fleet Maintenance Assets

6.1.1 Echeloned Maintenance Levels

Organizational-level. Ship's force keeps the hull, mechanical, and electrical plants safe and combat-ready by executing preventative checks, battle damage control, and light repairs with organic tools, allowing the platform to remain as self-sufficient as possible while deployed [99, 100].

Intermediate-level. Intermediate Maintenance Activitys (IMAs) and tenders augment the crew with certified artisans (e.g., welders, riggers, calibration labs) who can disassemble components, correct corrosion, and return equipment to specification without docking the ship; Navy Afloat Maintenance Training Strategy (NAMTS) builds those Navy Enlisted Classifications (NECs), and so strike groups can recover faster when forward [99, 100].

Depot-level. Naval Shipyards (NSYs), private yards, and SUPSHIPs conduct heavy industrial work such as reactor refueling, shaft and underwater body repairs, and major modernizations that exceed shipboard capability; every task is tied to the Class Maintenance Plan (CMP) and engineered work instructions [100, 101, 102].

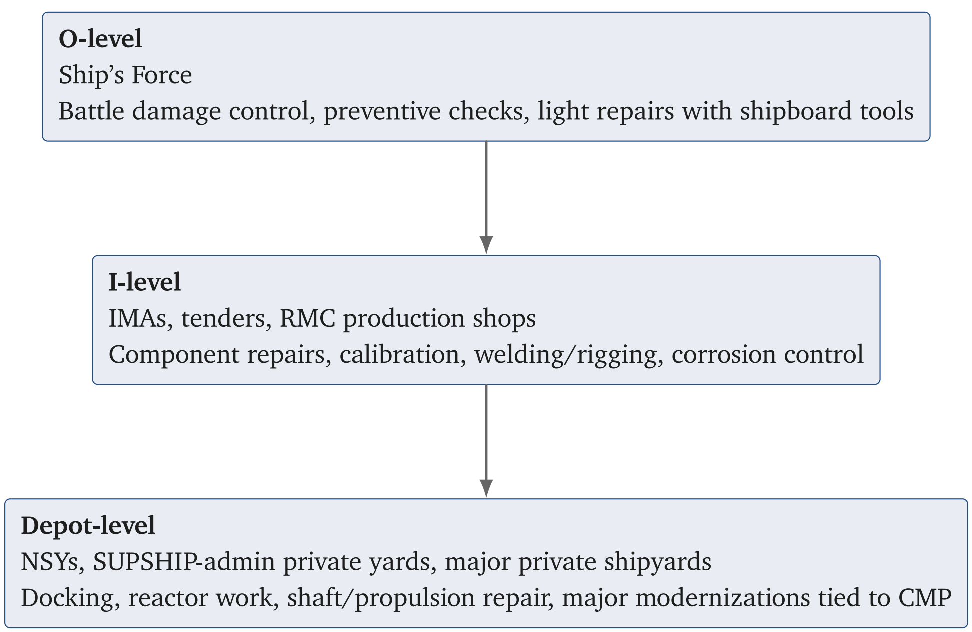


Figure 6.1. Maintenance echelon coverage and typical executing organizations from organizational-level through depot-level work.

Note

Board cue: reference at least one example task in each echelon (e.g., IMA balancing pump rotors vs. depot replacement of a main reduction gear) to show command of maintenance depth.

6.1.2 Shore Maintenance Organizations and Roles

NSYs. Public yards (e.g., Norfolk, Pearl Harbor, Puget, Portsmouth) report to NAVSEA 04 and integrate nuclear workforces, planning codes, and lifting/handling departments that mirror Naval Sea Systems Command Instruction (NAVSEAINST) 5450.14D standard organization [101].

Regional Maintenance Centers (RMCs). Regional Maintenance Centers deliver voyage repairs, execute CMP work, and act as the local TA for availabilities; they synchronize contracting, production shops, and Total Ship Readiness Assessment (TSRA) under Commander, Navy Regional Maintenance Center (CNRMC) [100, 102].

SUPSHIP. Supervisors of Shipbuilding administer new construction and conversion/modernization contracts as ACOs, coordinate budget execution, and provide technical oversight for nuclear refueling overhauls (e.g., CVN RCOH) *(Appendix C.2: RDML Jonathan J. Jett-Parmer serves as NAVSEA Deputy Commander, Supervisor of Shipbuilding; RDML Robert J. Dodson (USNR) is the reserve deputy; Commander, Naval Sea Systems Command (COMNAVSEA) is ADM William J. Houston (Submarine officer, non-EDO))* [9, 100, 102].

Private shipyards. Huntington Ingalls, Bath Iron Works, General Dynamics National Steel and Shipbuilding Company (NASSCO)/Electric Boat, Fincantieri Marinette Marine, and Austal execute modernization packages, submarine maintenance, and surge depot work as part of the industrial base portfolio directed by OPNAV and PEO contracts [100].

6.1.3 Regional Maintenance Command Structure

Flag oversight (EDO). CNRMC is an EDO flag billet (RDML Dan L. Lannamann) reporting to COMNAVSEA; CNRMC sets standards and performance goals for all RMCs and aligns with TYCOM production priorities *(Appendix C.2: COMNAVSEA is VADM James P. Downey; CNRMC is RDML Dan L. Lannamann)*[102, 103].

RMC reporting. RMCs (e.g., Mid-Atlantic, Southeast, Southwest, Japan, and Puget Sound Naval Shipyard and Intermediate Maintenance Facility (IMF) detachments) report administratively to CNRMC and operationally to the supported TYCOM, with competency-aligned codes (100 fleet support, 200 availability management, 300 production shops, 400 quality assurance/technical authority) to standardize execution[102].

Work control touchpoints. Regional Maintenance and Modernization Coordination Office (RMMCO), availability work package managers, and contracting officers integrate with ship's force, Port Engineers, and TYCOM maintenance officers; EDO Port Engineers and RMC production officers bridge technical authority to the TYCOM readiness owner[100, 102].

6.1.4 Command, Funding, and Technical Authority Chains

Operational chain. CNO delegates readiness to USFF and PACFLT, who assign Type Commanders to own production schedules and risk acceptance for their forces [99].

Technical chain. NAVSEA 21/04 issue modernization and maintenance standards, while warfare centers and TAs publish CMP-derived work packages; disputes escalate through the Technical Authority chain independent of the operational chain [99, 102].

Funding chain. Fleet maintenance (Operations and Maintenance, Fleet (OC&MF)) flows from OPNAV N82 to BSOs and on to execution activities, while modernization is forecast via Fleet Modernization Design (engineering) funding (FUNDS-D)/Fleet Modernization Alteration funding (FUNDS-ALT)/Fleet Modernization Procurement funding (FUNDS-K) and, in limited pilots, OPN funds for selected private-yard availabilities to accelerate contract awards [27, 100].

Maintaining clarity between supported/supporting chains prevents the common failure mode where operational priorities outpace technical warrant approvals or funding authority, triggering rework and availability churn.

6.1.5 Workforce Readiness and Assessments

Readiness assessments. TSRA events and Condition Found Reports feed the Current Ship's Maintenance Project (Current Ship's Maintenance Project (CSMP)) so planners can refresh the CMP and future availability work packages [100].

Training pipelines. Programs such as NAMTS, SurgeMain reservists, and civilian apprentice pipelines grow targeted skills (e.g., non-skid application, inside machinist) to close gaps identified in Fleet readiness reviews [100].

Digital enablers. Class maintenance databases, Navy Data Environment (NDE) instances, and tool control systems give RMCs and NSYs shared configuration visibility that underpins quality assurance and auditability [102].

Timely feedback from these mechanisms is what allows SURFMEPP/SUBMEPP to rebalance work between the organizational-, intermediate-, and depot echelons before a ship misses a CNO availability window.

6.2 Navy Modernization Program

6.2.1 Modernization vs. Maintenance

Modernization. Improves capability or readiness above the baseline by installing approved Ship Alterations (SHIPALTs), Modernization Work Orders, and Fleet modernization program items that change the ship's configuration, safety posture, or mission systems [99, 104].

Maintenance. Restores the system to its original performance and reliability; it cannot change capability without an approved modernization directive per OPNAVINST 4700.7M [104, 105].

Repair. Corrects degradations to return the item to the current baseline; repairs that change form, fit, or function require an approved modernization document to avoid unauthorized configuration changes under OPNAVINST 4700.7M [99].

EDOs must articulate which work package element they are defending, because it dictates the funding source (e.g., FUNDS-D/FUNDS-ALT for modernization versus OC&MF for maintenance) and who holds approval authority.

6.2.2 Governance, Roles, and Documentation

Modernization governance. The Navy Modernization Program (NMP) integrates OPNAV resource sponsors, PEOs, Participating Acquisition Resource Managers (PARMs), Warfare Centers, and Fleet forces through the NMP Executive Steering Group and Sub-Class Modernization Coordinating Groups; the process is codified in SL720-AA-MAN-030, TS9090.310, and the Joint Fleet Maintenance Manual (JFMM) [105, 106, 107].

Key documents. Modernization Change Documents capture the requirement, Justification and Cost Forms allocate funding, SHIPALT Records (Record of Change) provide lifecycle traceability, installation drawings direct work, and the Alteration Equivalent to Repair (Alteration Equivalent to Repair (AER)) process keeps urgent fixes legal [104, 107].

Execution agents. Alteration Installation Teams (AITs), RMMCOs, and SUPSHIPs integrate modernization tasks into availabilities while Naval Supervising Activities verify that work followed the approved package *(Appendix C.2: SUPSHIP is led by RDML Jonathan J. Jett-Parmer with RDML Robert J. Dodson (USNR) as reserve deputy (Appendix C.2))*[106, 107].

Table XVI: Modernization document ownership and funding tie points

Document	Primary Owner	When Used	Funding / Approval Tie
Change Initiation Notice	Fleet sponsor or PARM	Starts the change process; screens technical need and logistics impacts before engineering begins	Drives resource sponsor endorsement and PARM tasking; no funding commitment yet
Justification and Cost Form (JCF)	PARM and resource sponsor	Allocates modernization dollars and prioritizes hulls; accompanies Change Install planning	Aligns FUNDS-D (engineering) and FUNDS-K (procurement); documents budget line and phasing
ShipAlt Record of Change	Alteration Engineering Agent	Provides authoritative configuration record across hulls and life of the change	Required for configuration audits, warranty, and to show work stayed within authorized scope
Installation Drawing / LAR	Alteration Engineering Agent and installer	Directs production work and testing during the scheduled availability or pier-side install	Must match funded scope; links to MOA/RMMCO package and turn-over tests before closeout

6.2.3 Process Flow and Funding Integration

1. **Identify need.** Fleet feedback, reliability data, or new threats drive a Change Initiation Notice, which is screened for technical feasibility and logistics impacts [104].

2. **Engineer and budget.** The Alteration Engineering Agent develops Ship Check data, installation drawings, and metrics while the PARM aligns FUNDS-D (lifecycle engineering) and FUNDS-K (procurement) appropriations; pilot efforts now allow limited OPN use for private-yard execution when approved by OPNAV N9/N83 [104, 106].

3. **Select availability.** Modernization Work Package Integration (Work Package Integration (WPI)) events match each change to a specific CNO availability, Continuous Maintenance availability, or pier-side AIT window based on lead time material and platform readiness [107].

4. **Install and certify.** AITs execute under a Memorandum of Agreement (MOA) that spells out safety, quality, and turnover requirements; completion data is entered into the NDE to update the authoritative configuration record [106, 107].

Failure to follow this sequence is why unauthorized equipment has been discovered during Board of Inspection and Survey (INSURV) events—a career ender for the responsible EDO.

6.2.4 Execution Enablers and Lessons Learned

Configuration management. Every modernization line item in the Availability Work Package (AWP) must trace back to an approved change document, logistics support analysis, and allowance parts list to avoid sparing gaps [105].

MOA discipline. The MOA between the AIT, RMMCO, the Naval Supervising Activity (NSA), and Ship's Force defines boundary isolations, tag-out responsibilities, testing, and habitability; lessons learned from Consolidated Afloat Networks and Enterprise Services (CANES) and Aegis Weapon System (AWS) installs show that unsigned MOAs invite production and safety conflicts [104, 107].

AIT employment. Use AITs when the task requires repeat installations across multiple hulls, unique Original Equipment Manufacturer (OEM) skills, or when it is more cost-effective to mobilize the capability than to train an entire shipyard trade [106].

6.3 Fleet Maintenance Policy and Implementation

6.3.1 Policy Baseline

OPNAVINST 4700.7M. Establishes the maintenance policy for United States Navy ships: maintain the fleet at the highest practical readiness, perform work at the lowest capable level, and apply Reliability-Centered Maintenance (RCM)/Condition-Based Maintenance (CBM) methods to manage Total Ownership Cost [99].

Office of the Chief of Naval Operations Notice (OPNAVNOTE) 4700. Publishes CNO availability intervals, durations, man-day targets, and Repair Days for each platform, providing the authoritative baseline for the Long-Range Maintenance Schedule (Long-Range Maintenance Schedule (LRMS)) [108].

JFMM and Condition-Based Maintenance Plus (CBM+). USFF/PACFLTINST 4790.3 (JFMM) assigns responsibilities to TYCOMs, RMMCOs, and NSAs, while OPNAVINST 4790.16 issues CBM+ policy so sensors and analytics can replace purely time-based maintenance when justified [105, 109].

Table XVII: Maintenance policy map at a glance

Document	Owner	Decisions Touched	Compliance Talking Points
OPNAVINST 4700.7M	CNO (N9)	Sets fleet maintenance policy baseline (lowest capable level, RCM/CBM, color-of-money discipline)	Use as the anchor for deferral justifications, craft risk acceptance to its hierarchy, and cite when defending public/private yard mix and maintenance levels
OPNAVNOTE 4700	CNO (N9)	Publishes class-by-class availability intervals, durations, manday targets, and repair days that drive the Long-Range Maintenance Schedule	Treat as the authoritative schedule baseline; deviations require TYCOM/CNO alignment and supporting maintenance metrics
JFMM (COMFLTFORCOMINST 4790.3)	USFF/PACFLT	Defines execution standards for work control, assessments, certifications, and availability gate reviews	Align DFS/TSO closure, QA evidence, and RMMCO coordination to JFMM chapters; reference for watch team and AIT roles
CBMP (OPNAVINST 4790.16)	CNO (N4)	Governs condition-based maintenance approvals, data pedigree, and sensor analytics used to replace time-directed tasks	Document the analytics package (sensor health, thresholds, reliability data) and secure CBMP authority approval before removing directed tasks

The entire maintenance enterprise hinges on tying ship assessments to these references; citing a different standard during a board (e.g., OEM manual) without reconciling it to the JFMM and policy hierarchy is a red flag.

6.3.2 Strategy Frameworks

Optimized Fleet Response Plan (OFRP). OPNAVINST 3000.15 drives Fleet Response Plans that align readiness, training, and maintenance phases; maintenance windows in the OFRP billet plan are how TYCOMs defend availability lengths and sequencing [110].

Surface Force Readiness Manual (SFRM). Naval Surface Force Atlantic (SURFLANT)/Naval Surface Force Pacific (SURFPAC) 3502-series guidance aligns training milestones (Basic, Advanced, Integrated) with maintenance health metrics so a ship cannot progress if critical maintenance or material conditions remain open [111].

Naval Sustainment System—Shipyards (NSS-SY). NSS-SY injects commercial best practices (Theory of Constraints, value-stream management, digital dashboards) into the four public yards with the goal of 80% on-time Attack Submarine, Nuclears (SSNs) deliveries and no idle submarine backlog by FY26 [112].

6.3.3 Availability Approval and Execution Discipline

1. **Baseline work.** SURFMEPP/SUBMEPP compile CMP work, TYCOMs repairs, and modernization candidates into the Baseline Availability Work Package (BAWP) at A-36 months; work then flows through the CNO scheduling conference for final approval [108, 112].
2. **Directed vs. condition-based maintenance.** Directed Maintenance Tasks (Directed Maintenance Task (DMT)) must be executed regardless of condition (e.g., structural checks mandated by Submarine Safety Program (SUBSAFE)), while CBM tasks rely on evidence such as oil analysis, smart sensors, or TSRA discoveries to justify deferral [99, 109].
3. **Approval boards.** TYCOMs Availability Work Package Reviews validate that all Departure from Specifications (DFSs) and Temporary Standing Orders (TSOs) have closure plans before endorsing the package for CNO approval [105].
4. **Metrics.** Schedule adherence, manday performance, and quality escape metrics (e.g., repeat maintenance, Objective Quality Evidence (Objective Quality Evidence (OQE)) rejects) appear in USFF/PACFLT readiness briefs and feed the next LRMS update [112].

6.3.4 Trends, Risks, and Talking Points

- **Industrial base mix.** Navy policy seeks roughly a 50/50 public-private mix for surface depot work, with CVNs and SSNs/SSBNs remaining predominantly public-yard to retain nuclear competencies [99].
- **Manpower constraints.** NSS-SY reviews highlight persistent shortages in key trades (nuclear welders, radiological controls, rigging) that drive docking delays; referencing NSS-SY corrective actions shows awareness of current risk mitigations [112].
- **Digital thread.** The NDE modernization module is now the authoritative source for installed configuration, allowing planners to reconcile modernization with maintenance data without resorting to ad-hoc spreadsheets [107].
- **Fiscal discipline.** Maintenance execution must honor color-of-money boundaries (OC&MF vs. OPN/SCN); unauthorized obligation adjustments are ADA violations [27, 51].

6.4 AIT Fundamentals and Lessons Learned

6.4.1 Purpose and Advantages

- **Concentrated expertise.** An AIT is a trained unit (government, contractor, or hybrid) directed by an AIT manager to install specific modernization changes that require unique skills, tooling, or repetitive production-line execution [107, 113].

- **Speed and repeatability.** Using the same team across hulls captures lessons learned, reduces quality escapes, and accelerates delivery of high-priority combat system upgrades (e.g., CANES, AWS, GPS-based Positioning, Navigation, and Timing Service (GPNTS)) without overloading the shipyard workforce [106].
- **Safety and configuration control.** Dedicated teams enforce technical work documents, boundary isolations, and logistics support for each alteration, reducing the risk of unauthorized changes that could violate SUBSAFE or Level I controls [107].

6.4.2 Roles and Responsibilities

Sponsor / PARM. Assigns and funds the AIT, prioritizes hulls, and ensures logistics (allowance updates, training, tech manuals) are resourced before work begins [106].

AIT manager. Leads planning, safety, schedule integration, and quality execution; serves as the single point of contact for the Naval Supervising Activity and Ship's Force [107].

On-Site Installation Coordinator (OSIC). Coordinates day-to-day shipboard actions, manages tag-outs and system turnovers, and resolves emergent issues between the AIT, NSA, and Ship's Force [113].

RMMCO. Screens alteration packages, aligns them with availability windows, and enforces waterfront integration and reporting requirements [107].

NSA/Lead Maintenance Activity (LMA). Controls work authorization, boundary isolations, tests, and Objective Quality Evidence for work accomplished under the availability contract [105].

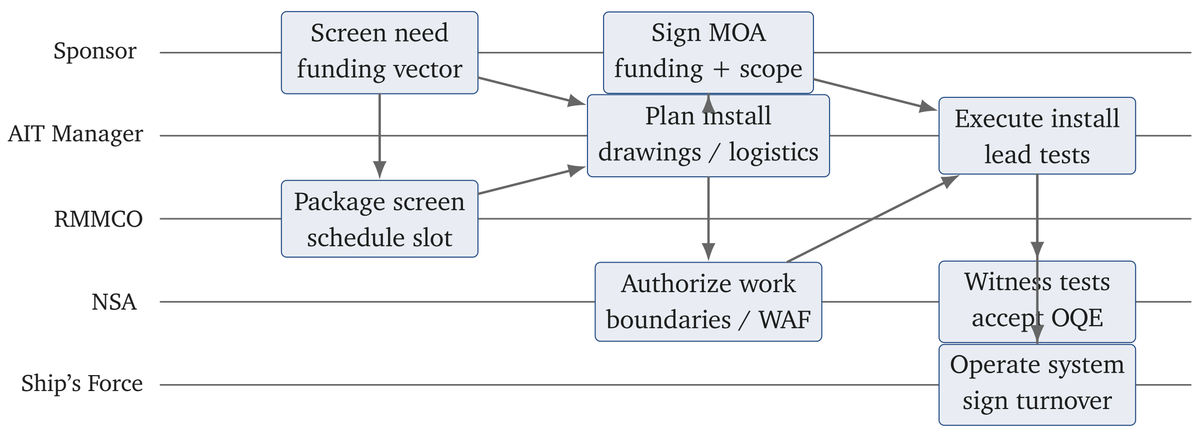


Figure 6.2. *AIT swim-lane showing sponsor, AIT manager, RMMCO, NSA/LMA, and Ship's Force interactions from screening through certification.*

6.4.3 Planning Controls and MOA Discipline

1. **MOA.** The MOA outlines scope, funding, schedule, safety, habitability, quality, testing, and turnover; it must be signed by the AIT sponsor, RMMCO, Ship's Commanding Officer, and NSA before work starts [107].
2. **Installation planning.** Each alteration requires a ship check, logistics certification, and completion of prerequisite changes; the NMP databases confirm that planned work matches the approved Change Document [106].

3. **Work Authorization Form (WAF)/tag-out alignment.** Integration with the Work Authorization Form process and Tag-out Users Manual (TUM) prevents conflicting isolations between the AIT and shipyard trades [114].

4. **Testing.** Entry/exit criteria, including pre-installation checkout, operational tests, and system certifications (e.g., Combat System Ship Qualification Test), are tailored per alteration and witnessed by NSA [107].

6.4.4 Lessons Learned for Board Prep

- **Late drawing changes drive rework.** Lock engineers early—AITs report that late drawing revisions, missing Interface Control Documents, or mismatched configuration data cause most delays [113].

- **Habitability and safety must be priced.** Many AITs arrive without berthing or industrial hygiene plans; ensuring the MOA captures services, hot work permits, and fall protection prevents emergent cost growth [106].

- **Document close-out.** Uploading OQE, system configuration updates, and post-install logistics changes to the NDE is mandatory—omitting them erodes Fleet supply support [107].

6.5 Waterfront Organizations, Work Controls, and Safety

6.5.1 Naval Propulsion Program Oversight

Naval Nuclear Propulsion Program (Naval Reactors). Established by executive order and statute, Naval Reactors owns cradle-to-grave authority for nuclear propulsion design, maintenance, training, and operations; they enforce uncompromising safety rules regardless of warfighting urgency [115].

Naval Reactors Representative Office (NRRO). Resident Naval Reactors Representative Offices monitor each nuclear-capable waterfront, enforce program standards, and can halt work if safeguards are threatened; they are independent of yard leadership to prevent production pressure from eroding safety [115].

Reactor Plant Contractor's Office (RPCO). Reactor Plant Contractor Offices represent the prime contractors (e.g., Newport News Shipbuilding) and ensure contract compliance and technical rigor during construction, refueling, and depot maintenance [115].

Joint Test Groups. Department of Energy (DOE) and Navy personnel lead integrated test teams that certify plant reconstitution before turnover; they align with NRRO, SUPSHIP, and Ship's Force to execute rigorous testing sequences [115].

Board insight: referencing the Thresher, Guitarro, and Miami investigations underscores why NRRO authority and independent QA cannot be waived, even under combat timelines.

6.5.2 SUBSAFE and Submergence Systems Controls

- **SUBSAFE.** NAVSEA 0924-062-0010 mandates design, material, process, and test controls for the seawater boundary and critical systems on submarines; any work that touches a SUBSAFE boundary requires certified Level I material, approved procedures, and documented OQE [116].
- **Submergence Systems Certification.** Submergence Systems Certification manual series (SS800) manuals reinforce submarine rescue and diver support equipment requirements; the Submergence Systems Certification Board mirrors SUBSAFE rigor for non-platform systems [117].
- **Level I / critical material.** NAVSUP/NAVSEAINST 4440.16 controls procurement, receipt, and storage of Level I and SUBSAFE material to prevent counterfeit or degraded items from entering the supply chain [118].
- **Industrial safety.** S0570-AB-SAF-010 (fire protection) and S9002-AB-MAN-010 (submarine industrial safety) govern hot work, confined space, and damage control readiness during availabilities [119, 120].

6.5.3 Work Authorization and Tag-out Discipline

1. **WAF.** The NSA/LMA generates WAFs to define scope, isolations, and boundary conditions; all trades and AITs must align their work with the active WAF or request revisions through the Project Manager [105].
2. **Tag-out Users Manual.** S0400-AD-URM-010 prescribes red/amber tags, double barrier requirements, and control authority; the Ship's Force owns the tag-out log even when a contractor executes the work [114].
3. **Special evolutions.** Diving, radiological work, and ordnance moves require additional approvals (e.g., Naval Sea Systems Command Supervisor of Salvage and Diving Supervisor of Salvage and Diving (SUPSALV), Naval Ordnance Safety); EDOs must ensure these are embedded in the availability plan [115, 121].

6.5.4 Waterfront Organization Insights

- **Unified waterfront teams.** Naval Shipyards, Regional Maintenance Centers, RMMCOs, and Ship's Force form integrated production and test teams; weekly Project Management Reviews should cover safety, quality, schedule, and funding status [115].
- **Culture of questioning attitude.** The Naval Propulsion Program teaches that any worker can stop a job; referencing the Miami (SSN-755) fire (shipyard worker's vacuum) shows how minor lapses escalate without that culture [115].
- **Documentation.** Work packages, technical work documents, and checklists must be updated in the NDE to preserve traceability for SUBSAFE and other certifications [116].

6.6 Battle Damage Assessment and Repair

6.6.1 Strategic Context

- **Global competition.** Renewed great-power competition with the People's Republic of China (PRC) and Russia (plus persistent Democratic People's Republic of Korea (DPRK) provocations) means distributed maritime operations must survive massed fires and cyber/electromagnetic attacks [122, 123].
- **Wartime Acquisition Response Plan (Wartime Acquisition Response Plan (WARP)).** SECNAV's WARP memo directs every SYSCOM/PEO to maintain plans, contracts, and trained people to pivot from peacetime acquisition to wartime recovery of ships and systems [124].
- **GAO findings.** GAO-21-246 warns that current Battle Damage Assessment and Repair (BDAR) capabilities are insufficient for contested logistics, highlighting gaps in expeditionary repair, pre-positioned materiel, and command-and-control [125].

6.6.2 BDAR Continuum

Damage Control. Ship's Force fights fires, floods, and casualties to stabilize the platform, preserve combat power, or keep it afloat until assistance arrives [123].

Rescue and Assistance (R&A). Military Sealift Command (Military Sealift Command (MSC)) tugs, Mobile Diving and Salvage Units (Mobile Diving and Salvage Unit (MDSU)), and SUPSALV assets conduct emergency towing, underwater repairs, object retrieval, and environmental mitigation [121].

Damage Assessment. Ship's Force, salvage teams, and ashore Damage Assessment Teams from RMCs, NSYs, or SEA 21 analyze hull integrity, stability, and mission systems to recommend disposition [123].

Expeditionary Repair. Forward Deployed Ship Repair Team (FDSRT) and Expeditionary Maintenance and Repair Facility (EMRF) deploy with Shop-in-a-Box (SIB) sets to restore mobility or key mission systems without returning to a depot [123, 126].

Depot Repair / Disposition. NSYs, RMCs, and private yards perform major repairs, partial restorations, or decide to scrap; choices depend on threat, capability needs, cost, and industrial capacity [123].

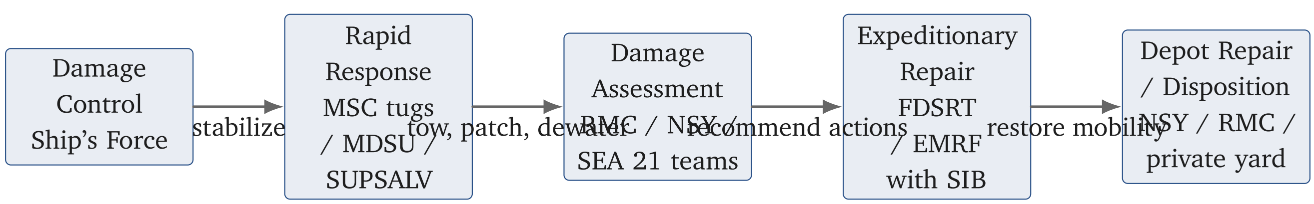


Figure 6.3. BDAR continuum from damage control through expeditionary repair to depot-level recovery, with primary responsible organizations at each step.

6.6.3 Command, Control, and Technical Authority

- **Operational control.** Fleet commanders designate BDAR task forces; Task Force logistics commanders orchestrate salvage, FDSRT, and EMRF employment while coordinating with theater Joint Logistics Enterprise nodes [127, 128].

- **Technical authority.** Wartime does not suspend Technical Authority—NAVSEA, Naval Reactors, and Warfare Centers still adjudicate waivers; however, WARP allows pre-planned delegated authorities and acceptance of graded risk when documented [124].
- **Reporting.** CASREP 4.0 timelines and Battle Damage Assessment reports flow through OPNAVINST 3040.2B to inform reconstitution priorities and Congressional notifications [129].

6.6.4 Logistics, Contracting, and Fiscal Readiness

Prepositioned materiel. Maritime Support Plan (MSP) force packages require pre-kitted SIBs, hull patches, and mobile power kits staged throughout the Indo-Pacific, United States European Command (USEUCOM), and United States Central Command (USCENTCOM) theaters [126].

Contracting. FAR Part 18 and DFARS Part 218 emergency authorities (e.g., oral solicitations, undefinitized contract actions, Defense Priorities and Allocations System ratings) enable rapid award, but commands must still document justification and maintain traceability for later definitization [51, 55].

Funding. BDAR relies on OMN for emergent repairs, SCN for major restorations, and Weapons Procurement, Navy (WPN) when combat system replacement constitutes procurement; Department of War (DoW) FMR Volume 3 prescribes reprogramming thresholds [27].

Supply chain resilience. The Resilient Maritime Logistics initiative (Project 33) aligns OPNAV N4, Fleet Logistics Centers, and Defense Logistics Agency to keep people, parts, and processes flowing under contested logistics [122, 123].

6.6.5 Maritime Support Plan Force Packages

1. **Afloat Augmentation.** 25–50 artisans augment the shipboard IMA to enable 24/7 casualty response within a strike group [126].
2. **Afloat Independent Maintenance Activity.** Tender-like capability distributed across vessels of opportunity that integrates EMRF, FDSRT, and logistics cells under Task Force Operational Control (OPCON) [126].
3. **Ashore Lean / Robust / Austere.** Scalable shore nodes that combine contractor oversight teams, NAVSEA contracting cells, and host-nation support to execute depot-equivalent work forward [126].

EDOs should be ready to explain which force package they would recommend for a given scenario, how long it takes to activate, and how technical authority, contracts, and funding are synchronized.

6.7 CNO Availability Planning

6.7.1 Balancing Requirements, Resources, and Readiness

Customers. Fleet commanders and TYCOMs drive capability requirements and availability length based on the Optimized Fleet Response Plan and readiness gaps [110, 130].

Planners. Shipyards, RMCs, and planning activities must sequence work, tooling, infrastructure, and LLTM so the schedule is executable [130].

Resources. Budgets (OC&MF, OPN, FUNDS-D/FUNDS-K), workforce, and pier/dock capacity must align with CNO-directed Repair Days [108].

EDOs should articulate how they reconcile conflicting demands—for example, deferring a lower-criticality modernization so mandatory SUBSAFE work fits within the allotted man-days.

6.7.2 Availability Work Package Development

1. **Baseline (A-36 to A-24).** SURFMEPP/SUBMEPP derive the BAWP from the CMP, TYCOMs repairs, and modernization candidates; Ship Checks, Condition Found Reports, and TSRA results refine the package [105, 130].
2. **Initial Planning (A-21).** Planning Activities (Carrier Planning Activity, Submarine Planning Activity, RMCs) reconcile the BAWP with Class Plan priorities and issue the Initial Availability Work Package (AWP) [130].
3. **Final Planning (A-12 to A-3).** Results of pre-availability tests (Pre-Availability Test (PATS)/Pre-Overhaul Test (POTS)), DFSs, TSOs, and new modernization approvals are adjudicated; the final AWP is issued no later than three months before A-0 [108, 130].
4. **Contracting.** For surface private-yard availabilities, Multiple Award Contract, Multi-Order (MAC-MO) constructs increase competition and flexibility; public yards rely on Advanced Industrial Management (AIM) processes [130].

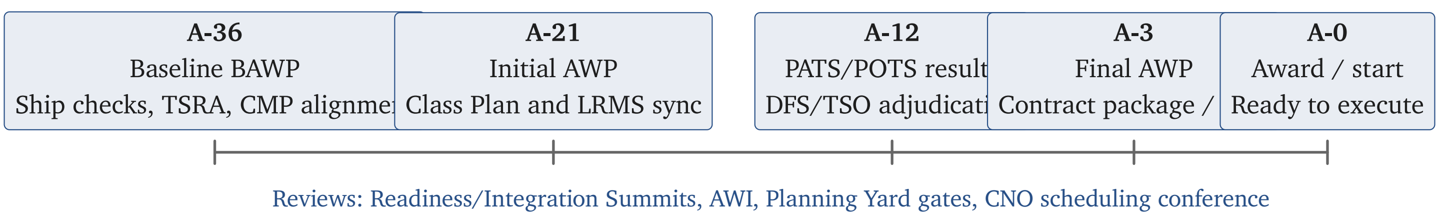


Figure 6.4. CNO availability planning timeline (A-36 through A-0) with key reviews and deliverables.

6.7.3 Planning Teams and Integration Forums

- **Carrier Team One.** Aligns NAVSEA 08/04/21, NAVAIR, Newport News Shipbuilding, and Fleet stakeholders to drive aircraft carrier availability excellence [130].
- **Surface Team One.** Unites SEA 21, SEA 05, SURFMEPP, TYCOMs, and PEOs to tackle systemic issues (e.g., late drawings, material delays) across destroyer/cruiser/amphibious availabilities [130].

- **Submarine Team One.** Led by PMS 392/SUBMEPP, integrates CNRMC, NAVSEA 08/21, and public yards to sustain SSNs/SSBNs availabilities and implement NSS-SY initiatives *(Appendix C.2: RDML Dan L. Lannamann holds the CNRMC billet)* [130].
- **Planning summits.** Readiness Summits, Availability Work Integration (Availability Work Integration (AWI)) events, and Integrated Production Planning (Integrated Production Planning (IPP)) meetings synchronize modernization, maintenance, logistics, and testing products [130].

6.7.4 Risk Controls and Board-Ready Talking Points

Critical path management. Identify the top three critical path evolutions (e.g., tank blasting/coating, propulsion plant maintenance, combat system rip-outs) and ensure early start requirements—barge services, LLTM, training—are executable before A-0 [130].

Material readiness. Demand 100% receipt of LLTM and government-furnished material before award; highlight mitigation plans for supply-chain risk (dual sources, cannibalization, additive manufacturing) [130].

Growth vs. new work. Explain how your team budgets anticipated growth work (historical allowances) versus truly new work (unique casualty), as well as the screening process for adding either to the contract [105].

Metrics. Use NSS-SY dashboards (Earned Value, manday burn, schedule adherence) and weekly risk reviews to show how you keep leadership informed and trigger help early [130].

6.8 CNO Availability Execution

6.8.1 Project Teams and Responsibilities

Shipyard team. The Project Superintendent (overall cost/schedule/safety owner), Project Manager (budget, earned value, planning integration), Assistant Project Superintendents (zone leads), Engineering & Planning (technical work documents, boundaries), and Business & Planning (funding, schedule coordination) form the core team [131].

RMC/private-yard team. The Port Engineer, Ship's Maintenance Manager (Ship's Maintenance Manager (SMM)), Ship's Force Maintenance Officer, TYCOM representative, Contracting Officer, and contractor Project Manager manage planning yard availabilities [131].

Ship's Force. Commanding Officer (CO)/Executive Officer (XO)/Chief Engineer (CHENG) maintain safety, tag-out, and configuration control; Combat Systems and Supply Officers coordinate tests, Shipboard Operational Verification Tests (SOVTs), and material receipt [105].

Specialists. Integrated Test Engineers, Project Support Engineers, Quality Assurance Specialists, Alteration Coordinators, and Administrative Contract Specialists provide focused oversight on testing, configuration, QA, and change management [131].

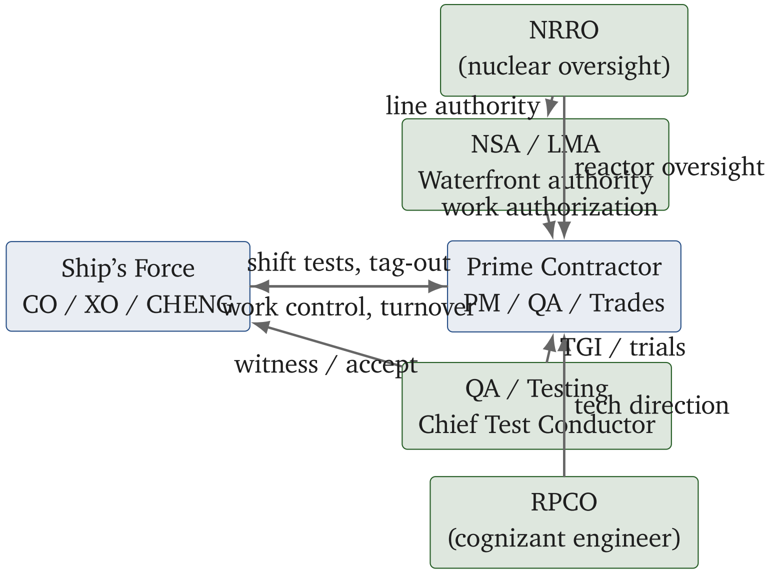


Figure 6.5. Nuclear waterfront project-team interactions across NRRO, RPCO, NSA/LMA, Ship's Force, and the prime contractor.

6.8.2 Execution Phases

1. **Nesting / Arrival.** Establish temporary services, fire watches, habitability, industrial hygiene controls, and turnover from Ship's Force to the Maintenance Activity; complete mandatory briefings (JFMM Chapter 2, Tag-out Users Manual) [131].
2. **Rip-out & Access.** Remove interferences, open systems, and conduct initial WAF reviews; late identification of hazardous material and access conflicts is a top schedule killer [105].
3. **Open & Inspect.** Execute inspections (tank, structural, piping) and document growth work; per NAVSEA Standard Item 009-01, inspections should be 100% complete by A+20% to preserve schedule margin [131].
4. **Repair & Production.** Complete component tasks, manage new vs. growth work approvals, and track critical path milestones using Task Group Instruction (TGI) [131].
5. **Testing, Trials, Certification.** Execute compartment turnover, Light-Off Assessment (LOA), SOVT, Combat System Ship Qualification Trial (CSSQT), Builder's/Acceptance Trials, and Sea Trials before Production Completion Date (PCD) [105].

Table XVIII: CNO availability execution phases

Phase	Key Deliverables	Controlling Documents	Responsible Leads
Nesting / Arrival	Temporary services, fire watch plan, habitability, turnover briefs, tag-out baseline	JFMM Chapter 2; Tag-out Users Manual; local safety manuals	Project Superintendent, Safety Officer, Ship's Force Department Heads
Rip-out & Access	Interference removal, WAF approvals, confined space/hazardous material plans	WAF/Tag-out procedures; hot work and gas-free engineering instructions	Zone Assistant Project Managers, Production Shop Leads, Gas Free Engineer

Continued on next page

TABLE XVIII: CNO AVAILABILITY EXECUTION PHASES **(Continued)**

Phase	Key Deliverables	Controlling Documents	Responsible Leads
Open & Inspect	Inspection reports, growth work candidates, material condition snapshots	Standard Item 009-01; QA/inspection instructions; class maintenance plans	Engineering & Planning, Quality Assurance, Class Maintenance Planner
Repair & Production	Completed work items, change control (JSNs/contract mods), schedule metrics, resource leveling	Contract/SOW; work package and change control processes; integrated master schedule standards	Project Manager, Port Engineer/KO, Business & Planning, Contractor PM
Testing, Trials, Certification	Test agendas, TGI alignment, LOA/SOVT/CSSQT completion, sea trial cards, turnover certificates	JFMM testing chapters; platform test manuals; INSURV/Builder's trial requirements	Chief Test Conductor, Ship's Force Test Coordinator, NSA/LMA witnesses

6.8.3 Governance, Quality, and Work Control

- **NSA / LMA.** Certifies that work is authorized, technically compliant, and properly tested before acceptance; reviews OQE prior to signing completion certificates [105].
- **Work control.** WAFs, TGI sequencing, and boundary tags must remain synchronized; growth work requires screening panels (Ship's Force, TYCOM, KO) to verify funding, schedule, and technical impacts [131].
- **New vs. growth work.** New work is undiscovered scope added post-award; growth work expands existing items. Both demand TYCOM approval, funding source identification, and contract mods (or Job Sequence Number (JSN) adjustments for public yards) [105].
- **Testing governance.** Chief Test Conductors manage test agendas; engineering agents and Ship's Force witness tests, and discrepancies must be adjudicated before certification events (e.g., LOA, CSSQT) [131].

6.8.4 Lessons Learned

- **Momentum matters.** Availability performance often collapses after undocking; maintain daily production meetings, emphasize housekeeping/safety, and enforce material staging to avoid second-half slumps [131].
- **Communication is a weapon system.** Cross-functional representatives (yard, Ship's Force, TYCOM, QA, Supply) must keep each other informed about configuration, testing, and material risks; unresolved disconnects manifest as trial failures [131].
- **Document everything.** Capture change incorporation, OQE, and lessons in the NDE immediately—waiting until the end invites errors and delays closeout [105].

7 Civilian Personnel

7.1 Civilian Personnel Fundamentals

7.1.1 Why CIVPERS management matters

Workforce weight. Civilian Personnel (CIVPERS) form the majority of the engineering and logistics talent at SYSCOMs and NWCs; how you engage and develop them directly sets Fleet readiness outcomes [132].

Human Resources Office (HRO) lanes. Core services are staffing, classification/compensation, employee relations, and labor relations; pull them in early for hiring plans, performance actions, and workplace changes [132].

Guidance stack. Primary references flow from Title 5, Office of Personnel Management (OPM) guidance, Department of the Navy policy, and local agreements; document assumptions and keep counsel/HRO informed to avoid missteps [132].

7.1.2 Workforce shaping and classification

Pay systems. Know the differences between General Schedule (GS), Wage Grade (WG), pay-band systems, and Senior Executive Service (SES) tiers; align the billet to the right system before recruiting [133].

Classification and Fair Labor Standards Act (FLSA). The Position Description (PD) drives pay plan, series, and grade/band; verify FLSA exempt/non-exempt status and ensure duties match the description to avoid backpay claims [133].

Filling vacancies. Compare reassignment/merit promotion, Direct Hire Authority (DHA), or competitive hiring; honor mandatory placement programs and capture recruiting timelines up front [133].

Downsizing levers. Sequence Reduction in Force (RIF), Separation Incentive Program (SIP), Voluntary Early Retirement Authority (VERA), and furlough options carefully; pair with workforce development to retain critical skills while meeting budget constraints [133].

7.1.3 Equal Employment Opportunity anchors

Bases and theories. The ten protected bases (race, color, religion, retaliation, sex, national origin, equal pay, age, disability, genetic information) and theories (disparate treatment, disparate impact, harassment, reprisal) frame every Equal Employment Opportunity (EEO) review [134].

Process discipline. Preserve timelines: timely contact with a EEO counselor, alternative dispute resolution where appropriate, investigation, and decision/hearing; supervisors must avoid reprisal and secure records [134].

Harassment prevention. Distinguish hostile environment from quid pro quo scenarios, act on any notice (verbal or written), and coordinate with HRO/Equal Employment Opportunity Commission (EEOC) liaisons for training and documentation [134].

7.1.4 Employee relations essentials

Work schedules and leave. Use standard, flexible, or Compressed Work Schedule (CWS) structures thoughtfully; apply annual/sick leave and leave without pay rules consistently and reinforce timekeeping discipline to avoid Absent Without Leave (AWOL) issues [135].

Performance cycle. Set clear standards early, conduct midpoint feedback, and close out on time; tailor awards (time-off, cash, quality step increases) to reinforce desired behaviors [135].

Conduct v.s. performance. Separate misconduct (e.g., AWOL, falsification) from poor performance; use counseling and a documented Performance Improvement Plan (PIP) before escalating to adverse action, and keep counsel/HRO informed [135].

7.1.5 Within-grade increases and GS step timing

Acceptable performance gates. Within-Grade Increases (WGIs) require an "acceptable level of competence" determination (typically a "Fully Successful" rating); maintain clear standards and timely appraisals so eligibility decisions are defensible [136].

Table XIX: GS within-grade increase waiting periods

Advancement to steps	Required waiting period at previous step
Steps 2, 3, and 4	52 weeks (1 year)
Steps 5, 6, and 7	104 weeks (2 years)
Steps 8, 9, and 10	156 weeks (3 years)

Assuming sustained acceptable performance, advancing from Step 1 to Step 10 takes 18 years (three years for Steps 1–4, six for Steps 4–7, and nine for Steps 7–10) [136, 137]. Exceptional performers may receive a Quality Step Increase (QSI), which moves them one step ahead of the normal schedule [136].

7.1.6 Performance v.s. conduct pathways

Supervisor baseline. Set and communicate standards, evaluate against them, and partner with HRO early; use coaching to reinforce desired behaviors [135].

Table XX: Differentiating performance and conduct issues

Aspect	Unacceptable performance	Misconduct
Definition	Failure to perform assigned duties at an acceptable level	Violation of a rule, law, or regulation (e.g., policy violations, absence without leave (AWOL), insubordination, security breaches)
Focus of action	Improve performance; not intended to punish	Correct behavior; penalties may be more severe depending on the offense
Standard of proof	Substantial evidence	Preponderance of the evidence (higher standard than performance)
Guiding principles	Often involves a structured performance improvement plan (PIP)	Penalties must be just and reasonable; apply the Douglas Factors when selecting a penalty

Handling unacceptable performance. Start with counseling and training, then use denial of a scheduled WGI if performance remains below standard. A formal PIP (often about 90 days) documents required improvements; persistent failure can lead to change to lower grade or removal, with HRO coordination at each step [135, 137].

Addressing misconduct. Apply progressive discipline calibrated to the offense: verbal or written counseling, reprimand, suspension, change to lower grade, or removal. The chosen penalty must be reasonable in light of the Douglas Factors (offense severity, job level, record, consistency, rehabilitation potential, and mission impact) [135, 137].

Employee rights and documentation. Honor the Weingarten right to union representation during investigatory interviews that could lead to discipline; if representation is requested, either provide it, postpone, or end the interview [138]. Keep actions non-discriminatory, coordinate on any EEO complaints, and document counseling, training, and decisions to support subsequent actions [134, 135].

7.1.7 Labor relations touchpoints

Bargaining unit scope. Confirm who is in the bargaining unit (non-supervisory, non-confidential employees) and what the collective bargaining agreement covers; track official time and steward access obligations [138].

Management rights v.s. bargaining. Management retains the right to assign work, direct employees, and determine budget; bargain procedures and impact when changing conditions of employment and document notice-to-union steps [138].

Past practice and change-of-command. There is no grace period for a new Officer in Charge (OIC); changes to established bargaining-unit conditions must satisfy the Federal Labor Relations Authority (FLRA) past-practice test (clear, consistent, long-standing, mutually accepted) and trigger notice-and-bargain steps before implementation. Use the required within-90-day Command Climate Assessment (CCA) to surface desired changes early with counsel/HRO to avoid implied acceptance of prior practices [139, 140].

Meetings and disputes. Provide the union a chance to attend formal discussions and Weingarten meetings; log official time and resolve Unfair Labor Practices (ULPs) through grievance procedures or FLRA channels [138].

Appendices

A Platinum Card

The *Platinum Card* is a primary study tool that will have most of the information required for your boards. Sources: EDO Coursebook Modules 3.1.1, 3.1.3, 3.1.4, and 3.1.6. Be able to draw and talk through Figures A.1–A.9.

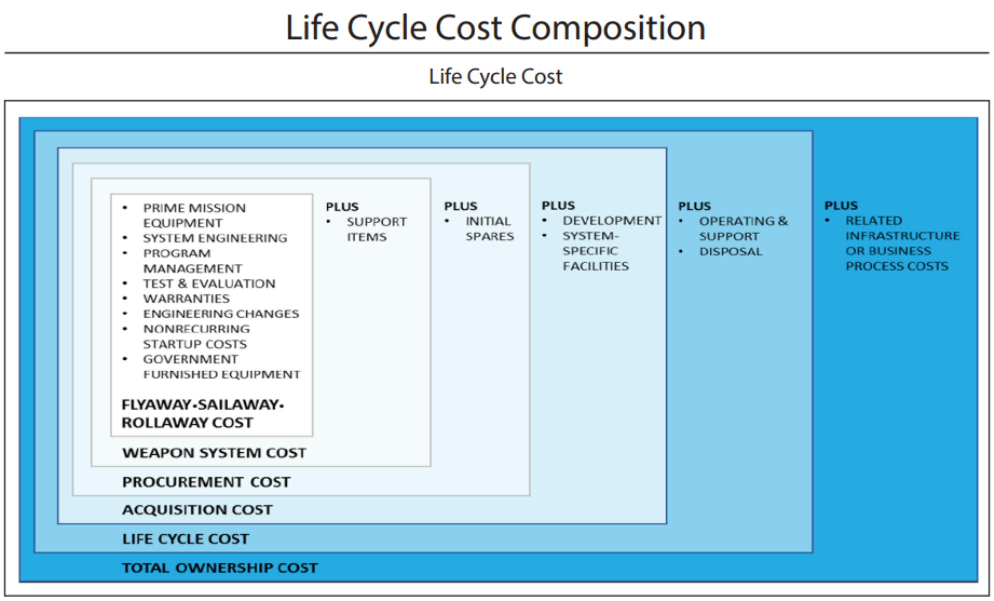


Figure A.1. *Platinum Card: Life Cycle Cost Composition*

Cost Estimating Requirements

	ACAT IB, IC & ID (MDAP)	ACAT IA
POE	Program initiation & all subsequent milestones, including FRP DR	Program initiation & all subsequent milestones - Includes Economic Analysis
CARD	MS A & all subsequent milestones including FRP DR - Draft 180 days prior to OIPT/milestone - Final 45 days prior to OIPT/milestone	MS A & whenever Economic Analysis is required - Draft 180 days prior to OIPT/milestone - Final 45 days prior to OIPT/milestone
CCE	MS A and all subsequent milestones including FRP DR	MS A and whenever an Economic Analysis is required

ICE Required by law for all MDAP programs *
- Prepared by OSD CAPE for ACAT ID, and ACAT IC at discretion of USD(A&S)
- Prepared by component cost agency (AFCAA, DASA-CE, NCCA) for ACAT IC and ACAT IB (if no CAPE estimate)
- In advance of any certification under Title 10, USC, Section 2366a (MS A) and Section 2366b (MS B)
- In advance of any decision to enter low-rate initial production (LRIP) (MS C) or full-rate production (FRP DR)
- In advance of any certification of MDAPs that experience critical cost growth (Title 10, USC, Sec 2433a)

*ICE statutory requirement (Title 10, US Code, Section 2434) and P.L. 111-23, May 22, 2009
Source: DoDI 5000.02 January 7, 2015, Incorporating Change 3, August 10, 2017 and Weapon Systems Acquisition Reform Act of 2009

ACAT II & ACAT III: POE (and, at MDA discretion, an independent cost estimate prepared by the component cost agency) at program initiation and all subsequent milestones

AFCAA = Air Force Cost Analysis Agency
ACAT = Acquisition Category
CAPE = Cost Assessment & Program Evaluation
CARD = Cost Analysis Requirements Description
CCE = Component Cost Estimate
DASA-CE = Deputy Assistant Secretary of Army (Cost & Economics)
FRP DR = Full Rate Production Decision Review
ICE = Independent Cost Estimate
MDA = Milestone Decision Authority
MDAP = Major Defense Acquisition Program
NCCA = Naval Center for Cost Analysis
OIPT = Overarching Integrated Product Team
POE = Program Office Estimate
USD(A&S) = Under Secretary of Defense (Acquisition and Sustainment)

Figure A.2. *Platinum Card: Cost Estimating Requirements*

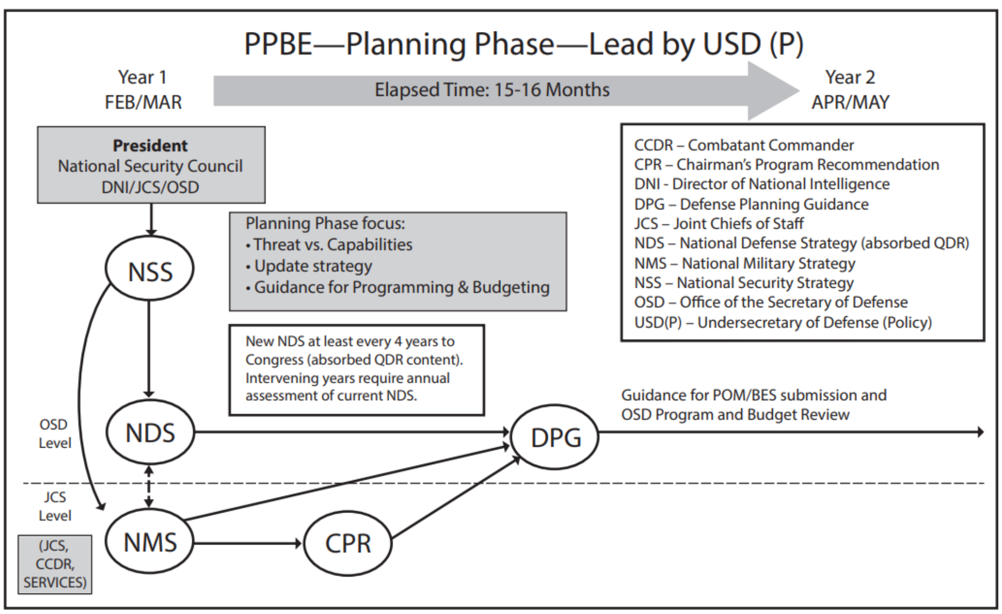


Figure A.3. *Platinum Card: PPBE Planning Phase (led by Under Secretary of Defense for Policy (USD(P)))*

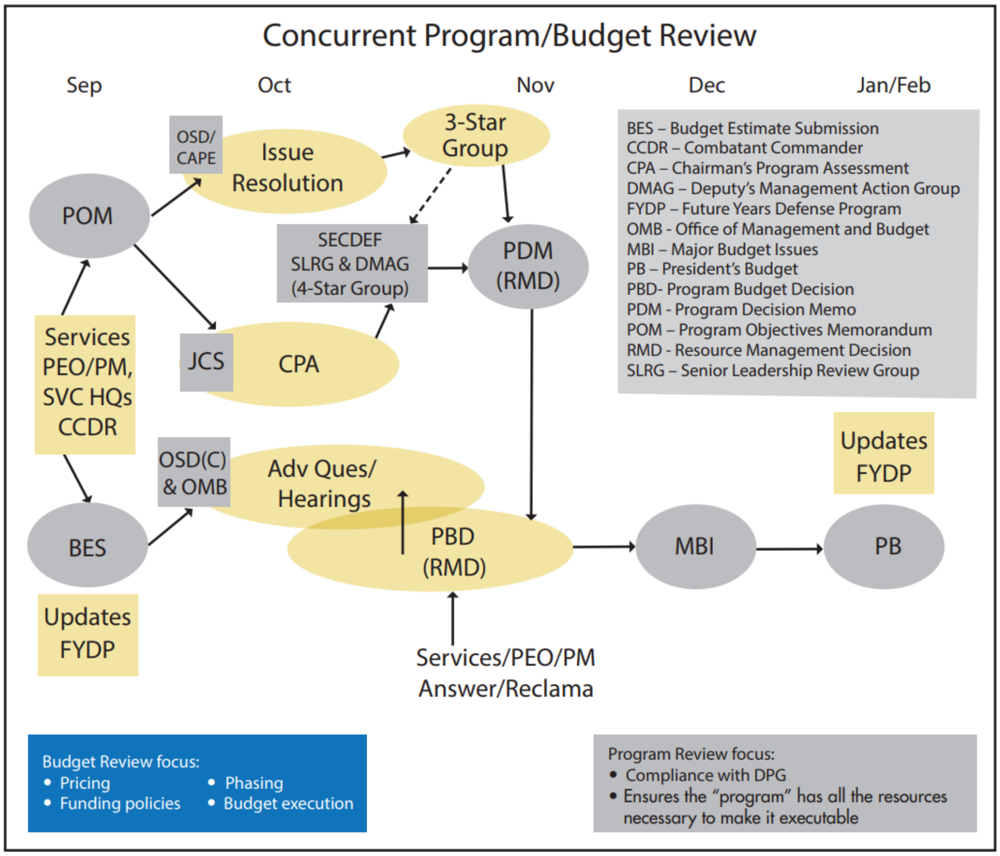


Figure A.4. *Platinum Card: Concurrent Program/Budget Review*

Below Threshold Reprogramming

Amounts are cumulative over Entire Period of Obligation Availability

APPRN	MAX INTO	MAX OUT	LEVEL OF CONTROL	OBL AVAIL
RDT&E	Lesser of +$15M or + 20%	Lesser of - $15M or - 20%	Program Element	2 Years
PROC	Lesser of + 15 M or + 20%	Lesser of - $15M or - 20%	Line Item	3 years SCN: 5 Years
O&M	+ $15M	- $15M	Budget Activity (or Defense Agency) Some Sub-Activity Limitations on Decreases (see reference below)	1 Year
MILPERS	+ $15M	- $15M	Budget Activity	1 Year
MILCON	Lesser of +$2M or + 25%	No Specific Congressional Restriction	Project	5 Years

Reference Sources: DoDFMR 7000.14-R, Volume 3, Chapter 6 (Sept 2015) and Chapter 7 (Mar 2011) Joint Explanatory Statement of Congress Making Appropriations for the Department of Defense for Fiscal Year 2024

Figure A.5. *Platinum Card: Below Threshold Reprogramming*

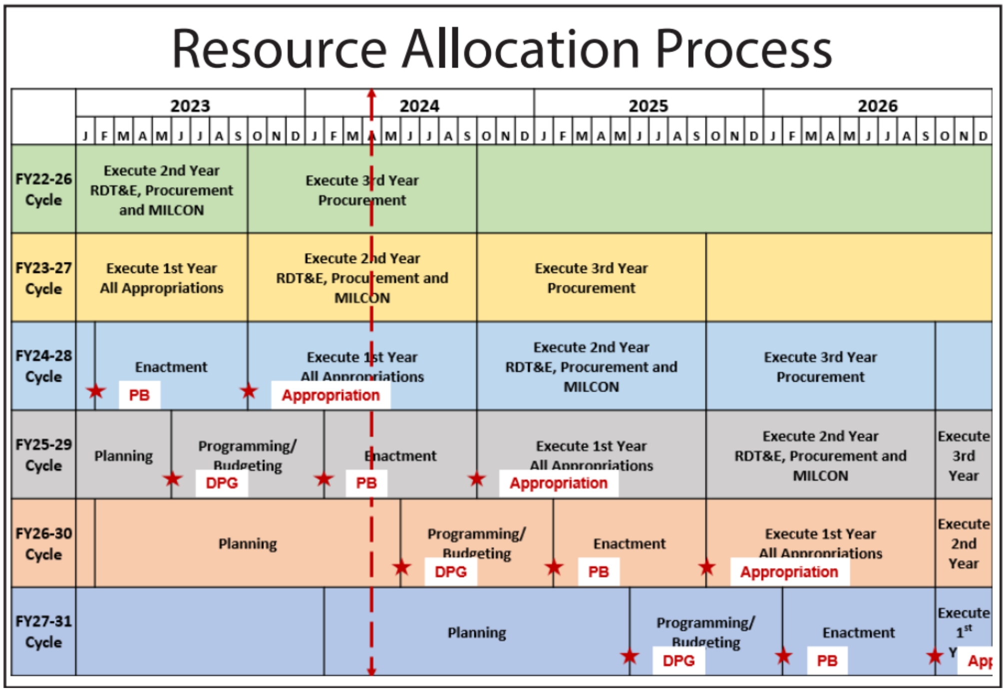


Figure A.6. *Platinum Card: Resource Allocation Process*

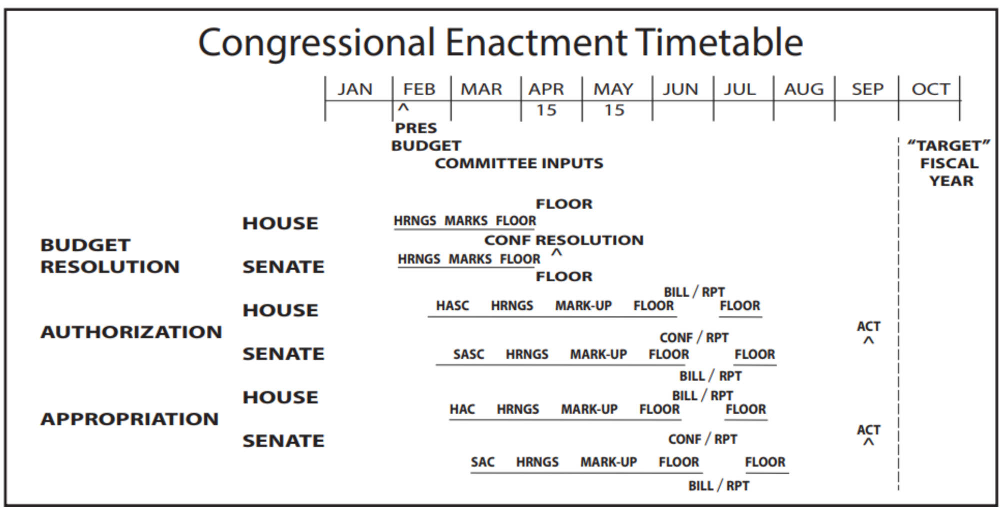


Figure A.7. *Platinum Card: Congressional Enactment Timeline*

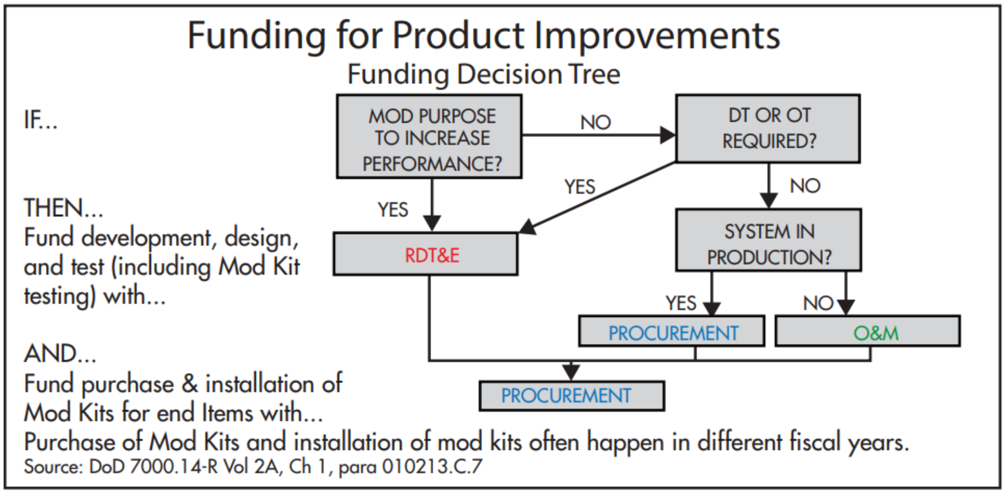


Figure A.8. *Platinum Card: Funding for Product Improvements*

RDT&E. Engineering changes requiring development/test.

Procurement. Production/installation of approved mods/end-items.

O&M. Minor mods/installation labor when authorized and not creating a new end-item.

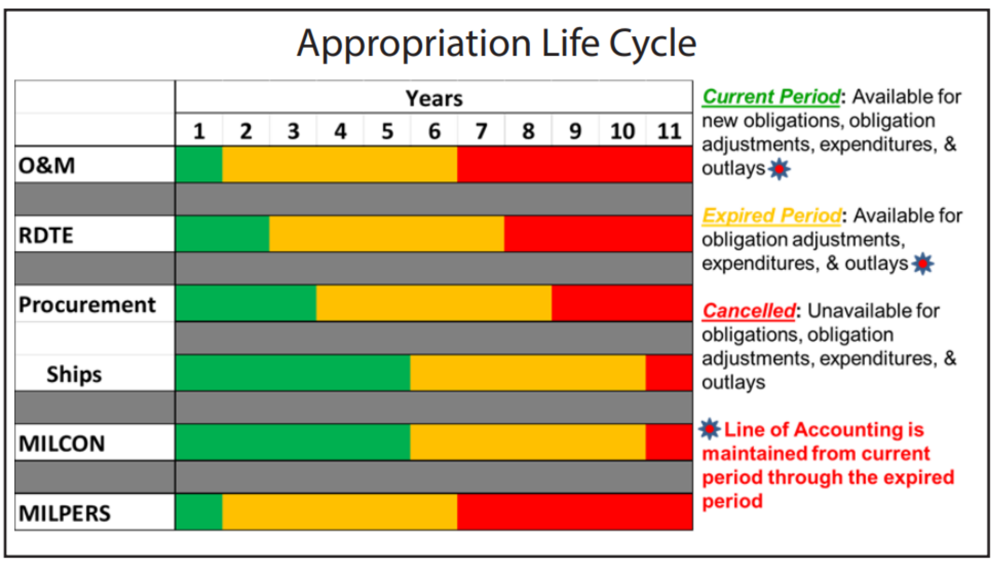


Figure A.9. *Platinum Card: Appropriation Life Cycle*

B Key Roles in Acquisition and Modernization

B.1 Key Roles in Acquisition and Modernization

Department acquisition governance.

Sources: SECNAVINST 5400.15D; SECNAVINST 5000.2G; DoDI 5000.85.

- ASN(RD&A), the SAE and CAE for the DON, sets acquisition policy, charters PEOs / DRPMs, and adjudicates milestone decisions when delegated by the Milestone Decision Authority.
- PEOs and DRPMs hold delegated program authority, establish baselines, and are accountable for translating capability needs into executable acquisition strategies.
- SYSCOM Commanders provide the workforce, infrastructure, and TA warrants that underpin PEO execution inside the Navy matrix construct.

Joint requirements governance.

Sources [80, 88].

- **JROC:** Validates joint capability requirements, prioritizes gaps, and issues JROCM direction that becomes binding input to Gate 2 and milestone documentation.
- **Joint Staff Gatekeeper/JCB:** Controls the KM/DS queue, assigns Joint Staffing Designators, and routes most Navy submissions for JCB-level validation to keep JROC time focused on the highest-risk issues.
- **FCBs:** Chair certification working groups (Net-Ready, Force Protection, System Survivability), monitor DOTMLPF-P implementation, and elevate Configuration Steering Board actions when requirement trades alter program baselines.
- **PAEs and CTCs:** The 2025 Acquisition Transformation Strategy empowers PAEs to convene Capability Trade Councils that record requirement/cost/schedule trades in real time so PEO teams keep the DAS synchronized with evolving demand signals.

Combatant command sponsors.

Sources [1, 88].

- **CCDRs** own warfighting problems, sign JUONs/JEONs, and direct lead Services or agencies to prosecute capability gaps that demand rapid solutions.

- **CCMD staffs** conduct mission-thread analysis and architecture sprints with Fleets, labs, and industry to translate operational risk into measurable capability requirements before the Gatekeeper accepts a package.
- Commanders remain voting participants during validation reviews and oversee fielding to ensure delivered capability solves the theater problem that generated the requirement.

Joint Chiefs leadership.
Sources [1, 88, 90].

- **CJCS:** Principal military adviser to the President/SECWAR, co-chairs the JROC, and leads joint concept and readiness development that frames every Capability-Based Assessment.
- **VCJCS:** Chairs the JCB, co-leads the DMAG, and arbitrates cross-Service requirement or resourcing disputes before they escalate to the Deputies or Secretary level.

OPNAV corporate leadership.
Source [1].

- **CNO:** Serves as Service Chief, chairs the Navy corporate board, and approves Navy positions for Gate 0/1 decisions before submissions move to the Joint Staff.
- **OPNAV resource sponsors (N8, N82, N9, N91):** Translate Fleet priorities into the POM/FYDP, enforce two-pass/seven-gate rigor, and synchronize requirements, budgets, and PEO execution.

Acquisition decision authorities.
Sources [3, 52, 67].

- **DAE / USW(A&S):** Issues acquisition policy, chairs the Defense Acquisition Board, and retains MDA for ACAT ID/IAM programs unless delegated.
- **SAE/ASN(RD&A):** Assigns Navy MDA for ACAT II and below, approves Service-tailored governance (e.g., two-pass/seven-gate), and charters PEOs/DRPMs.
- **Delegated MDA (PEO/DRPM):** Signs Acquisition Decision Memoranda, manages Configuration Steering Boards, and keeps KPP/APB trades synchronized with requirements direction and statutory certifications.

Program Authority chain.
Sources: SECNAVINST 5000.2G; DoDI 5000.85.

- **PEO/PA**: Owns program outcomes, charters PMs, approves acquisition strategies and baselines.
- **DRPM**: Direct-report program leads for special access or priority portfolios with the same milestone authority as PEOs.
- **Ship Program Manager (SPM)/PM**: Accountable for cost/schedule/performance, leads risk management, orchestrates IPTs.

- **Ship Acquisition Program Manager (SHAPM)**: Delivers new hulls; coordinates design/build/activation sequencing.
- **Ship Lifecycle Manager (SLM)**: Drives sustainment and modernization packages for in-service assets.
- **PARM/Ship Project Director (SPD)**: Sponsors platform-level upgrades and alteration packages.

Technical Authority chain.
Source: SECNAVINST 5400.15D.

- **UTA / Chief Engineer**: Issues enterprise technical policy, owns warrants.
- **TWHs**: Certifies design compliance, adjudicates departures/concurrence packages.
- **Engineering Agents (ISEA, DA, AEA, SIA, TDA)**: Execute lifecycle engineering; provide in-service engineering decisions under delegated authority.

Operational test oversight.
Sources [52, 69].

- **DOT&E:** Independent OSD authority that concurs on TEMPs, monitors operational test realism, and reports suitability/effectiveness outcomes to Congress.
- **Service Operational Test Agencies.** COMOPTEVFOR (COTF), AFOTEC, ATEC, and MCOTEA plan and execute IOT&E/FOT&E, render the OTRR recommendation, and provide Service suitability findings to the MDA.

Operational test readiness and fleet release.
Sources [71, 98].

- **COTF/COMOPTEVFOR:** Chairs OTRR events, certifies whether OT&E adequacy and logistics/training enablers support Fleet introduction, and can recommend restricted Fleet release when Cat I deficiencies remain.
- **MDA/PEO:** Disposition Cat I/II findings, resource fixes, and may defer FRPDR until OT&E adequacy and deficiency burn-downs meet Title 10 standards.
- **TYCOM:** Accepts Fleet risk, sets operating restrictions, and updates tactics and training based on OT&E and LFT&E findings.

Developmental test governance.
Sources [71].

- **DASW(T&E):** Designates Chief Developmental Testers and Lead Developmental Test and Evaluation Organizations for ACAT I programs, adjudicates test resource shortfalls, and co-chairs the Defense T&E Working-level Integrated Product Team (WIPT).
- **CDT:** Synchronizes the integrated test schedule, deficiency management cadence, and readiness evidence required for each TRR/milestone.

- **LDTO:** Provides instrumentation, modeling & simulation accreditation, range execution, and authoritative data packages for developmental tests.

Live-fire statutory leads.

Sources [71, 72].

- **DOT&E Live Fire Test Director:** Approves plans, monitors execution realism, and delivers the statutory lethality/vulnerability report to Congress.
- **Program Manager/Chief Engineer:** Funds threat-representative articles, integrates survivability design fixes, and aligns LFT&E data with the Acquisition Program Baseline and sustainment plans.
- **Service Safety Centers and Warfare Centers:** Provide casualty data, threat surrogates, and post-test forensic analysis that feed both live-fire reports and Fleet tactics updates.

NAVSEA headquarters roles.

Source: EDO Coursebook Module 2.1.2 (NAVSEA Organization, 2025 edition).

- **SEA 01 Comptroller**: BSO lead for NAVSEA financial governance and funds control.
- **SEA 02 Contracts**: Enterprise contracting authority, policy, and warrant management.
- **SEA 03 Cyber Engineering and Digital**: Drives cyber resiliency, digital engineering, and data transformation initiatives.
- **SEA 04 Industrial Operations**: Oversees public shipyards, maintenance execution, and quality assurance.
- **SEA 05 Chief Engineer**: Chief TA; issues warrants, certifies designs, and maintains technical standards.
- **SEA 06 Sustainment**: Manages product support strategies and lifecycle logistics integration.
- **SEA 07 Undersea Warfare**: Dual-hatted as PEO UWS; oversees undersea combat systems sustainment.
- **SEA 08 Nuclear Propulsion**: Three-hatted nuclear propulsion authority across DON and DOE roles.
- **SEA 09 Safety and Regulatory Compliance**: Aligns enterprise safety governance and reporting.
- **SEA 10 Total Force and Corporate Ops**: Manages workforce planning, corporate services, and governance.
- **SEA 21 In-Service Ships/CNRMC**: Leads surface-ship sustainment and modernization (PMS 321/326/339/421/443/451, SEA 21I, SURFMEPP).

NAVWAR enterprise directorates.

Source: EDO Coursebook Module 2.1.3 (NAVWAR Enterprise, 2024 edition).

- **Code 1.0 Comptroller**: Budget formulation/execution, BSO duties, and funds certification.
- **2.0 Contracts**: Contracting policy, strategy reviews, and award/administration oversight.

- **3.0 Counsel**: Acquisition law, ethics, protests, and claim resolution.
- **4.0 Logistics and Fleet Support**: Lifecycle logistics, technical data, and fleet distance support integration.
- **5.0 Chief Engineer/TA**: Enterprise architectures, interoperability, and cyber certification authority.
- **6.0 Program Management**: Portfolio governance, milestone preparation, and PEO integration.
- **7.0 Science and Technology**: S&T portfolio management, prototyping, and technology transition.
- **8.0 Corporate Operations**: Workforce, CIO services, facilities, security, and public affairs.
- **FRD-100 Fleet Support**: Deployed engineering assistance and sustainment response.
- **FRD-200 Installations**: Shore/afloat C4I installation planning and cutover execution.

Program Executive Offices (Navy portfolios).
Sources: SECNAVINST 5400.15D; EDO Coursebook Modules 2.1.4 and 3.1.5.

- **PEO CVN**: Designs, builds, and sustains nuclear-powered aircraft carriers (e.g., PMS 312/378/379).
- **PEO IWS**: Develops and sustains ship/submarine combat systems; mission-aligned IWS directorates cover sensors, weapons, C2, and allied integration.
- **PEO Ships**: Oversees surface combatant and amphibious ship construction and modernization.
- **PEO USC**: Leads Littoral Combat Ship (LCS), Guided-Missile Frigate (FFG) 62, expeditionary, and unmanned surface/undersea portfolios.
- **Team Submarines (PEO SSN, PEO SSBN, Program Executive Office, Columbia-Class Submarines (PEO Columbia), PEO UWS)**: Manages attack/strategic submarine acquisition, in-service support, and undersea combat systems.
- **PEO C4I**: Delivers fleet C4I; PMW 1XX focus on capability development, PMW 7XX on platform integration.
- **PEO Digital**: Provides enterprise digital services (e.g., Flank Speed) across the DON.
- **PEO MLB**: Modernizes manpower, logistics, and business IT systems.

Contracting authority (KO family).
Sources: FAR; DFARS; NMCARS; EDO Coursebook Module 3.2.1.

- **PCO**: Plans the acquisition, synchs with PMs before solicitation, awards and signs contracts/mods.
- **ACO**: Oversees post-award performance, surveillance, and payment/contract administration (often DCMA / NAVSEA field activities).
- **TCO**: Leads partial/full terminations, settlement negotiations, and equitable adjustments.

- **KO warrant**: Defines the dollar/authority limits; only the warranted KO can bind or obligate the Government.
- **Contracting Officer's Representative (COR) / assistant PM / engineering support**: Provide technical surveillance and acceptance recommendations; cannot direct work or obligate funds (FAR Parts 1 and 42).
- **HCA**: Approves actions such as letter contracts and high-value single-award IDIQs; may delegate no lower than flag/Senior Executive levels within the DON per FAR 1.601 and DFARS/NMCARS supplements.
- **SAE, ASN(RD&A)**: Signs D&Fs for multiyear contracting, extraordinary relief, and other actions reserved to the Service Acquisition Executive under FAR Subparts 1.7 and 17.1.

Program Manager / KO partnership.

Sources: DoDI 5000.85; EDO Coursebook Module 3.2.1.

- PM integrates warfighter need, technical baseline, and budget; KO CICA, incentives, and change management before RFP release or modification execution.

Source selection governance.

Sources: FAR Parts 1, 5, and 15; NAVSEA Source Selection Guide (2022); EDO Coursebook Module 3.2.2.

- **SSA**: Senior official who approves the Source Selection Plan, receives SSAC/SSEB recommendations, and signs the best-value decision memorandum.
- **SSAC**: Advisory council that synthesizes SSEB findings, compares proposals across factors, and briefs the SSA on trade-offs.
- **SSEB**: Multi-disciplinary evaluators (technical, management, past performance, cost/price) who rate proposals against Section M factors.
- **Small Business Professional / Competition Advocate**: Confirms set-aside decisions, reviews subcontracting plans, and endorses synopsis waivers.
- **Cost/Price Analyst & Legal Counsel**: Validate reasonableness determinations, alignment of Sections L/M, and clause sufficiency before release.
- **Award Fee Determining Official & Award Fee Board**: Plans award-fee periods, chairs performance reviews, and signs the determination memo authorizing or withholding fee payments (FAR Part 16; NAVSEA Source Selection Guide).

Fiscal control & certification.

Source: Department of War Financial Management Regulation (DoW FMR) 7000.14-R Volume 3.

- **Funds Certifying Official/Comptroller**: Verifies purpose/time/amount before obligation; ADA safeguard.
- **Program/Budget Analyst**: Tracks execution, monitors reprogramming thresholds, prepares obligation/expenditure burn-down.

- **Resource allocation chain**: OUSW(C) apportions to the Navy; OPNAV (N82/FMB) allocates to BSOs; SYSCOM comptrollers (e.g., SEA 01) issue suballocations to executing activities.

PPBE resource sponsors.

Source: EDO Coursebook Module 3.1.1 (PPBE, 2025 edition).

- **CAPE**: Independent analytic staff supporting Program Review decisions and ICE/POE for major programs [25, 28].
- **N8**: Navy resource integrator (“checkbook”): builds a balanced Navy POM across portfolios.
- **N80**: Runs the POM build and integration; adjudicates sponsor submissions.
- **N81**: Provides force-structure and analytic assessments supporting POM trades.
- **N82**: Budget formulation/execution chain (FMB); manages allocation, execution controls, and reprogramming posture.
- **N83**: Campaign analysis and future force design; anchors wargames and mission-level analytics that frame POM decisions.
- **N84**: Capability integration and resources; aligns PEO execution phasing, offsets, and two-pass/seven-gate products with budget marks.
- **N89**: Advanced concepts and future force design cross-checks; wargaming and architecture integration to keep sponsor asks coherent with POM toplines.
- **N9**: Requirements sponsor (“demand signal”): validates/advocates warfare requirements.
- **N91**: Cross-domain integration; mission-level issue resolution (often labeled N9I).
- **N95/N96/N97/N98**: Expeditionary/surface/undersea/air warfare sponsors under N9.
- **N99**: Unmanned warfare systems sponsor; integrates unmanned air/surface/undersea capability portfolios and defends their investments in PPBE.
- **R3B/NCB**: Navy governance boards that align requirements and resourcing decisions feeding the POM.

Congressional resource chain.

Source: EDO Coursebook Module 3.1.2 (Congressional Enactment, 2025 edition).

- **HASC / SASC**: Authorize defense programs and policy in the annual NDAA.
- **HAC / SAC**: Produce appropriations bills that provide BA to execute Navy programs.
- **HBC / SBC**: Set topline guidance and 302 allocations through the budget resolution process.
- **CBO, CRS, GAO**: Supply independent scoring, research, and oversight that shape committee deliberations.

Cyber/Authorizations (when IT/C2 in scope).
Source: SECNAVINST 5000.2G.

- **Authorizing Official (AO)**: Grants Authority to Operate; enforces Risk Management Framework (RMF) controls that drive design and integration requirements.

B.2 Combatant Commands (Reference)

AOR: The geographic (or astrographic) area assigned to a CCDR for missions and forces. (Title 10)

Geographic (7):

- United States Africa Command (USAFRICOM)
- USCENTCOM
- USEUCOM
- United States Indo-Pacific Command (USINDOPACOM)
- United States Northern Command (USNORTHCOM)
- United States Southern Command (USSOUTHCOM)
- United States Space Command (USSPACECOM)[1]

Functional (4):

- United States Cyber Command (USCYBERCOM)
- United States Special Operations Command (USSOCOM)
- United States Strategic Command (USSTRATCOM)
- United States Transportation Command (USTRANSCOM)

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[1] USSPACECOM is treated by DoW as a **geographic (astrographic)** command with an AOR beginning at the Kármán line (∼100 km)

C EDO Distribution

C.1 EDO Billet Distribution (Nov 2025 Main Slate)

Purpose.

Quick reference to where EDOs have billets on the November 2025 main slate, grouped by community and sorted by location with billet counts by rank. Current as of 2025-11-01.

Method.

Parsed the "SLATE_EDO_MAIN" worksheet in *EDO Main Slate November 2025.xlsx* by mentor group (community), location (AHPORT), and billet rank (BRANK); rank order follows flag to junior officer seniority (VADM, RADM, RDML, CAPT, CDR, LCDR, LT, LTJG).

Location expansions.

Location codes were expanded best-effort based on billet context (e.g., ANNAP → Annapolis, MD; MONTEY → Monterey, CA). Confirmations are appreciated for potentially ambiguous codes listed after the table.

TABLE XXI: EDO BILLET DISTRIBUTION BY COMMUNITY, LOCATION, AND BILLET RANK (NOVEMBER 2025 MAIN SLATE)

Community	Location	Billet ranks (count)
ANY	Annapolis, MD	CAPT 1 / CDR 1 / LCDR 2 / LT 4
ANY	Cambridge, MA	CAPT 1 / LCDR 1
ANY	Crane, IN	CAPT 1
ANY	MacDill AFB (Tampa, FL)	CAPT 1 / CDR 1
ANY	Millington, TN	CAPT 1 / LCDR 3
ANY	Monterey, CA	CDR 1
ANY	Port Hueneme, CA	CAPT 1 / CDR 1 / LCDR 2
CC	Adelaide, Australia	CDR 1
CC	Arlington, VA	CDR 1
CC	Bath, ME	LCDR 1
CC	China Lake, CA	LCDR 1

Continued on next page

Table XXI: EDO billet distribution by community, location, and billet rank (November 2025 main slate) **(Continued)**

Community	Location	Billet ranks (count)
CC	Dahlgren, VA	CAPT 4 / CDR 6 / LCDR 7
CC	Dam Neck, VA	CDR 1
CC	Fort McNair, DC	CDR 1
CC	Moorestown, NJ	CAPT 2
CC	Norfolk, VA	CDR 3 / LCDR 2 / LT 3
CC	Pascagoula, MS	LCDR 1
CC	Pearl Harbor, HI	CDR 1
CC	Point Mugu, CA	CAPT 1 / CDR 2
CC	Port Hueneme, CA	CAPT 1 / CDR 4 / LCDR 3 / LT 2
CC	San Diego, CA	LCDR 1
CC	Tucson, AZ	CDR 1 / LCDR 1 / LT 1
CC	Wallops Island, VA	CAPT 2
CC	Washington Navy Yard, DC	LCDR 2
CC	Washington, DC	CAPT 12 / CDR 8 / LCDR 9 / LT 1
CC	White Sands, NM	CDR 1
CVN-C	NAS North Island (San Diego, CA)	CAPT 1 / CDR 3 / LCDR 1
CVN-C	Newport News, VA	CAPT 1 / CDR 4 / LCDR 5 / LT 5
CVN-C	Norfolk, VA	CAPT 1 / CDR 3 / LCDR 3
CVN-C	Point Mugu, CA	CAPT 1 / CDR 1
CVN-C	Portsmouth, VA	CAPT 1
CVN-C	San Diego, CA	CAPT 2 / CDR 5 / LCDR 3
CVN-C	Washington, DC	CAPT 4 / CDR 4 / LCDR 3
CVN-C	Yokosuka, Japan	LCDR 1
DIVER	Little Creek, VA	LCDR 1
DIVER	Manama, Bahrain	CDR 1 / LT 1
DIVER	NAS North Island (San Diego, CA)	LCDR 1
DIVER	Panama City, FL	CDR 1 / LT 1
DIVER	Pearl Harbor, HI	CDR 1 / LCDR 1
DIVER	Rota, Spain	LT 1

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Table XXI: EDO billet distribution by community, location, and billet rank (November 2025 main slate) **(Continued)**

Community	Location	Billet ranks (count)
DIVER	Washington, DC	CAPT 1 / LCDR 4 / LT 1
FLAG	Pearl Harbor, HI	RDML 1
FLAG	San Diego, CA	RDML 1
FLAG	Washington Navy Yard, DC	VADM 1
FLAG	Washington, DC	VADM 1 / RADM 2 / RDML 5 / LCDR 1
IMG	Agana Heights, Guam	CDR 3
IMG	Apra Harbor, Guam	CDR 1
IMG	Bangor, WA	CDR 1
IMG	Bremerton, WA	CAPT 4 / CDR 5 / LCDR 6 / LT 13
IMG	Fort Monroe, VA	CAPT 1
IMG	Groton, CT	CDR 1
IMG	Kaneohe Bay, HI	CDR 2
IMG	Keyport, WA	CAPT 1
IMG	Norfolk, VA	RADM 1 / CAPT 2 / CDR 3 / LCDR 2 / LT 1
IMG	Pearl Harbor, HI	CAPT 6 / CDR 5 / LCDR 6 / LT 10
IMG	Perth, Australia	LCDR 1
IMG	Point Loma, CA	CDR 1
IMG	Point Mugu, CA	CAPT 1 / CDR 1
IMG	Portsmouth, NH	CAPT 4 / CDR 3 / LCDR 8 / LT 8
IMG	Portsmouth, VA	CAPT 4 / CDR 5 / LCDR 5 / LT 11
IMG	Washington, DC	CAPT 3 / CDR 2 / LCDR 1
IWE	Agana Heights, Guam	LCDR 1
IWE	Bremerton, WA	LCDR 1
IWE	Chantilly, VA	CDR 1 / LCDR 6 / LT 7
IWE	Charleston, SC	CAPT 1 / CDR 1 / LCDR 1 / LT 3
IWE	Coronado, CA	LCDR 1
IWE	Falls Church, VA	CAPT 1
IWE	Fort Meade, MD	LT 1
IWE	Manama, Bahrain	LCDR 1

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Table XXI: EDO billet distribution by community, location, and billet rank (November 2025 main slate) **(Continued)**

Community	Location	Billet ranks (count)
IWE	Naples, Italy	CDR 1
IWE	New Orleans, LA	CDR 1
IWE	Norfolk, VA	CAPT 1 / CDR 1 / LCDR 2
IWE	Pearl Harbor, HI	CAPT 1
IWE	Point Mugu, CA	LCDR 1
IWE	Portsmouth, VA	LCDR 4 / LT 2
IWE	San Diego, CA	RADM 1 / CAPT 7 / CDR 14 / LCDR 21 / LT 5
IWE	Suffolk, VA	LT 1
IWE	Washington Navy Yard, DC	CDR 1
IWE	Washington, DC	CDR 2 / LCDR 3
IWE	Yokosuka, Japan	CDR 1 / LCDR 2
NR	Arlington, VA	CAPT 4 / CDR 4 / LCDR 9 / LT 9 / LTJG 1
NR	Bremerton, WA	LT 1
NR	Groton, CT	LCDR 1
NR	Pearl Harbor, HI	LCDR 2
NR	Washington, DC	CDR 1
NR	Yokosuka, Japan	LCDR 1
RESERVE	Bremerton, WA	LCDR 1
RESERVE	Pearl Harbor, HI	LCDR 1
RESERVE	Portsmouth, VA	LT 1
RESERVE	Washington, DC	CAPT 1 / LT 1
SMMR	Al Jubayl, Saudi Arabia	LCDR 1
SMMR	Arlington, VA	CAPT 1
SMMR	Coronado, CA	CAPT 1 / CDR 2 / LCDR 6 / LT 2
SMMR	Everett, WA	CAPT 1 / CDR 1 / LT 3
SMMR	Little Creek, VA	LCDR 1
SMMR	Manama, Bahrain	CDR 2 / LCDR 1 / LT 1
SMMR	Mayport, FL	CAPT 2 / CDR 2 / LCDR 6 / LT 2
SMMR	Naples, Italy	CAPT 2

Continued on next page

Table XXI: EDO billet distribution by community, location, and billet rank (November 2025 main slate) **(Continued)**

Community	Location	Billet ranks (count)
SMMR	Norfolk, VA	CAPT 6 / CDR 9 / LCDR 10 / LT 9
SMMR	Pearl Harbor, HI	CAPT 1 / LCDR 2
SMMR	Point Mugu, CA	CDR 1
SMMR	Riyadh, Saudi Arabia	CDR 1
SMMR	Rota, Spain	CDR 1 / LT 1
SMMR	San Diego, CA	CAPT 4 / CDR 5 / LCDR 5 / LT 6
SMMR	Sasebo, Japan	CDR 2 / LCDR 1 / LT 3
SMMR	Singapore	CDR 1
SMMR	Washington, DC	LCDR 3
SMMR	Yokosuka, Japan	CAPT 1 / CDR 3 / LCDR 4 / LT 5 / LTJG 1
SSP	Cape Canaveral, FL	CDR 1
SSP	Kaneohe Bay, HI	CAPT 1 / CDR 1 / LCDR 1 / LT 1
SSP	Little Rock, AR	CAPT 1 / CDR 1 / LCDR 2 / LT 1
SSP	Magna, UT	CDR 1
SSP	Pittsfield, MA	CAPT 1 / LCDR 1
SSP	Silverdale, WA	CAPT 1 / CDR 1 / LCDR 1 / LT 1
SSP	Sunnyvale, CA	LCDR 1
SSP	Titusville, FL	LCDR 1
SSP	Uniondale, NY	LT 1
SSP	Washington Navy Yard, DC	CAPT 5 / CDR 8 / LCDR 7 / LT 2
SURFPK	Bath, ME	CAPT 1 / CDR 1 / LCDR 1
SURFPK	Bethesda, MD	CAPT 1
SURFPK	Marine Corps Base Camp Lejeune, NC	CDR 2 / LCDR 3 / LT 2
SURFPK	Mobile, AL	CDR 1 / LCDR 2
SURFPK	New Orleans, LA	CDR 2 / LCDR 1 / LT 1
SURFPK	Norfolk, VA	LCDR 2
SURFPK	Pascagoula, MS	CAPT 2 / CDR 6 / LCDR 4 / LT 4
SURFPK	Philadelphia, PA	CAPT 1 / LCDR 1
SURFPK	Point Mugu, CA	CDR 1 / LCDR 1

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Table XXI: EDO billet distribution by community, location, and billet rank (November 2025 main slate) **(Continued)**

Community	Location	Billet ranks (count)
SURFPK	San Diego, CA	CDR 1 / LCDR 1 / LT 1
SURFPK	Washington, DC	CAPT 10 / CDR 22 / LCDR 7
UEDO	Arlington, VA	LCDR 1
UEDO	Groton, CT	CAPT 1 / CDR 4 / LCDR 10 / LT 4
UEDO	Newport News, VA	CDR 2 / LCDR 3 / LT 5
UEDO	Norfolk, VA	LCDR 1
UEDO	Point Mugu, CA	CDR 1
UEDO	Portsmouth, NH	CAPT 1
UEDO	San Diego, CA	LCDR 1
UEDO	Washington, DC	CAPT 7 / CDR 9 / LCDR 18

Unconfirmed locations. Please confirm or correct these best-effort expansions: ALJUBA (assumed Al Jubayl, Saudi Arabia), CAMBRI (assumed Cambridge, MA), FINEGA (assumed Finegayan, Guam), LITTLE (assumed Little Rock, AR), MAGNA (assumed Magna, UT), MARINE (assumed Marine Corps Base Camp Lejeune, NC), NHPA (assumed Naval Hospital Pensacola, FL), PNT (assumed Point Mugu, CA), and UNDALE (assumed Uniondale, NY).

C.2 EDO Flag Officers (snapshot)

Verify names/billets the week of your board. Current as of: 05 Dec 2025 (sourced from Navy flag officer biographies).

Non-EDO billets (for awareness only). The NAVSEA December 2025 org chart shows several key billets led by non-EDO officers: NAVSEA 08 (ADM William J. Houston, Submarine officer), PEO IWS (RDML Thomas J. Dickinson, Surface Warfare AP), PEO MLB (RDML Kevin P. Biehn, Surface Warfare AP), NAVSEA 05 (RDML Jay A. Seif, Submarine AP), PEO UWS (RDML Douglas J. Adams), PEO SSBN/SEA 07 (RDML Todd Weeks), and PEO USC (currently SES; RDML Kevin R. Smith was relieved in May 2025)[9].

Vice Admirals

VADM Seiko Okano.
PMILDEP to ASN(RD&A); former COMNAVWAR and PEO C4I / Project Overmatch lead[4].

Rear Admirals (Upper Half)

RADM Brian A. Metcalf.
PEO Ships; delivers surface combatants, auxiliaries, and amphibious ships[6].

RADM Casey J. Moton.
PEO CVN; leads CVN new construction and in-service nuclear availability portfolios[7].

RADM Jonathan E. Rucker.
PEO SSN; oversees attack submarine acquisition and lifecycle support[8].

RADM J. Kurt Rothenhaus.
COMNAVWAR; commands NAVWAR and its PMWs delivering C4I/cyber capabilities[5].

RADM Douglas L. Williams.
Director for Test, MDA; synchronizes joint missile defense test campaigns[141].

Rear Admirals (Lower Half)

RDML Peter D. Small.
NAVSEA Chief Engineer and Deputy Commander for Naval Systems Engineering; Commander, NSWC/NUWC[142].

RDML Dan L. Lannamann.
CNRMC; oversees regional maintenance centers and depot availabilities[103].

RDML Dianna Wolfson.
Director, Fleet Maintenance, USFF; integrates TYCOMs readiness and ship maintenance execution[143].

RDML Daniel W. Ettlich.
Director, Fleet Maintenance (N43), PACFLT; leads west-coast and forward-deployed maintenance readiness[144].

RDML Scott M. Brown.
Deputy Commander, Industrial Operations, NAVSEA; oversees shipyard and industrial base sustainment[145].

RDML Huan Nguyen.
Deputy Commander for Cyber Engineering, NAVSEA; integrates cybersecurity and digital engineering into ship programs[146].

RDML Jonathan J. Jett-Parmer.
Deputy Commander, Supervisor of Shipbuilding, Conversion and Repair (NAVSEA); leads SUPSHIP technical authority for new construction and overhauls[147].

Reserve EDO Flags

RDML Robert J. Dodson (USNR).
Deputy Commander, Supervision of Shipbuilding (NAVSEA); aligns reserve expertise to construction oversight[148].

RDML Michael S. Richman (USNR).
Deputy Director, Regional Strike Systems, Strategic Systems Program (SSP); supports strategic weapons sustainment[149].

D Community Rosters and Job Pyramids

Distribute responsibly: rosters may contain sensitive personal information.

D.1 Cannon Cocker

D.2 IWE

D.2.1 Roster (IWE Round-Up, 12 Sep 2025)

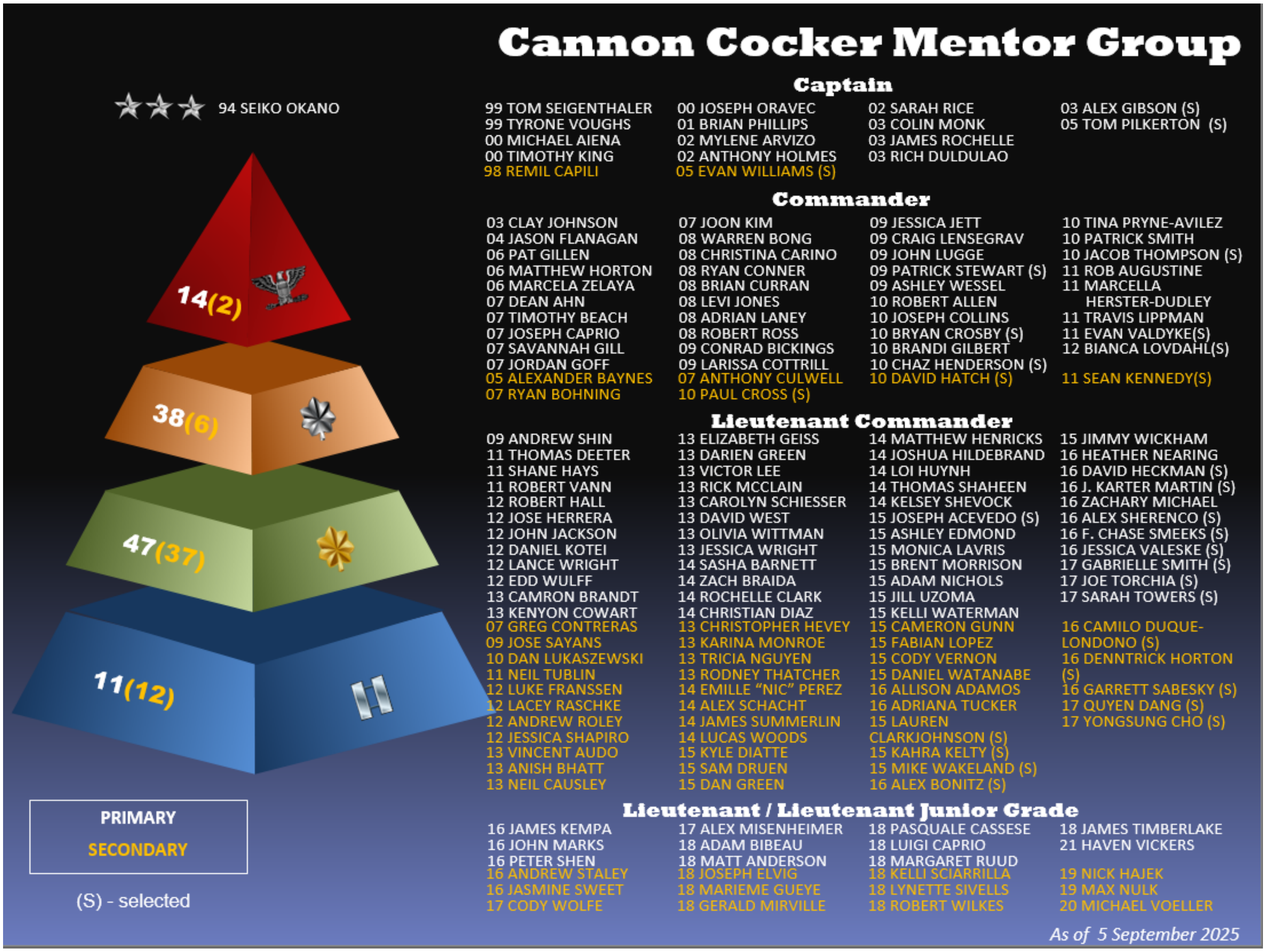

Figure D.1. Cannon Cocker roster.

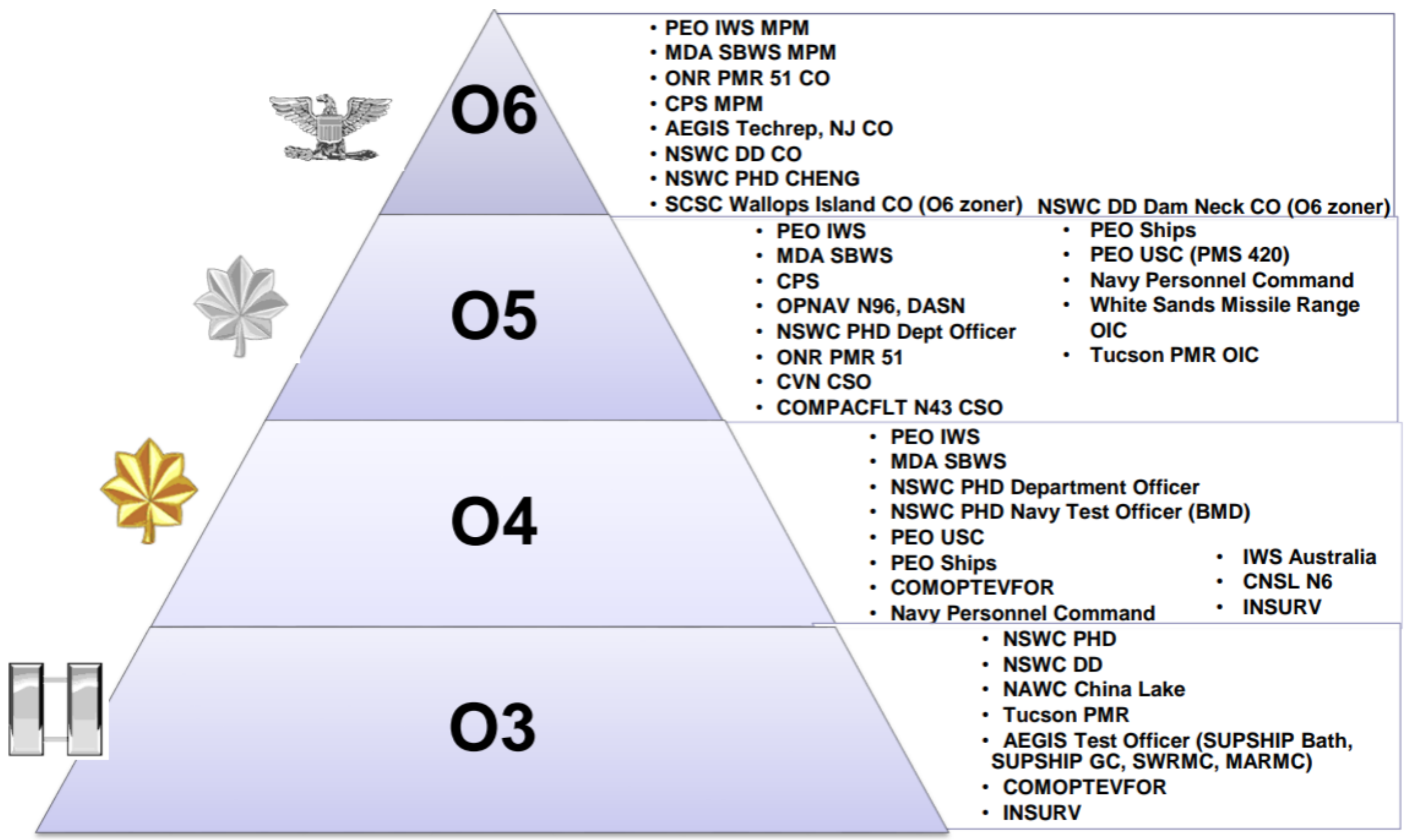

Figure D.2. *Cannon Cocker pyramid.*

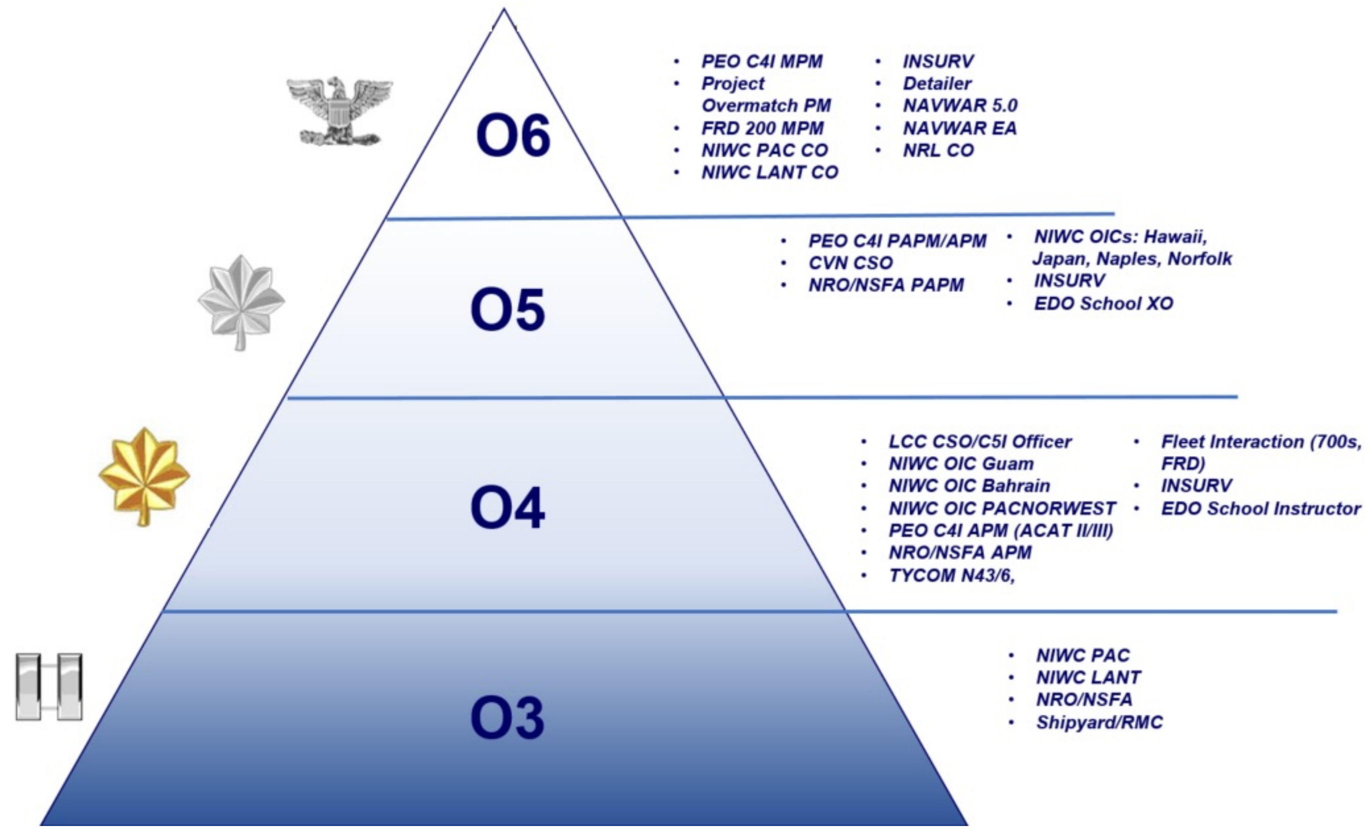

Figure D.3. *IWE job pyramid.*

NIWC Pacific EDOs

Distribute Responsibly - May Contain Sensitive Personnel Info

– CDR Jarrod Ozereko
- FIO Military Lead Manager
- Military Group Head for Code 4.0
- EDQP Counseling Officer

– LT Marieme Gueye
- FIO Project Officer
  - USS CAPE ST. GEORGE, USS CHARLESTON, USS COMSTOCK, USS HARPERS FERRY USS J.P.MURTHA, USS MOBILE, USS OAKLAND, USS MOBILE, USS SAVANNAH

– LT Gerard Mirville
- FIO Project Officer
  - USS CHOSIN, USS MAKIN ISLAND, USS BOXER, USS LAKE ERIE, USS ESSEX, USS TRIPOLI, USS GREEN BAY, USS AMERICA, USS PRINCETON, USS PEARL HARBOR

– LT Se-hoon Kim
- FIO Project Officer
  - USS MOMSEN, USS PAUL HAMILTON, USS SOMERSET, USS CURTIS WILBUR, USS FITZGERALD, USS LENAH H SUTCLIFF-HIGBEE, USS MICHAEL MONSOOR

– LT Quyen Dang
- Automated Digital Network Systems (ADNS) Tactical Transport Engineering (ATTE)
- Deputy Integrated Product Team Lead for Software Define Network Engineering Evolution (SDNE2)

– LT Lynette Sivells
- FIO Project Officer
  - USS STETHEM, USS ASHLAND, USS MANCHESTER, USS ANCHORAGE, USS CHUNG HOON, USS O'KANE, USS OMAHA

– LT Cody Wolfe
- FIO Project Officer
  - USS CHAFEE, USS HALSEY, USS PINCKNEY, USS RUSSELL, USS SPRUANCE, USS STERETT, USS STOCKDALE

8

Figure D.4. *IWE roster (NIWC Pacific EDOs) from IWE Round-Up (12 Sep 2025).*

NSFA EDOs

Distribute Responsibly - May Contain Sensitive Personnel Info

- **CAPT Steve Werner**
  - Program Manager, SIGINT
- **CDR Patrick Gillen**
  - Mission Services, GEOINT
- **CDR Sade Jurgensen**
  - Branch Chief, GEOINT
- **CDR Alex Williams**
  - Branch Chief, SIGINT
- **CDR(Sel) Mark Mueller**
  - Systems Engineer
- **LCDR Neil Causley**
  - COTR, COMM
- **LCDR Neil Tublin**
  - IPT Lead, COMM
- **LCDR Anna Peterson**
  - Project Officer
- **LCDR Daniel Green**
  - Branch Chief, GEOINT
- **LCDR Victor Lee**
  - Project Officer, GEOINT
- **LT Alexandra Sherenco**
  - Space Systems COTR, SIGINT
- **LT Margaret Ruud**
  - Project Officer, COMM
- **LT Kelli Sciallia**
  - Project Officer, GEOINT
- **LT Robert Wilkes**
  - Project Officer, SIGINT
- **LT Alex Misenheimer**
  - Project Officer, SIGINT
- **LT Zachary Michael**
  - Project Officer, SIGINT

AS&T – Advanced Systems and Technology Directorate
COMM – Communications Systems Directorate
GED – Ground Enterprise Directorate
GEOINT – Geospatial Intelligence Systems Acquisition Office
RSPO – Radar Systems Program Office
SAO – Survivability Assurance Office
SED – Systems Engineering Directorate
SIGINT – Signals Intelligence Systems Acquisition Office

13

Figure D.5. IWE roster (NSFA EDOs) from IWE Round-Up (12 Sep 2025).

E CNO Strategic Messaging

E.1 CNO 34 Day One Message (Foundry–Fleet–Fight)

NAVADMIN 180/25; issued upon assuming duties as the 34th CNO [150].

Mission.

The Navy shall be organized, trained, and equipped to deliver peace through strength, securing our national interests and prosperity. We exist for prompt and sustained combat at sea, preserving freedom of navigation, safeguarding global commerce, and deterring those who would challenge America's values and sovereignty.

Vision.

We stand at an inflection point—an era marked by great power competition, proliferating threats, rapid technological convergence, and an increasingly contested maritime domain. To prevail, we must build and sustain a Navy that is ready to fight and win—today, tomorrow, and well into the future. At the center of this vision is the Sailor. Our men and women are the Navy's main weapon system—their skill, resilience, and courage remain our greatest asymmetric advantage. We will put them first by relentlessly pursuing full-spectrum readiness, deepening cross-domain integration, harnessing innovation at speed, and strengthening our warrior ethos. Our Navy must be resilient, agile, globally present, and combat credible. We will align tightly with national defense priorities, expand our ability to integrate with Joint Force and Allied partners, and preserve our enduring advantage at sea. In doing so, we will guarantee freedom of navigation, deter aggression, and project power where and when it matters most.

Priorities.

We will view everything we do through an operational lens focused on the Foundry, the Fleet, and the way we Fight.

Foundry.

The Foundry represents the enduring sum of our total force, shore infrastructure, maintenance depots, schoolhouses, industrial base, and intellectual capital required to generate, sustain, and modernize naval power. Every ounce of sea power we wield originates in the Foundry; it is the bedrock of our Navy. We will not neglect the global network of infrastructure and capacity that enables our Sailors to excel. From new ship construction to depot-level repair, from weapons production to the training pipelines that develop warfighters, the Foundry underwrites combat credibility. To remain the world's most formidable fighting force, we will operationalize readiness, accelerate shipbuilding and repair at scale, and modernize our force-generation model. This means strengthening public-private partnerships, investing in resilient supply chains, expanding workforce development, and ensuring our training systems deliver Sailors fully prepared for the high-end fight. Simply put—without a strong Foundry, there is no Fleet.

Fleet.

The Fleet is forged in the Foundry and is our Nation's decisive instrument of maritime power. Comprised of people, platforms, and payloads, it is designed to deter aggression, project power, and secure our national interests on and from the sea. No image symbolizes American resolve more than a U.S. Navy ship on the horizon—lethal, resilient, and ready to fight. The Fleet's unique mobility, persistence, and global reach enable us to impose costs on adversaries, reassure allies, and guarantee the free flow of commerce. Tempered by our resolve, our Fleet is world-class, battle-ready, and second to none. With our Sailors at the helm, supported by strong Navy families, there is no challenge we cannot overcome. We will continue to build an integrated, all-domain Fleet—one that fuses maritime (surface, air, and undersea), cyber, space, and strategic capabilities while leveraging Joint Force and Allied partners to dominate across the competition continuum. Interoperability and interchangeability with allies and partners will be our force multiplier. By enhancing collective maritime security, we will magnify our forward presence and strengthen deterrence worldwide.

Fight.

Fighting is our ultimate honor and sacred responsibility—delivering violence on behalf of the Nation when called. For nearly 250 years, the U.S. Navy has honed its toughness, tenacity, and ingenuity to secure victory. Those who challenge us in combat will be met with overwhelming force and sent to the bottom. But we must fight differently in the future. In today's rapidly evolving landscape, we will innovate, adapt, and integrate cutting-edge technologies at the speed of relevance. We will deliver and refine the Navy Warfighting Concept and develop a new Navy Deterrence Concept, both anchored in peace through strength. We will accelerate the integration of artificial intelligence, robotic and autonomous systems, resilient Command, Control, and Communications (C3) architectures, and synchronized effects—ensuring decision advantage, survivability, and combat overmatch. We will ruthlessly pursue a Future Fleet Design—comprised of unmatched conventionally manned ships, deeply integrated with asymmetric hedge forces that expand our reach and impose costs more effectively across every domain. Our charge is clear: To ensure the Navy remains the most lethal, survivable, and globally dominant maritime force.

Theory of victory.

At the end of my tenure as CNO, success will be measured by results:

- Platforms delivered and repaired on time.
- Fully manned and combat-ready ships.
- Ordnance production meeting contracted demand.
- Backlogs in repair parts eliminated.
- Sailors trained to the highest levels of mastery.

Through disciplined execution, we will increase our lethality and preserve our sustained advantage. To achieve this, we are calibrating our sights, refining our course, and transforming our approach of solving our Navy's toughest problems. Guided by the Navy Warfighting and Deterrence Concepts, we will develop and execute a holistic Campaign Plan that delivers the warfighting capability and capacity necessary to win in the 21st century. To maintain transparency and unity of effort, I will release a series of "CNO Notes"—or "C-Notes"—providing guidance and direction to the Fleet. These will be grounded in my vision, informed by your feedback, and driven by operational necessity. They will require your trust, ownership, and commitment to be successful. I am proud to serve alongside you—America's Sailors—as we embark on this next chapter in our Navy's history. Built in the Foundry—Tempered in the Fleet—Forged to Fight.

Daryl L. Caudle
Admiral, United States Navy
34th Chief of Naval Operations

E.2 CNO 34 C-Notes and Charge of Command

NAVADMINs 186/25, 203/25, 241/25, and 225/25. [151, 152, 153, 154]

E.2.1 Executive summary of priorities

Overview.

ADM Caudle's initial set of C-Notes and his Charge of Command reinforce the Foundry–Fleet–Fight construct: care for Sailors, strengthen the Navy–industrial team, and deliver a combat-credible fleet that can deter and win. Leaders are directed to build trust, enforce standards, and maintain a visible warfighting mindset across every command.

Overview.

ADM Caudle's initial set of C-Notes and his Charge of Command reinforce the Foundry–Fleet–Fight construct: care for Sailors, strengthen the Navy–industrial team, and deliver a combat-credible fleet that can deter and win. Leaders are directed to build trust, enforce standards, and maintain a visible warfighting mindset across every command.

C-Note #1: Sailors First.

The Fleet's advantage is a trained, supported Total Force. Priorities include modern housing and galley upgrades, streamlined uniform requirements, resilient digital connectivity ashore, and pay/benefits accuracy so every Sailor can focus on the fight.[151]

C-Note #2: Foundry Always.

The Foundry—people, infrastructure, and materiel—forges naval power. Caudle calls for relentless training (Career Training Continuum), expanded live-virtual-constructive integration, stronger shipyard shifts, revitalized intermediate maintenance activities, and modernized shore platforms and supply chains to deliver ready forces at scale.[152]

C-Note #3: World Class Fleet.

The Fleet is the decisive instrument of national power, integrating people, platforms, and payloads across all domains. The note introduces a Hedge Strategy (tailored offsets and tailored forces) to outpace threats, modular payload investments, resilient Command and Control (C2)/Artificial Intelligence (AI)/unmanned integration, clarified administrative/operational control, and undersea C2 realignments (Commander, Task Force (CTF)-54/74).[153]

Charge of Command.

Command is an action and a privilege. Commanding Officers are accountable for warfighting readiness, disciplined risk management, mission command, continuous learning, and visible ownership of their Sailors' welfare and performance. Immediate superiors will review the Charge with subordinate COs to ensure shared understanding and mentorship.[154]

E.2.2 Message texts

C-Note #1: Sailors First (NAVADMIN 186/25).

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CLASSIFICATION: UNCLASSIFIED/
ROUTINE
R 031453Z SEP 25 MID180002011151U
FM CNO WASHINGTON DC
TO NAVADMIN
INFO CNO WASHINGTON DC
BT
UNCLAS
NAVADMIN 186/25
MSGID/NAVADMIN/CNO WASHINGTON DC/CNO/SEP//
SUBJ/C-Note #1: Sailors First//
RMKS/1. The lifeblood of our Navy is the Sailor: well-trained, connected,
supported and fit to fight. In every ocean around the world, our national
security interests are challenged by increasingly complex threats.
Underpinned by our warrior ethos, we will rise to that challenge. For 250
years, we have been a world-class fighting Navy, and our strength lies in our
Sailors. Our teamwork and esprit de corps are strengthened by our core
values and devotion to our mission.
2. We must resource the Navy our Nation Needs. I am committed to the well-
being of our Sailors and their families - providing them with state-of-the-
art platforms, world-class facilities and dependable support through our
Total Sailor and Family Action Plan.
We will leverage the insights and lessons learned from the past to improve
and rebrand our Culture of Excellence initiative to its new focus - Total
Sailor: Fit to Fight.
3. Today, I have directed your Navy leadership to take the steps necessary
to ensure that No Sailor Will Live Afloat. We will invest in unaccompanied
and family housing in addition to optimizing our new BAH authority to ensure
every Sailor resides in housing that is clean, comfortable and safe.
4. During my term, we will transform our galley facilities to ensure they
are designed to meet the nutritional needs and choices of our Sailor's
modern-day lifestyle. It is paramount that we modernize food service to
deliver healthy and flexible options.
5. Effective immediately, the Secretary of the Navy has directed we convene
a focused assessment of your uniforms and seabag. We will streamline uniform
requirements to ensure you have affordable, available and durable service
clothing that you are proud to wear.
The goal will be to reduce the number of uniforms while respecting our hard-
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earned traditions that makes our Navy so special.
6. You have grown up in a digitally connected world where disconnection
means isolation. Therefore, I have mandated efforts to improve cell service
on our installations and bring free Wi-Fi to all of our barracks.
7. I have also ordered changes to the NAVADMIN process ensuring our Sailors
are able to find clear and correct information in one singular location. We
will not use NAVADMINs to work around the effort required to properly change
the base instructions any longer.
8. Additionally, I've directed Navy leadership to revisit prior efforts such
as Sailor pay to ensure our previous corrective actions are still working so
that each Sailor is receiving the correct amount of pay and entitlements in a
timely manner.
9. You are my priority and I need you focused on the fight. This is the
first of many efforts to improve your Quality of Life - so, stand by for
what's coming next.
10. Built in the Foundry - Tempered in the Fleet - Forged to Fight.
11. ADM Daryl Caudle, 34th Chief of Naval Operations sends.
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CLASSIFICATION: UNCLASSIFIED/
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C-Note #2: Foundry Always (NAVADMIN 203/25).

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CLASSIFICATION: UNCLASSIFIED/
ROUTINE
R 222218Z SEP 25 MID120012080004U
FM CNO WASHINGTON DC
TO NAVADMIN
INFO CNO WASHINGTON DC
BT
UNCLAS
NAVADMIN 203/25
MSGID/NAVADMIN/CNO WASHINGTON DC/CNO/SEP//
SUBJ/C-NOte #2: Foundry Always//
RMKS/1. The Foundry is the engine that drives our warfighting advantage. It
transforms raw inputs and forges them into lethal
outputs: ready warfighters, platforms and sustained readiness, at the speed
and scale necessary to win.
2. The Foundry builds, generates, sustains and modernizes naval power
through three critical components: PEOPLE, INFRASTRUCTURE and MATERIEL.
3. People: Our Total Force, comprised of Sailors, civilians and the
industrial workforce, is our most decisive warfighting advantage.
To prevail in conflict, we must recruit, train, educate and retain a
technically proficient and resilient team. Investing in our people's
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development and well-being is not separate from combat power, it is the very foundation of our Warrior Ethos, which is not exclusive to those in uniform. It's a mindset marked by honor, integrity, resilience and a relentless commitment to excellence and service.

That ethos must live across the spectrum: from our schoolhouses and combat centers to the workforce that lays our keels and develops our intellectual property. Every role our Total Force fills is vital by design.

4. To keep pace with the proliferation of technology and the global evolution of threats, we are enhancing the quality of our curricula. Relentless improvement, technical competence and tactical mastery are all tenets demanded of a confident warfighter. Empowered by timely action and responses from our Navy's top leaders, we will inculcate a culture of continuous learning that delivers world-class, modernized and battle-ready Sailors. Therefore, we will rename Ready Relevant Learning to "Career Training Continuum" (CTC), the Navy's enterprise program of record for individual training and career-long learning.

5. Through a series of sprints, we will also expand the Military Learning Continuum and Civilian Workforce Development. With built-in feedback loops and mechanisms in place, we will rapidly identify, assess, respond to and correct gaps in our Sailor's training journey.

I have also directed that our Live-Virtual-Constructive (LVC) efforts be expanded past the current use cases (e.g., integrated C2X training, Fallon airwing events, etc.) to include a more generalized view of LVC that spans from unit level training to schoolhouse utilization to tailored wargaming.

6. Battle effectiveness is not achieved by only afloat platforms or commands. Simply put, without a strong Foundry, there is no effective Fleet. Therefore, I am working to rewrite the Battle "E"
instruction to recognize shore installations with exceptional readiness and performance.

7. Infrastructure: Shore platforms generate naval power. Our global network of infrastructure, shipyards, airfields, barracks, electrical grids and piers are foundational to projecting that power. Modernization ensures our readiness can scale across the spectrum of conflict. We will prioritize and strengthen our shore platforms to support Fleet-wide operations and ensure our People take pride and ownership in the places that they work in and live.

8. Expanding the Fleet's deployable platforms is a whole-of-Navy evolution, and while we are not short on ideas to get there, our execution has not met expectations. The levers: innovation, oversight, optimization and recapitalization are known, but the fastest gains in production will come from our People and their elbow-grease. To meet operational timelines, we must prioritize workforce development, labor capacity and world-class project management to increase our efficiency.

9. Accordingly, we will strengthen second and third shifts in our public shipyards to increase throughput and on-time delivery. Only by improving

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shipyard training, workplace conditions and organizational support can we
reduce attrition, accelerate experience and retain the workforce, both
private and public, responsible for delivering our Fleet.
10. I will release a future C-NOte on Esprit de Corps Ashore, emphasizing
the critical role our infrastructure plays in forging warfighting readiness.
11. Materiel: Reliable supply chains, accurate Coordinated Shipboard
Allowance Lists (COSALs), comprehensive maintenance capabilities and
coordinated logistics are instrumental to keeping our Fleet in the fight.
Winning in conflict demands a strong Navy-Industrial Base partnership,
grounded in transparency, common understanding and intellectual honesty.
12. Our current maintenance structure lacks the right balance of Sailors,
civilians, artisans and engineers. Years of depot and activity level
consolidation have reduced opportunities for our Sailors to master the skill
sets needed to repair and sustain at sea.
13. For these reasons, I am pursuing the stand-up of Shore Intermediate
Maintenance Activities (SIMA) in Norfolk and San Diego.
These facilities will provide our Sailors with hands-on training in advanced
ship repair and expose them to modern capabilities such as AI/ML, advanced
manufacturing (e.g., 3-D printing, CAD/CAM, etc.), workflow monitoring
technologies and robotic systems.
14. We are also reviewing Class Maintenance Plans. To maintain a combat
surge-ready posture, our maintenance models must evolve.
Introducing shorter, more frequent and efficient availabilities will
increase our readiness, improve on-time completion and reduce risk at a
comparable cost.
15. Foundry is more than a metaphor; it is the process, standard and
discipline that underpins our Navy's ability to deliver ready, lethal and
resilient forces, on time, on cost and at scale.
16. Built in the Foundry - Tempered in the Fleet - Forged to Fight.
17. ADM Daryl Caudle, 34th Chief of Naval Operations sends.//
BT
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CLASSIFICATION: UNCLASSIFIED/
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C-Note #3: World Class Fleet (NAVADMIN 241/25).

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CLASSIFICATION: UNCLASSIFIED/
ROUTINE
R 012011Z DEC 25 MID120012255382U
FM CNO WASHINGTON DC
TO NAVADMIN
INFO CNO WASHINGTON DC
BT
UNCLAS
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NAVADMIN 241/25
MSGID/NAVADMIN/CNO WASHINGTON DC/CNO/DEC//
SUBJ/C-Note #3: World-Class Fleet//
RMKS/1. History proves that great nations are built on the strength of great navies. And today, our Fleet stands as the world's premier warfighting force - ready, capable, and unmatched. But we are more than ships, aircraft, and weapons. Our Sailors - battle ready, trained, resilient, and forged in the Foundry are the engine that empowers our Fleet. Through their strength, we command the seas, deter aggression, defend our homeland, and when called, deliver decisive combat power. This is what makes us a world-class Navy - second to none.
2. The Fleet is our most decisive instrument of national power and serves as the differentiated value our Navy brings to the Joint Force: expeditionary reach, unmatched mobility, and persistent presence that promotes our vital interests, protects the global commons, and wins our Nation's wars. Comprised of People, Platforms, and Payloads, the Fleet spans across all domains. To our Allies, it serves to reassure. To our Adversaries, it deters and shapes behavior. To all, it is a signal that freedom of the seas is non-negotiable and will never be surrendered.
3. As threats evolve and the world grows more complex, we will continue to build an integrated, all-domain Fleet - capable of conducting sea denial and, when needed, sea control within the battlespace and at decisive global chokepoints. This means fusing maritime, cyber, space, and undersea capabilities, and integrating Joint and combined forces to dominate across the spectrum of conflict. In short, we will not only design our forces to fight and win, but to outpace emerging threats through smart and cost-effective "hedges" to lower the risks faced by our general purposes forces while increasing survivability and lethality.
4. To that end, building a Fleet to cover every possible threat is too expensive, unrealistic, and sub-optimized. Instead, I have directed the development of a new Navy Strategy. Built on the fundamentals from our Navy Warfighting and Deterrence Concepts, this new effort introduces an enduring "Hedge Strategy" - designed as an approach that reduces operational risk by investing in Tailored Offsets and Tailored Forces as a way to amplify and synergize our general purposes forces to more effectively address low probability-high consequence contingencies.
5. At its core, our Navy is already a collection of "hedge" forces and concepts: risk-informed decisions about where to invest and how to fight while weighing the probability of occurrence in terms of severity and loss. For example, "Hellscape" is a use case of the generalized Hedge Strategy. Naval Special Warfare is our hedge against low intensity and irregular warfare threats. SSBNs and E-6Bs are our hedge that deters strategic attack with less than eight percent of the Navy. And our Navy-Marine Corps team itself, has and will continue to serve as the Joint Force's global hedge - forward deployed, postured, and ready to respond to a variety of threats at a

time and place of our choosing.
6. Tailored Forces are logical groupings of tailored offsets that strengthen our general purpose forces. The Global Maritime Response Plan (GMRP) and associated Readiness Conditions (RESCON) provide tailored force packages that are Combat Surge Ready (CSR) in support of our Combatant Commanders through tailored certifications to execute specific missions. This is in run today as we continue to work to achieve our 80 percent CSR goals for all platforms.
7. Further, we will also invest in platforms, sensors, and weapon systems built with modularity that are scalable and ready for rapid upgrade. We will accelerate the delivery of integrated, networked capabilities - including unmanned systems, Artificial Intelligence, and resilient Command and Control (C2) architectures to ensure decision advantage and guarantee our overmatch in contested environments around the world.
8. Many of our Navy's toughest challenges cannot be solved in one year or even across one five-year Future Years Defense Program (FYDP) budgeting cycle. These types of longer-range problems (e.g., spare parts, critical munitions, foundry improvements, shipbuilding, and depot repair enhancements) require time horizons and funding profiles with trajectories that balance fiscal realities to operational risk.
Therefore, I have directed the development and use of multi-year/multi-FYDP "funding curves" to ensure sustained investment in the capabilities at the required capacity our warfighters need.
With clear "commitment to the curve" - I will prioritize the required sustained investments necessary to solve many of the problems that have hampered our Navy for decades.
9. The Administrative Chain of Command (ADCON) is comprised of subject matter experts who form the best force generators in the Navy. Over time we have drifted into commingling force generation and force employment with overlapping C2 structures that lack clear accountability. Therefore, we will review existing C2 structures from an operational task perspective and align C2 to the risk owner.
Additionally, we will review and align force generation responsibilities to ensure we are following the "Standard Model" of excellence. This will ensure that we clearly delineate the role of the appropriate Type Commander (TYCOM) to the Fleet Commander with clean and accountable hand-offs between force generation and force employment of their forces.
10. Additionally, to most effectively counter the pacing threat that the People's Republic of China presents, CTF-54 will be separated from CTF-74 allowing each to hone their focus on specific undersea mission sets, particularly in our most challenging threat environments. CTF-54 will be dual-hatted as Commander, Submarine Squadron 21 in Bahrain. To support the new CTF-54, primary broadcast control authority and submarine operating authority duties (e.g., waterspace management) will be shifted to Commander, Submarine Group 8 and CTF-69 in Naples, Italy. To support CTF-74 more fully across the Western Pacific Area of Responsibility, we will be assessing the

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establishment of Task Groups to enable clear and unambiguous C2 lines of
operational accountability.
11. Fleet Improvement Offices (FIOs) are driving a culture of continuous
improvement, intellectually honest assessments, disciplined execution, as
well as improved operational risk management and accountability. We are
codifying this through a new instruction, under the Office of Warfighting
Advantage (OWA), to scale FIO practices across our Foundry, Fleet, and Fight
priorities.
12. Built in the Foundry - Tempered in the Fleet - Forged to Fight.
13. ADM Daryl Caudle, 34th Chief of Naval Operations sends.//
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Charge of Command (NAVADMIN 225/25).

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CLASSIFICATION: UNCLASSIFIED/
ROUTINE
R 302016Z OCT 25 MID120012171056U
FM CNO WASHINGTON DC
TO NAVADMIN
INFO CNO WASHINGTON DC
BT
UNCLAS
NAVADMIN 225/25
MSGID/NAVADMIN/CNO WASHINGTON DC/CNO/OCT//
SUBJ/CHARGE OF COMMAND - ANNOUNCEMENT TO THE FLEET//
RMKS/1. As the 34th Chief of Naval Operations (CNO), I've released my Charge
of Command for officers selected for command, emphasizing the unique
responsibilities and privileges inherent in leading warfighters at a
strategic inflection point. Commanding Officers (COs) hold the highest
operational and moral accountability for their units, and their leadership
directly shapes combat readiness, culture, and mission success.
2. In the Charge of Command, I characterize command as "an action, not a
position," highlighting the essential attributes of integrity, courage,
humility, and ownership. The Charge emphasizes that the Navy's decisive
advantage is its Sailors, and COs must invest in their people with the same
intensity as operational readiness, fostering trust, accountability, and a
warfighting culture that thrives under pressure.
3. Command inherently involves risk. The Charge directs COs to manage
operational risk deliberately, provide clear Commander's intent, and empower
teams under Mission Command principles.
Disciplined risk-taking, sound judgment, and adaptability are required to
ensure mission success while cultivating initiative and learning throughout
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the command.
4. The Charge further stresses the importance of preparation, assessment,
and continuous learning. COs are expected to plan, rehearse, test
assumptions, and innovate faster than potential adversaries. Excellence is
achieved through initiative, critical thinking, disciplined failure as a
learning pathway, and fostering a culture of resilience and innovation.
5. COs are the single accountable officer for their units and are expected
to lead decisively, mentor future leaders, and uphold the Navy's enduring
values and warfighting mindset. COs and their commands must always be ready
to Fight Tonight. As Fleet Admiral Chester Nimitz once said, "When you're in
command, command."
6. Immediate Superior in Command commanders are expected to review the
charge with their subordinate COs, ensuring they understand, internalize, and
apply its principles. This review supports mentorship, reinforces
accountability, and strengthens unit readiness across the fleet.
7. The full Charge of Command letter, including detailed guidance and
reference materials are available at: https://www.navy.mil/
All personnel are encouraged to read, understand, and support the principles
outlined to strengthen their commands and the Navy as a whole.
8. Built in the Foundry - Tempered in the Fleet - Forged to Fight!
9. ADM Daryl Caudle, 34th Chief of Naval Operations sends.//
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CLASSIFICATION: UNCLASSIFIED/
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F Current Events

F.1 Current Events Relevant to EDOs and Acquisition

F.1.1 Project Overmatch and Digital Modernization

Five Eyes Project Arrangement.

In Oct 2024 Project Overmatch signed a Project Arrangement with Australia, Canada, New Zealand, and the United Kingdom, embedding Cooperative Project Personnel in San Diego to co-develop resilient coalition kill-chain networks ahead of RIMPAC 26; Capt. Remil Capili (PM), VADM Seiko Okano (steering chair/PMILDEP), Shayna Bond (international deputy), and David Flowers (international lead) direct the effort while a coalition lab stands up to support continuous testing[22].

Open DAGIR pilot.

By 18 Jun 2025 Project Overmatch, NAVWAR, the Chief Digital and Artificial Intelligence Office, and Defense Innovation Unit (DIU) were onboarding commercial AI/ML applications through the Open DAGIR pipeline so Fleet commanders could visualize readiness data ("Maven Smart Systems") across the hybrid fleet in weeks instead of years [23].

Leadership signal.

VADM Seiko Okano continues to serve as the direct reporting program manager and steering committee chair while serving as PMILDEP to ASN(RD&A), keeping Overmatch digital architecture tied to Program Decision Meetings and coalition governance[3, 22].

F.1.2 EDO Community Updates

NAVWAR change of command.

Rear Adm. (now VADM) Seiko Okano relieved Rear Adm. Doug Small on 9 Aug 2024, assuming command of NAVWAR and direct oversight of Project Overmatch; she emphasized information warfare dominance and continued alliance engagement with industry and academia [155].

Role of the PMILDEP.

As PMILDEP to ASN(RD&A), Okano now chairs Program Decision Meetings when delegated and synchronizes PAEs and SYSCOM technical authority, a critical linkage as Capability Trade Councils and portfolio scorecards begin replacing JCIDS artifacts [3, 80].

Tycom maintenance focus.

USFF and PACFLT maintenance flag officers continue to drive execution of regional maintenance center reforms and the Naval Sustainment System–Shipyards initiatives highlighted in the 2025 coursebook; these billets remain key board discussion topics because they link EDO leadership to depot performance [66].

F.1.3 Acquisition Policy and Industrial Base Highlights

WAS.

The Acquisition Transformation Strategy replaces JCIDS with Capability Portfolio Management and Capability Trade Councils, empowering PAEs to adjudicate cost/schedule/performance trades in real time and document decisions in portfolio scorecards [80].

Industrial base actions.

SECWAR directed the Department to expand multi-year procurements, establish an Industrial Base Consortium, and reform bid protest incentives so the Munitions Acceleration Council and similar bodies can stabilize demand and attract private capital [80].

PPBE integration.

USW(C) and USW(A&S) are piloting portfolio budgets and flexible execution authorities so PAEs can move funds inside approved controls while maintaining transparency with OMB and Congress—a major shift boards will expect candidates to explain [80].

Security cooperation realignment.

DSCA and DTSA now fall under USW(A&S), aligning exportability, Foreign Military Sales, and industrial-base planning so allied orders and U.S. production trades can be brokered inside Capability Trade Councils [84].

National Security Capital Forum.

Section 1092 of the FY 2025 NDAA created the NSCF, chaired by the Office of Strategic Capital, to coordinate public/private financing decisions that affect defense materiel transactions—EDOs should highlight how their programs leverage or are scrutinized by the forum [85].

F.2 Shipbuilding and Sustainment Updates

F.2.1 Columbia-Class (SSBN) Delays

The Columbia-class ballistic missile submarine program, the Navy's highest acquisition priority, experienced significant schedule pressure in 2025. Navy officials reported an estimated 12- to 17-month delay in the delivery of the lead boat, United States Ship (USS) *District of Columbia* (SSBN-826), pushing its first planned patrol into 2030 [156]. The delays were attributed to a combination of factors, including shipyard workforce shortages, supply chain disruptions for critical components like the bow dome and main turbine generators, and the inherent complexities of a first-of-class build [156, 157]. By late 2025, General Dynamics Electric Boat reported the lead vessel was approximately 60% complete, but the tight schedule margins remain a significant concern for maintaining the nation's continuous strategic deterrent capability [158].

F.2.2 Amphibious Warship Block Buy (LPD-17)

In a significant move to stabilize the amphibious ship industrial base, the Navy executed a multi-ship “block buy” procurement for four amphibious warships from Huntington Ingalls Industries (HII) Ingalls in late 2024 and early 2025 [159]. This procurement, valued at nearly $1 billion in savings, includes three San Antonio-class amphibious transport docks (Landing Platform Dock (LPD)-33, 34, and 35) and one America-class amphibious assault ship (LHA-10) to be built between FY25 and FY29 [159]. The block buy provides workload stability for the shipyard and its suppliers, enabling more efficient production and ensuring the Navy meets the legally mandated 31-ship amphibious force structure [160]. This decision followed congressional pressure and signals a commitment to the amphibious fleet after a period of debate over future requirements.

F.2.3 Shipyard Infrastructure Optimization Program (SIOP) Status

The Shipyard Infrastructure Optimization Program (Shipyard Infrastructure Optimization Program (SIOP)), a 20-year, $21 billion effort to modernize the Navy’s four public shipyards, reached a critical phase in 2025. A key milestone was the award of a $377.7 million contract for seismic upgrades to Dry Dock #4 at Puget Sound Naval Shipyard, which is essential for servicing nuclear submarines, including the future Columbia-class SSBNs [161]. However, the program faces significant cost growth due to inflation and unforeseen construction complexities, with annual construction cost escalation exceeding 6% [162]. In July 2025, program officials announced a comprehensive review to revise cost and schedule estimates for the entire SIOP portfolio [162]. Despite these challenges, the Navy continues to invest approximately $2.7 billion annually, with the goal of reducing submarine maintenance periods by up to 15% and returning ships to the fleet faster [162].

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List of Acronyms

Acronym	Definition
COTF	Commander, Operational Test Force
DCAA	Defense Contract Audit Agency
DCMA	Defense Contract Management Agency
KO	Contracting Officer
RADM	Rear Admiral (Upper Half)
RDML	Rear Admiral (Lower Half)
RFP	Request for Proposal
ROA	Return on Assets
ROS	Return on Sales
SAE	Service Acquisition Executive
VADM	Vice Admiral